Rel-Sortability of Time-Unrolled Causal Graphs

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Abstract

Time-unrolled directed acyclic graphs with strongly connected summary graphs have been proposed as causal-discovery benchmarks on which sorting variables by their number of relatives (rel-sortability) is uninformative. We compute the exact rel-sortability of every edge of a time-unrolled graph, for every summary graph with self-loops and every horizon. Strong connectivity does not protect at benchmark horizons: a directed ring is perfectly rel-sortable up to a horizon equal to its size. Much of the excess comes from the fixed start: after a latent burn-in as long as the largest finite distance in the summary graph, the value is fixed by the source strongly connected components, and this burn-in cannot be shortened. These results are machine-checked in Lean 4. With time labels hidden, the fixed start raised the order accuracy of a relatives-sorting baseline by 0.13 in simulations; the exact values predict the sign and rank the conditions, and a post hoc detectability calculation reproduces the size. The fixed start also makes the data var-sortable; on raw data DAGMA and NOTEARS beat FLOP on in-trees but lost to it after a burn-in, a ranking change we had not predicted. With time labels the start identifies nothing beyond the reduced form, and under a fixed start the time order is identifiable, given faithfulness, up to explicit exceptions. Finite-sample results give an effective sample size and show that a latent driver saturates relatives counts; on a river benchmark the count was saturated; a physical testbed gave partial support. All failed tests are reported.

Keywords: causal discovery, sortability, time series, benchmark simulation, formal verification, finite-sample theory

1 Introduction

Simulated benchmarks drive much of causal discovery, and several of their regularities can be exploited without any causal reasoning. Marginal variances tend to grow along

the causal order (*var-sortability*; Reisach, Seiler, & Weichwald, 2021), so a method that sorts by variance does well on data from common generators, and it explains the strong benchmark performance of some continuous-optimisation methods, a family that includes NOTEARS (X. Zheng, Aragam, Ravikumar, & Xing, 2018). The coefficient of determination R^2 has the same property and is scale invariant (Reisach, Tami, Seiler, Chambaz, & Weichwald, 2023); internally standardised structural models are not var-sortable by construction and, empirically, mostly not R^2-sortable (Ormaniec, Sussex, Lorch, Schölkopf, & Krause, 2025); and Herman, Wahl, Ninad, and Runge (2025) relate the R^2 pattern to colliders and confounders and propose a generator that avoids it. Most recently, Reisach, Chambaz, Blanchard, and Weichwald (2026) defined *rel-sortability*, which orders a node by the number of its *relatives* (nodes joined to it by an open path, marginally dependent on it under faithfulness). Relatives are a property of the graph alone, so the criterion cannot be removed by reweighting. For random DAGs it is high; for *time-unrolled* DAGs built from a summary graph with self-loops, Reisach et al. (2026, Thm. 4) show that it tends to 1/2 if every weakly connected component of the summary graph is strongly connected, and they propose such graphs as benchmarks. Only sufficiency is shown in the arXiv version; they do not analyse the value beyond this case.

Three questions remain open, and they matter for real time series, whose summary graphs are often feed-forward (rivers, supply chains, signalling cascades): (Q1) what is the rel-sortability of a time-unrolled DAG when the summary graph is not strongly connected; (Q2) does it depend on how the series is started and observed; and (Q3) can a relatives count be estimated from one finite, autocorrelated record, so that a method could actually exploit it?

Contributions. We read the exact values as statements about benchmark generators. **(1) Short horizons.** The exact rel-sortability of every edge of the time-unrolled DAG G_T, for every summary graph with self-loops and every T (Theorem 1); its limit $\frac{1}{2} + |E_\times|/(2|E_\mathrm{S}|)$ with the rate $\min(\Delta, T-1)/(T-1)$ (Theorem 2); and necessity of the condition of Reisach et al. (2026, Thm. 4) (Corollary 3). Strongly connected summary graphs are not protected at the horizons proposed for benchmarks (a ring has $s_\rho(G_T) = 1$ for all $T \leq p$). **(2) Burn-in.** The excess for chains and out-trees is a boundary effect of the deterministic start (Proposition 4); after a latent burn-in of $B \geq \Delta$ slices the value is exact for every T and fixed by the *source* strongly connected components (Theorem 5 and Corollary 6), and $B \geq \Delta$ cannot be lowered (Remark 21). (1) and (2) are machine-checked in Lean 4 (Section 7). **(3) Benchmark effect** (Section 4). With time labels hidden, the fixed start inflates rel-sortability (the exact values give the sign and the ranking of conditions, not the size) and var-sortability. On raw data DAGMA and NOTEARS overtake FLOP on in-trees under the fixed start and fall behind it after a burn-in, which we had not predicted; FLOP remains the best label-free method on average. We attribute this ranking change to var-sortability, as a hypothesis that our theory does not cover. The experiment, its replication and the extension to competitive learners each had a time-stamped plan; a survey of deployed generators shows which are exposed. **(4) Finite samples, aggregation and real data.** Limits for directed Erdős–Rényi summary graphs proved on paper (Proposition 7); an effective sample size and the saturation of relatives counts

by a latent driver (Section 6); and real-data studies with time-stamped analysis plans, reported with every failed test (Section 8). Block means and subsamples are treated in Section 5. **(5) Time labels and identifiability** (Sections 4.2 and 4.3; paper proofs, not machine-checked). For a method that has the time labels, the start law carries no information about the mechanism beyond the reduced form (Proposition 8), so the fixed start is an artefact for label-free methods only. For those, the slice of every column is a graph invariant of the unrolled DAG (Proposition 9), its Markov equivalence class is explicit (Proposition 10), and under the fixed start and faithfulness the unrolled DAG is identified from second moments up to the reversal of isolated units (Proposition 11), with three exact non-identifiability results and one open conjecture for the burn-in window (Proposition 12 and Conjecture 13).

The graph results of (1) are short consequences of a walk-length lemma and of Reisach et al. (2026, Lemma 1). Their value is that they determine the value for every summary graph, horizon and start, and change what a benchmark designer should conclude: much of the rel-sortability of time-unrolled benchmarks is an artefact of how the simulation is started. The finite-sample results use textbook tools. We see the paper as a completion of the unrolled-graph analysis of Reisach et al. (2026, Section 5) plus a finite-sample and real-data layer, not as a new theory of sortability.

Related work. Varsortability (Reisach et al., 2021) and R^2-sortability (Reisach et al., 2023) have been analysed for standardised SCMs (Ormaniec et al., 2025), for unit-free generators with a time-series extension (Herman et al., 2025), and, for time series, by Lohse and Wahl (2025), who adapt both criteria to summary graphs, note that cyclic pairs push the signal towards 1/2, and measure them on river-flow and Causal Chambers data; they give no theorem and do not study relatives or observation schemes. Reisach et al. (2026) introduce rel-sortability and the time-unrolled DAG, and prove that strong connectivity of every weak component of the summary graph is sufficient for the limit 1/2. Aggregation and subsampling change time-series models (Breitung & Swanson, 2002; A.A. Weiss, 1984) and identifiability (Gong, Zhang, Schölkopf, Glymour, & Tao, 2017; Gong, Zhang, Schölkopf, Tao, & Geiger, 2015); their effect on sortability is in Section 5. The size distortion of a Pearson test with the nominal sample size under autocorrelation is classical (Bartlett, 1935; Clifford, Richardson, & Hémon, 1989; Roy, 1989) and is handled in practice by effective sample sizes (Afyouni, Smith, & Nichols, 2019; Runge, Nowack, Kretschmer, Flaxman, & Sejdinovic, 2019); testing the independence of two series through lagged cross-correlations goes back to Haugh (1976). Residual confounding by unreliable covariates is also known (Westfall & Yarkoni, 2016). Formal libraries for causal inference exist (Tan & Syrgkanis, 2026). Time-series discovery methods such as PCMCI and PCMCI+ (Runge, 2020; Runge, Nowack, et al., 2019), VAR-LiNGAM (Hyvärinen, Zhang, Shimizu, & Hoyer, 2010) and DYNOTEARS (Pamfil et al., 2020) use time order, and benchmarks for them include CauseMe (Runge, Bathiany, et al., 2019), the Causality 4 Climate competition (Weichwald et al., 2020) and CausalTime (Cheng et al., 2024); our finite-sample results concern only the marginal tests that define relatives. Robustness of score-based learners to violated assumptions is studied by Montagna et al. (2023). Continuous-optimisation learners are developed and compared largely on simulated DAGs, which is why regularities of the generator matter for their

evaluation. This holds for the linear learners tested in Section 4 (NOTEARS, DAGMA, Bello, Aragam, & Ravikumar, 2022, and GOLEM, Ng, Ghassami, & Zhang, 2020) and for nonlinear additive-noise learners such as SPADE (Dang, Le, & Dinh, 2026), a B-spline (Kolmogorov–Arnold) DAG learner that was compared with DAGMA, GraN-DAG, GOLEM, NOTEARS, SCORE, NoGAM, PCMCI and VAR-LiNGAM on simulated nonlinear DAGs, with mixed results against NoGAM, and whose graph learned from a financial panel in levels collapsed in a re-analysis on stationary returns: an instance, on real data, of a learned graph driven by non-stationarity and not by the mechanism.

2 Setting

For a directed graph, $\mathrm{anc}(y)$ and $\mathrm{desc}(y)$ contain y; the relatives of y are $\mathrm{rel}(y) = \mathrm{desc}(\mathrm{anc}(y)) = \{z : \mathrm{anc}(y) \cap \mathrm{anc}(z) \neq \emptyset\}$ and $\rho(y) = |\mathrm{rel}(y)|$. For an edge (x, y), $s_\rho(x, y) = 1, \frac{1}{2}, 0$ if $\rho(x) <, =, > \rho(y)$, and $s_\rho(G)$ is the mean over *edges*, as in Reisach et al. (2026, Eq. (4)) (quoted verbatim in Appendix B). A value $1/2$ means the criterion carries no orientation information.

A *summary graph* $G_\mathrm{S} = (V, E_\mathrm{S})$ has $p \geq 2$ nodes and a self-loop at every node. The *time-unrolled DAG* G_T has nodes $v(t)$, $t \in \{1, \ldots, T\}$, and an edge $(i(t), j(t{+}1))$ for each $(i, j) \in E_\mathrm{S}$ and $t < T$, so every summary edge has $T - 1$ copies, all at lag one (a VAR(1)-shaped unrolling; other lags are discussed after Theorem 5). $d(u, v)$ is the shortest-path length in G_S (self-loops ignored; $d(v, v) = 0$; ∞ if unreachable), and anc_S, rel_S are taken in G_S. SCC means strongly connected component. $E_\times \subseteq E_\mathrm{S}$ is the set of edges between different SCCs and $E_\circ = E_\mathrm{S} \setminus E_\times$. A *source* SCC has no incoming edge from another SCC, and $\mathrm{Src}(v)$ is the set of source SCCs that reach v.

3 Exact Rel-Sortability of Time-Unrolled DAGs

Theorem 1 (every edge, every T) *For $(i, j) \in E_\mathrm{S}$ and $1 \leq t < T$, the edge $(i(t), j(t{+}1))$ of G_T has $s_\rho = 1$ if some $u \in V$ satisfies $d(u, j) \leq t \leq d(u, i)$, and $s_\rho = 1/2$ otherwise. The value does not depend on T.*

Lean: edge_score_dist, edge_score, strictCond_iff_dist.

Sketch. Every edge of G_T advances time by one, so $u(s)$ is an ancestor of $v(t)$ iff $d(u, v) \leq t - s$ (pad a shortest path with self-loops). Hence the roots of $v(t)$ are $\{u(1) : d(u, v) \leq t - 1\}$. Since $i(t)$ is a parent of $j(t{+}1)$, $\mathrm{rel}(i(t)) \subseteq \mathrm{rel}(j(t{+}1))$, and the inclusion is strict iff $j(t{+}1)$ has a root that $i(t)$ lacks (Reisach et al., 2026, Thm. 1, Lemma 1). Full proof in Appendix B. (The Lean proof characterises rel in G_T directly and does not use their Lemma 1.)

Theorem 2 (exact limit and rate) *For every $T \geq 2$, with $L(G_S) = \frac{1}{2} + \frac{|E_\times|}{2|E_S|}$ and $\Delta \leq p-1$ the largest finite shortest-path distance in G_S,*

$$0 \leq s_\rho(G_T) - L(G_S) \leq \frac{\min(\Delta, T-1)\,|E_\circ|}{2(T-1)|E_S|} \leq \frac{p-1}{2(T-1)},$$

so $\lim_{T\to\infty} s_\rho(G_T) = L(G_S)$.

Lean: sRho_unrolled_exact, sRho_unrolled_bounds_maxFinDist, tendsto_sRho_GT.

Sketch. For a crossing edge, j does not reach i; take $u = j$ in Theorem 1: all $T-1$ copies score 1. For an edge inside an SCC, i and j are reached by the same nodes, which forces $t \leq d(u,i) \leq \Delta$, so at most Δ copies score 1.

Corollary 3 (necessary and sufficient condition) $\lim_T s_\rho(G_T) = 1/2$ *iff every weakly connected component of G_S is strongly connected. Otherwise $s_\rho(G_T) \geq \frac{1}{2} + \frac{1}{2|E_S|}$ for every T. The "if" half is Reisach et al. (2026, Thm. 4). If G_S is acyclic apart from self-loops, with m other edges, $L = \frac{1}{2} + \frac{m}{2(m+p)}$ whatever its shape (chain, in-tree, out-tree, collider).*

Lean: tendsto_half_iff, crossEdges_eq_empty_tfae, sRho_unrolled_ge_of_cross, reisach_theorem4, limitValue_dag.

Proposition 4 (the excess is a boundary effect) *For $t \geq p$, $\rho(v(t)) = T\,|\mathrm{rel}_S(v)| - \sum_{w\in\mathrm{rel}_S(v)} d(\mathrm{anc}_S(v), w)$.*

Lean: rho_unrolled_large_t, rho_unrolled_eq.

So for a crossing edge with $\mathrm{rel}_S(i) = \mathrm{rel}_S(j)$ (root to child in an out-tree, a chain), the strict order rests on $O(1)$ extra relatives out of $\Theta(T)$, all located in the first p slices. Figure 1 shows the smallest case. A sample-based relatives count on a long record cannot see such a difference, which motivates the stationary version.

Theorem 5 (stationary start) *Observe the window of the last $T \geq 2$ slices of G_{B+T} with a latent burn-in $B \geq \Delta$ (or of the infinite-past graph on $V \times \mathbb{Z}$), with relatives computed in the full graph. Then every window node has $\rho_W(v(t)) = T|\mathrm{rel}_S(v)|$, and for every T*

$$s_\rho^W = \tfrac{1}{2} + \frac{|E_\uparrow|}{2|E_S|},\quad E_\uparrow = \{(i,j) \in E_S : \mathrm{Src}(i) \neq \mathrm{Src}(j)\}.$$

Lean: rhoBurn_eq_maxFinDist, sBurn_eq_maxFinDist, rhoZ_eq, sZ_eq, relSet_eq_iff_src; sharpness: edgeLoops_burn_sharp.

Sketch. After Δ slices every finite distance is absorbed, so $w(s) \in \mathrm{rel}(v(t))$ iff $\mathrm{anc}_S(v) \cap \mathrm{anc}_S(w) \neq \emptyset$. For an edge (i,j), $\mathrm{rel}_S(i) \subseteq \mathrm{rel}_S(j)$ with equality iff $\mathrm{Src}(i) = \mathrm{Src}(j)$. The burn-in cannot be shorter in general: for one edge with self-loops ($\Delta = 1$), $B = 0$ leaves $s_\rho^W \geq 2/3$ for every T (Remark 21).

Scope: unit lags. Theorems 1, 2 and 5 are stated, and machine-checked, for VAR(1)-shaped unrollings. If each summary edge (i,j) instead acts at a lag $\tau_{ij} \in$

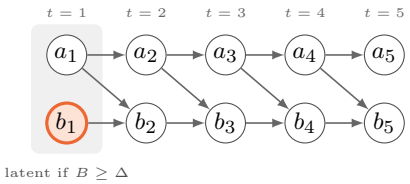


Fig. 1 Summary chain $a \to b$ unrolled to $T = 5$. Every a_t has the relatives $\{a_s\} \cup \{b_s : s \geq 2\}$ ($2T - 1$ nodes); every b_t with $t \geq 2$ also has b_1 ($2T$ nodes). So each edge $a_t \to b_{t+1}$ is strictly sorted (finite limit $L = 2/3$), but only because of the root b_1 at the fixed start. With a latent burn-in slice, both counts are $2T$ and $s_\rho^W = 1/2$.

Table 1 Rel-sortability of 3-node summary graphs (with self-loops, $|E_S| = 5$). Finite: $\lim_T s_\rho(G_T)$ (Theorem 2); stationary: s_ρ^W (Theorem 5). Lean: chain3_limit, chain3_stat, fork_limit, fork_stat, collider_limit, collider_stat; 3-cycle by reisach_theorem4.

summary graph	finite G_T (limit)	stationary
chain $a \to b \to c$	0.7	0.5
fork $a \leftarrow b \to c$	0.7	0.5
collider $a \to c \leftarrow b$	0.7	0.7
3-cycle (6 edges)	0.5	0.5

$\{1, \ldots, \tau_{\max}\}$ (self-loops at lag one), the same proofs go through with d replaced by the lag-weighted distance d_τ: the copy $(i(t), j(t+\tau_{ij}))$ is strict iff some u has $d_\tau(u, j) - \tau_{ij} + 1 \leq t \leq d_\tau(u, i)$, the limit is unchanged, and the stationary result needs a burn-in $B \geq \Delta\tau_{\max}$ (paper proof, Appendix N; not machine-checked). Contemporaneous (lag-0) edges are not covered.

Corollary 6 (stationary condition; graph classes) $s_\rho^W = 1/2$ *iff every weakly connected component of* G_S *has exactly one source SCC. Strongly connected graphs, out-forests, chains and forks give* $1/2$. *In an in-forest (a river network),* $s_\rho^W = \frac{1}{2} + \#\{\text{edges into confluences}\}/(2|E_S|)$, *which exceeds* $1/2$ *iff there is a confluence.*

Lean: sBurn_eq_half_iff_maxFinDist, sZ_eq_half_iff, stat_half_of_outForest, statValue_inForest, statValue_inForest_gt_iff, collider_stat, fork_stat.

Table 1 shows the two versions for 3-node graphs: chain and fork have limit 0.7 and stationary value 0.5, the collider 0.7 in both. In the finite generator every summary DAG with the same m/p has the same value, so a benchmark built this way rewards relatives-sorting on chains and forks, where the stationary process gives no signal; for long stationary records the stationary value is the relevant one. All statements of this section were checked by exhaustive enumeration ($p \leq 4$ and 10^4 random graphs with $p \leq 7$; 46,724 cases, 0 failures; Appendix C).

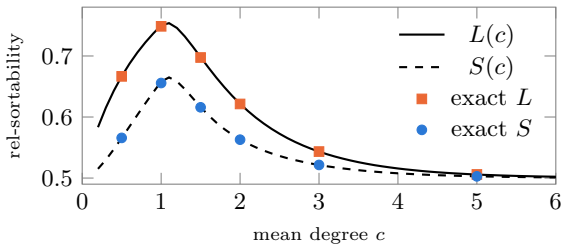


Fig. 2 Proposition 7 (lines) against exact computation on seeded graphs ($p = 2 \cdot 10^4$ to 10^5; points, means over seeds). The standard deviations over seeds are at most 0.0022 (Appendix O).

Without self-loops (Appendix M). Every edge of G_T still scores 1/2 or 1, independently of T, by an explicit walk-length criterion (machine-checked), and Reisach et al. (2026, Thm. 4) still holds, but the closed form and the converse of Corollary 3 fail: for the single edge $0 \to 1$ without loops $s_\rho(G_T) = 1/2$ for every T (with loops the limit is 2/3). The loop assumption is therefore necessary for the closed forms.

3.1 Random summary graphs

Let G_S be the directed Erdős–Rényi graph $G(p, c/p)$ plus a self-loop at every node, and x the largest root of $x = 1 - e^{-cx}$ ($x = 0$ for $c \leq 1$). By Theorems 2 and 5 only $|E_\times|$ and $|E_\uparrow|$ matter.

Proposition 7 (ER limits; proved on paper) *Let* $L(G_S) = \lim_{T\to\infty} s_\rho(G_T)$ *at fixed* p *and* $S(G_S) = s_\rho^W$. *In probability and in* L^1 *as* $p \to \infty$, $L(G_S) \to \frac{1}{2} + \frac{c(1-x^2)}{2(1+c)}$ *and* $S(G_S) \to \frac{1}{2} + \frac{c\,F_\uparrow}{2(1+c)}$ *with* $F_\uparrow = (1-x)(1-e^{-c}) + x(1-x)\big(1 - e^{-c(1-x)d}\big)$, *where* d *is the smallest fixed point in* $[0,1]$ *of* $d = 1 + e^{-c} - e^{-c(1-x)d}$ ($d = 1$ *for* $c \leq 1$).

Lean: only the deterministic witness lemmas for $E_\uparrow$ *(upEdge_witness, upEdges_subset_confluenceNL); the probabilistic proof (Appendix O.1) uses textbook Poisson-coupling, Chernoff and Galton–Watson facts (van der Hofstad, 2016). Not proved: rates, the window* $c = 1 + \lambda p^{-1/3}$, *a.s. convergence, fixed* T *as* p *grows.*

The first formula says that the fraction of non-loop edges inside an SCC tends to x^2 (giant strongly connected component, Dorogovtsev, Mendes, & Samukhin, 2001; Dovgal, de Panafieu, Ralaivaosaona, Rasendrahasina, & Wagner, 2020). $L(1) = 3/4$, L peaks at 0.754 near $c = 1.07$ and then decays like e^{-c}; S peaks at 0.665 near $c = 1.09$ (Figure 2). Random summary graphs are thus rel-sortable only near or below criticality, as the giant SCC removes the signal. For in-forests (rivers) the identities are exact (Appendix O).

What the value means. Show an edge in random order with the two relatives counts; the rule "orient towards the larger count, flip a coin on ties" has accuracy exactly s_ρ, never errs on an edge it decides, and no pairwise rule on the two counts does better (a few lines from Reisach et al.'s Theorem 1, Appendix P). So $s_\rho = 0.7$ caps such a method at 70% orientation accuracy. Rules that also see node degrees can exceed it.

Table 2 B1, heterogeneous noise, raw data; means over 180 data sets per family. OA: order accuracy of rel- and Var-SortnRegress, G1 / G2. ΔOA: G1−G2 for rel, with 95% graph-cluster bootstrap CI. SC: strongly connected.

family	exact s_ρ	rel OA	ΔOA rel	var OA
symER (SC)	.78/.50	.63/.50	+.126 [.115,.137]	.64/.50
ring (SC)	.95/.50	.69/.51	+.180 [.167,.193]	.73/.50
chain	.89/.50	.75/.53	+.217 [.209,.226]	.93/.70
out-tree	.84/.50	.69/.55	+.144 [.127,.161]	.94/.71
in-tree	.79/.66	.74/.67	+.069 [.057,.081]	.95/.72
ER(c=1)	.77/.56	.67/.59	+.085 [.068,.103]	.86/.65
ER(c=2)	.83/.52	.70/.59	+.109 [.092,.127]	.74/.59

Lean: accRule_sortRule, accRule_le_sRho, edgeAcc_sortRule_strict.

4 What the Start Artefact Does to Benchmarks

4.1 Benchmark experiments

Design (B1). We unrolled 420 summary graphs (30 per family and $p \in \{5, 10\}$; families in Table 2; details in Appendix Q) over $T \in \{3, 5, 10\}$ and drew $n = 1000$ linear-Gaussian trajectories (positive weights, heterogeneous noise) either with the fixed start of Reisach et al. (2026) (G1) or after a latent burn-in of 100 steps (G2; an exact stationary start reproduced G2 to within 0.013). Methods saw the pT columns shuffled, with time labels hidden: rel-, R^2- and Var-SortnRegress (Reisach et al., 2026, 2023), FLOP (Wienöbst, Henckel, & Weichwald, 2026), PC (Spirtes, Glymour, & Scheines, 2000) in causal-learn (Y. Zheng et al., 2024) and a random order; SID by gadjid (Henckel, Würtzen, & Weichwald, 2024). The analysis plan was committed before any algorithm was run (Appendix Q). CIs below resample whole summary graphs.

Rel-sortability: the exact values give sign and ranking. The fixed start raised rel-SortnRegress order accuracy in all five non-strongly-connected families (pooled over these five +0.125, 95% CI $[0.116, 0.134]$; Holm $p \leq 1.1 \cdot 10^{-10}$ per family; +0.133 over all seven). Because this order accuracy is essentially the estimated rel-sortability, the hypothesis tests whether the estimated counts track the exact values, not algorithm impact. They track, at about half the size: across the 42 (family, p, T) cells, Spearman 0.80 with the exact gap $s_\rho(G_T) - s_\rho^W$ (graph-cluster permutation $p \leq 10^{-4}$; 0.62 over graph–horizon units), slope 0.53. So the exact gap predicts the sign of the gain and ranks the cells, but not its size. A post hoc calculation (Figure 3, Appendix S) reproduces the size: the expected outcome of the same thresholded-correlation sort under the exact start-dependent window covariances, with the probability that each correlation is detected at $n = 1000$, matches the gain with cell $R^2 = 0.97$ and slope 1.03. This is the analytic expectation of the algorithm, so it checks that the simulated gain has the size its covariances imply; it does not separate the structural mechanism of Section 3 (pairs that are relatives under one start only)

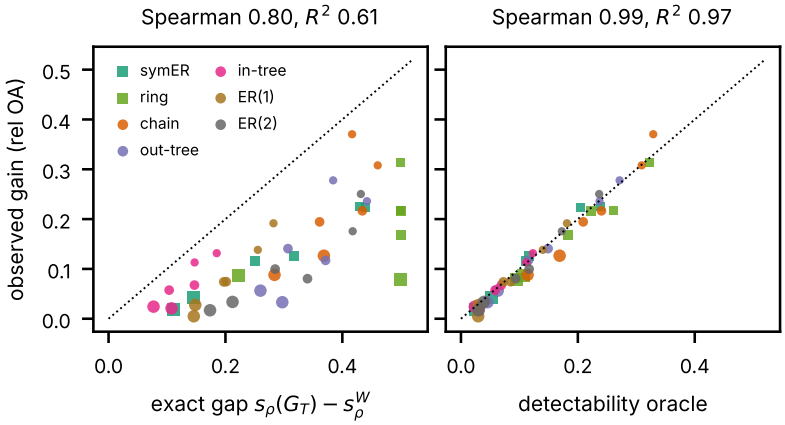


Fig. 3 B1: observed G1−G2 gain in rel-SortnRegress order accuracy per (family, p, T) cell (marker size grows with T; squares: strongly connected) against the exact population gap (left) and against the post hoc expectation of the algorithm under the exact window covariances (right). Dotted: identity.

from a general change of correlation strength during the transient. In a joint regression the exact gap adds nothing to it (coefficient −0.05). On rings with $p = T = 10$, for example, the gap is 0.50 but only 40% of the relatives have a detectable correlation; the calculation gives +0.094 and we observed +0.079. The gain does not come from the latently confounded first window slice: without edges leaving it, it is +0.089 $[0.079, 0.100]$. Strong connectivity did not protect short horizons: on rings and symmetrised ER graphs the gain was +0.13 to +0.31 in all six cells with $T < p$ and +0.08 to +0.22 in four of the six cells with $T \geq p$, as the exact values predicted (rings: $s_\rho(G_T) = 1$ for $T \leq p$). A separately planned replication (B1-NL, Appendix R; new seeds) with nonlinear additive dynamics, signed weights, and signed weights with uniform noise reproduced the gain in all 126 cells (pooled +0.108, +0.137, +0.137; 5/5 families significant in each; Spearman 0.68–0.82, slopes 0.34–0.56).

Sort-based baselines. The start changed the size of most method differences (Holm-significant G1×G2 interactions in 294 of 595 pair×cell SHD tests) and flipped the sign of 11 (Holm over 1,190 tests), all among sort-based baselines: e.g. on chains rel-SortnRegress beat R^2-SortnRegress under G1 and lost under G2. These baselines are no better than the empty graph on SHD under G2 (SHD/$|E|$ = 1.39–2.04; FLOP 0.46 and PC 0.64 never exceed 1), so SID is the more informative metric for them. In B1-NL the swap hypothesis was supported by the plan's rule under nonlinear dynamics, but only through one swap between two weak baselines, and not under signed weights; we discount it. The interactions themselves replicate (in-tree Var-SortnRegress vs FLOP on SHD, SID and orientation in all three parametrisations; chain rel- vs R^2-SortnRegress on SID and orientation, and on SHD in the short-horizon cells); sign reversals on SHD did not recur because the B1-NL level differences are larger.

Competitive learners: the second artefact. The fixed start also makes the window var-sortable, because variance accumulates from the noise-only first slice (Var-SortnRegress +0.20 raw, +0.002 standardised; rel-SortnRegress is scale invariant). A third planned experiment (Appendix S; 384 new data sets, families ring, chain, in-tree,

Table 3 Competitive learners (Appendix S), raw data unless noted; 96 graphs, G1 / G2. OA: ancestral order accuracy; Δ: G1−G2 with graph-cluster 95% CI. GOLEM-NV at $p = 5$ only.

method	SHD/$\lvert E\rvert$	OA	ΔOA
FLOP	.19 / .52	.89 / .77	–
DAGMA	.35 / .90	.80 / .60	+.197 [.177,.216]
NOTEARS	.47 / .89	.74 / .59	+.156 [.136,.175]
GOLEM-NV	.39 / .80	.79 / .56	+.232 [.205,.258]
DAGMA, std.	.87 / .87	.35 / .41	−.064 [−.079,−.048]
NOTEARS, std.	.76 / .84	.49 / .46	+.036 [.022,.049]
in-trees: DAGMA	.08 / .87		
NOTEARS / FLOP	.13 / .76; .30 / .56		
PCMCI+ (labels)	.02 / .02	1.0 / 1.0	–

ER(c=2), $T \in \{3, 5\}$) ran DAGMA (Bello et al., 2022), NOTEARS (X. Zheng et al., 2018) and GOLEM (Ng et al., 2020) with default settings on raw and standardised data (Table 3). On raw data all four variants improved sharply under G1 (order accuracy +0.16 to +0.23, SHD/$\lvert E\rvert$ −0.41 to −0.55; Holm $\leq 4 \cdot 10^{-8}$). This changed their ranking against FLOP: on in-trees DAGMA and NOTEARS beat FLOP under G1 and lost under G2 (both reversals Holm-significant and repeated on SID and order accuracy; DAGMA's also with first-slice pairs excluded; GOLEM only drew level with FLOP under G1), and a continuous method had the lowest mean SHD/$\lvert E\rvert$ among label-free methods in 3 of 8 (family, p) cells under G1 (one of them by 0.09 against 0.10, untested) against 0 of 8 under G2. The plan had predicted no rank change against FLOP; that prediction failed. It had predicted that standardisation would reduce the gain without removing it. By the plan's rule this held, except that DAGMA's SHD gain was removed; but the order-accuracy effect of the start became negative for DAGMA and GOLEM (−0.06 to −0.10), which the plan had not expected, and NOTEARS kept a small significant gain (+0.036; SHD/$\lvert E\rvert$ −0.077). We read the ranking changes as a var-sortability effect, but this is a hypothesis, and the evidence for it is weak. Standardised DAGMA and GOLEM are below chance in order accuracy under both starts (0.31–0.42), so they cannot show whether a scale-free part remains; standardised NOTEARS shows that some does; the fixed start also raises R^2-sortability, which is scale free; and the var-sortability gain is the same on rings and chains while the learners' gains are not. A control that matches marginal variances across the two starts has not been run. In any case Theorems 2 and 5 do not predict these ranking changes; they cover the scale-free gain of the sort baselines (rel-SortnRegress +0.19 again here). PCMCI+ (Runge, 2020), which uses the time labels, recovers the window under both generators (SHD/$\lvert E\rvert = 0.02$). Holm across all 21 confirmatory claims of B1, B1-NL, B3 and this experiment keeps every claim except the B1-NL swap claim under signed weights ($p = 0.108$).

Which generators are exposed. Two conditions are needed. First, the slices must be given to a method that does not use time order; a single record given to a time-series method is outside this threat model. Second, the start must matter: for the graph artefact of Section 3 this means a burn-in shorter than $\Delta\tau_{\max}$ steps, and for the

variance artefact it means a start that has not mixed in distribution, which depends on the spectral radius and which we did not compute for any deployed generator. We read the source code of common time-series generators (Appendix S). The time-unrolled construction of Reisach et al. (2026) meets both conditions: it starts at $t = 1$ with no burn-in and gives every slice to the method as a separate variable (its data code is not public). The others produce one record for time-series methods, so they fail the first condition as shipped. On the second: an exact stationary start (Herman et al., 2025) removes both artefacts; burn-ins of 100 to 1000 steps (causalnex, the Neural-GC simulators (Tank, Covert, Foti, Shojaie, & Fox, 2021)) remove the graph artefact and the variance artefact only if the process mixes within them; tigramite's `structural_causal_process` (Runge, 2026) drops $\lfloor 0.2T \rfloor$ steps, its older `var_process` none, the CDML generator (Lawrence, Kaiser, Sampaio, & Sipos, 2021) $2\tau_{\max}$, and CausalTime's demo rolls out 20 steps (Cheng et al., 2024), so windows cut from their first steps would be exposed. We could not check how the CauseMe data were initialised.

4.2 With time labels the start identifies nothing more

The experiments above hide the time labels. One may ask whether the fixed start is also a source of information for a method that has them, for example through the changing covariance of the transient. It is not.

Proposition 8 (no gain from the transient) *Let $X_t = \mathsf{B}X_t + AX_{t-1} + e_t$ for $t = 1, \dots, T$, with $I - \mathsf{B}$ invertible, e_t i.i.d. and independent of the start $X_0 \sim \nu$ (ν may be a point mass). Put $M = (I - \mathsf{B})^{-1}A$ and $\eta_t = (I - \mathsf{B})^{-1}e_t$. Then the law of $(X_0, \dots, X_T)$ depends on $(\mathsf{B}, A,$ law of $e)$ only through $(M,$ law of $\eta)$. Two structures with the same reduced form are therefore observationally equivalent for every start law, every T and every number of independent replicates. If $\Sigma_t = \mathrm{Cov}(X_t)$ is nonsingular for some $t \leq T - 1$, then $M = \mathrm{Cov}(X_{t+1}, X_t)\Sigma_t^{-1}$ and $\Omega = \mathrm{Cov}(\eta_t) = \Sigma_{t+1} - M\Sigma_t M^\top$ are identified.*
Lean: *not machine-checked (paper proof).*

Proof $X_t = MX_{t-1} + \eta_t$ is a Markov chain whose transition kernel depends on $(M,$ law of $\eta)$ only and whose initial law is ν. The two formulas follow from $\mathrm{Cov}(X_{t+1}, X_t) = M\Sigma_t$ and $\Sigma_{t+1} = M\Sigma_t M^\top + \Omega$. □

The covariance trajectory Σ_t is itself a function of (Σ_0, M, Ω). Identification through heteroskedasticity (Rigobon, 2003) needs the *innovation* covariance to change between regimes; in a transient with fixed mechanisms only the state covariance changes. In the lag-one model of this paper ($\mathsf{B} = 0$) the transition matrix is identified under either start as soon as one Σ_t is nonsingular, in agreement with the labelled reference above (PCMCI+, SHD$/|E| = 0.02$ under both generators). The fixed start is therefore an artefact for label-free methods only. When each replicate is observed at a single time, so that only the marginal laws are seen, identification of the reduced form from these marginals has been established for linear stochastic differential equations (Guan et al., 2026; Wang, Geng, Huang, Huang, & Gong, 2023); this is consistent with Proposition 8 and is not our contribution.

4.3 Is the time order identifiable without labels?

Reisach et al. (2026) end by raising the question to what extent knowledge of the time order can and should be used for causal discovery on independent trajectories of short time-unrolled DAGs. A prior question is how much of the time order is hidden when the labels are removed. The statements below are population statements with paper proofs (Appendix U); none is machine-checked, and we do not propose a learning method.

Setting. The columns of the data are the pT cells (unit j, slice t) of $X_{t+1} = AX_t + \varepsilon_t$ ($A_{ji} \neq 0$ iff $(i,j) \in E_S$, ε_t with diagonal covariance D, the same A and D for all t) in an unknown order; p and T are known. The *column DAG* is G_T on the columns. The *fixed-start law* has a diagonal covariance of the first slice (not necessarily equal to D), the *free law* an arbitrary one, which covers a burn-in window and an exact stationary start. The height of a node is one plus the number of edges of a longest directed path ending in it. A unit is *loop-only* if its only parent in G_S is itself, and *isolated* if in addition it has no child other than itself; m_0 is the number of loop-only units. A *representation* of a covariance Σ is a parameter of the same model class (an assignment of columns to cells, A' with non-zero diagonal, diagonal D', a start covariance of the given law) that reproduces Σ; it is *sparsest* if no representation has fewer non-zero entries in A'. Markov equivalence is equality of skeleton and unshielded colliders (Spirtes et al., 2000).

Proposition 9 (slices are a graph invariant) *(a) In G_T, the height of $v(t)$ is t. Hence T, p and the partition of the columns into slices are determined by the column DAG. (b) The partition of the columns into units is not: for the directed ring with self-loops, relabelling the cell (i,t) as unit $i - t \pmod p$ gives the reversed ring with self-loops and the same column DAG. (c) Without self-loops (a) fails: for the single edge $0 \to 1$ the unrolled graph is a set of disjoint edges, and the slice of an edge is not determined.*

Lean: not machine-checked (paper proof; (a) uses the walk-length argument behind edge_score_dist).

Proposition 10 (Markov equivalence class of G_T) *An edge of G_T is reversible within its Markov equivalence class iff it is an edge $j(t) \to j(t+1)$ of the self-chain of a loop-only unit j. The class has exactly T^{m_0} members.*

Lean: not machine-checked (paper proof).

So under the fixed start a perfect label-free learner that uses conditional independence only is left with T^{m_0} graphs: $m_0 = 0$ for a ring, 1 for a chain or an out-tree, and the number of leaves for an in-tree.

Proposition 11 (fixed start, given faithfulness) *Let the columns be Gaussian with covariance Σ from the fixed-start law, $T \geq 2$, and let their distribution be faithful to G_T. (a) Every sparsest representation of Σ has the true column DAG, except that the self-chain of an isolated unit may be reversed as a whole. (b) For $T \geq 3$ the chain of an isolated unit j with coefficient $a = A_{jj}$, noise variance d and start variance v_1 can be reversed iff its start is stationary, $v_1(1 - a^2) = d$. For $T = 2$ it can always be reversed. (c) Hence for $T \geq 3$ the column DAG is identified from Σ iff no isolated unit has a stationary start.*

Lean: not machine-checked (paper proof; faithfulness is assumed, not shown to be generic).

Proposition 12 (three exact non-identifiability results) *(a)* $T = 2$, free law. *For the summary edge $a \to b$ with self-loops and a free start covariance, the labelling with slices $\{a_1, b_2\}, \{a_2, b_1\}$ is a representation with the same number of edges and a different column DAG ($a_1 \to a_2$, $a_1 \to b_1$, $b_2 \to b_1$). (b)* Isolated units, stationary start. *Reversing the chain of an isolated unit with a stationary start leaves Σ unchanged, for every T (Peters, Janzing, Gretton, & Schölkopf, 2009; G. Weiss, 1975); in particular $p = 1$ is never identified from a stationary window. (c)* Symmetric dynamics. *If $A = A^\top$, $D = \sigma^2 I$ and the start is stationary, reversing all slices leaves Σ unchanged for every T, and the reversed column DAG is a sparsest representation.*

Lean: not machine-checked (paper proof).

Conjecture 13 (burn-in window) *Under the free law with $T \geq 3$, $p \geq 2$, a weakly connected summary graph, faithfulness to the graph that adds a complete DAG on the first slice, and a time-reversed innovation covariance that is not diagonal, every sparsest representation has the true column DAG.*

We can reduce Conjecture 13 to a finite combinatorial statement (the skeleton admits only one partition into T layers of size p that is compatible with an unrolling) and have checked it by hand only for the summary edge $a \to b$ with $T = 3$ (Appendix U).

Reading. Under the fixed start, hiding the labels hides little in the population: the slices are a graph invariant, and the column DAG is identified up to isolated units. This is a third regularity of the time-unrolled construction, next to rel- and var-sortability, and unlike them it comes from the shape constraint (the graph is an unrolling), which a burn-in does not remove; whether it also survives a burn-in is Conjecture 13. Three cautions apply. The statements assume faithfulness, and the tied parameters of an unrolled model form a null set among all linear Gaussian models on G_T, so the usual genericity argument does not apply; Boege, Kubjas, Misra, and Solus (2025) show that coloured Gaussian DAG models can have no faithful distribution at all. They concern the global optimum of a sparsity criterion, not a computable procedure. And they say nothing about finite samples: at $n = 1000$ FLOP is far from the labelled reference (Table 3). Dynamic Bayesian networks with a start network and a tied transition network are classical when the labels are known (Friedman, Murphy, & Russell, 1998).

5 Aggregation versus Subsampling

Real records are often block means or subsamples of a faster process. This section states what these observation schemes do to the three sortability criteria; proofs are in Appendix D.

For a stable linear SVAR with independent innovations, nodes with disjoint summary ancestor sets are independent at all lags, and so are any per-node, time-invariant transforms, including block means and subsamples; otherwise the lag-0 covariance is nonzero for almost every parameter (Proposition 14). So the population relatives of

contemporaneous values are rel_S for every observation scheme, and the population relatives sort is the stationary value of Theorem 5 (on non-self-loop edges), not the finite-G_T value. Var- and R^2-sortability behave differently (Theorem 15): block means converge, at the textbook Fejér rate for averages of a stationary series (Brockwell & Davis, 1991, Ch. 7), to the long-run structural model of Gong et al. (2017), whereas subsamples keep the snapshot covariance. As a consequence, for large block size, block-mean R^2-sortability is 0 on out-stars and 1 on in-stars with at least two arms for all weights; on out-stars subsampled data are R^2-sorted correctly on average (0.84) while block means drift to 0. We claim only these sortability consequences.

Assumptions: $X(t) = \sum_{\tau=1}^{L} A_\tau X(t-\tau)+\varepsilon(t)$, $t \in \mathbb{Z}$, i.i.d. innovations with diagonal positive-definite covariance Σ, stable (all roots of $\det(I - \sum_\tau A_\tau z^\tau)$ outside the unit disc), stationary solution; the summary graph has $i \to j$ iff $(A_\tau)_{ji} \neq 0$ for some τ. $\Gamma(h)$ is the autocovariance, $A(1) = \sum_\tau A_\tau$, and for a block size D the block means are $\bar{Y}_s = \frac{1}{D}\sum_{u<D} X(sD+u)$ and the subsamples $Z_s = X(sD)$.

Proposition 14 (relatives survive any per-node filter) *If* $\mathrm{anc}_S(i) \cap \mathrm{anc}_S(j) = \emptyset$ *then* X_i *and* X_j *are independent at all lags, and so are any per-node, time-invariant (possibly nonlinear) transforms of them, including* $\bar{Y}$ *and* Z *for every* D. *If the ancestor sets intersect,* $\Gamma(0)_{ij}$, $\mathrm{Cov}(\bar{Y})_{ij}$ *and the long-run covariance are nonzero for Lebesgue-almost every parameter.*

Lean: *not machine-checked (needs the SVAR moving-average representation and real-analytic genericity).*

Theorem 15 (block means vs subsampling) *(a)* $\|D\,\mathrm{Cov}(\bar{Y}) - S_{LR}\| \le \frac{1}{D}\sum_h |h|\|\Gamma(h)\|$, *with* $S_{LR} = (I - A(1))^{-1}\Sigma(I - A(1))^{-\top}$. *(b) If* G_{S} *is acyclic apart from self-loops,* S_{LR} *is the covariance of a static SEM (this long-run SEM is due to Gong et al., 2017, Eqs. (5), (10)) on* G_{S} *with weights* $A(1)_{ji}/(1-\lambda_j)$ *and noise variances* $\sigma_j^2/(1-\lambda_j)^2$, $\lambda_j = A(1)_{jj} < 1$. *(c)* $\mathrm{Cov}(Z_s) = \Gamma(0)$ *for every* D. *(d) Hence, for large* D, *block-mean* R^2*-sortability is* 0 *on out-stars and* 1 *on in-stars with* ≥ 2 *arms, for all weights; under subsampling every contemporaneous criterion keeps its* $\Gamma(0)$ *value.*

Lean: *(a) counting identity fejer_identity; (b) longrun_factor, selfloop_sum_lt_one; the probabilistic steps, the rate and (c)–(d) are not machine-checked.*

Part (a) is the standard Fejér/Cesàro computation for the variance of the mean of a stationary series (Brockwell & Davis, 1991, Ch. 7), written for block means of a vector process; (b) is due to Gong et al. (2017); (c) is stationarity. We claim only the sortability consequences (d).

The R^2 statement for stars strengthens the three-node analysis in Herman et al. (2025, App. D), whose qualitative message (that R^2 reflects connectivity, with an asymmetry between colliders and confounders) we credit. Numerical observations (exact population moments, not proved): mean block-mean R^2-sortability over 150 in-trees moves from 0.853 ($D = 1$) to 0.834 ($D = 4$–8) and 0.880 ($D = 256$), so it is not monotone in D; on out-stars subsampled data are R^2-sorted at 0.842 on average while block means give 0.546 ($D = 4$), 0.101 ($D = 16$) and 0 in the limit.

6 Finite-Sample Theory

We now ask whether the population relatives can be estimated. X is a zero-mean stationary Gaussian process with absolutely summable auto- and cross-covariances $\gamma_{ab}(h)$, standardised so that $\gamma_{aa}(0) = 1$, observed at n times. Two nodes are declared relatives if the Fisher-z test of the sample correlation $\hat{r}_{ab}$ rejects at level α with Bonferroni over $m = d(d-1)/2$ pairs, i.e. $|\hat{r}_{ab}| > \tanh(z_c/\sqrt{n'-3})$, $z_c = \Phi^{-1}(1-\alpha/2m)$; $\hat{\rho}(v) = 1 + \#\text{declared partners}$. The *nominal* test uses $n' = n$. Let $B_{ab} = \sum_h |\gamma_{aa}(h)\gamma_{bb}(h) + \gamma_{ab}(h)\gamma_{ab}(-h)|$ (for an independent pair $B_{ab} = \sum_h |\rho_a(h)\rho_b(h)|$, at least the Bartlett sum $W_{ab} = \sum_h \rho_a(h)\rho_b(h)$, with equality iff all products are nonnegative; e.g. 4.56 against 0.22 for AR(1) coefficients 0.8 and -0.8).

F1: recovery and effective sample size. That the nominal-n test has the wrong size under autocorrelation is classical: effective degrees of freedom go back to Bartlett (1935), with spatial and multivariate versions in Clifford et al. (1989) and Roy (1989), and the variance with exactly our summand B_{ab} is in Afyouni et al. (2019). We add a non-asymptotic exact-recovery bound for the relatives set (Theorem 27, Appendix E): if the thresholds separate the population correlations by more than explicit deviations, the declared set and hence $\hat{s} = s_\rho^{\text{pop}}$ are recovered with failure probability decaying as $\exp(-n\min(\tau, \rho_{\min} - \tau)^2/(2(\sqrt{B} + \rho_{\max}\sqrt{D_\gamma})^2))$, so n/B acts as an effective sample size. For i.i.d. snapshots $B = 1 + \rho^2$; under strong persistence B can be in the tens. This bound is for a fixed threshold; for the Bonferroni Fisher-z threshold $\tau_n \asymp z_c/\sqrt{n'}$ that we use, the failure probability it certifies tends to $d(d-1)e^{-z_c^2/(2B)} > 0$ (0.089 for i.i.d. data, $d = 5$, $\alpha = 0.01$; Proposition 31), and recovery is instead guaranteed asymptotically (Proposition 30).

Proposition 16 (size distortion of the nominal test) *In this setting, a non-relative pair is declared by the nominal test with probability* $\to 2\bar{\Phi}(z_c/\sqrt{W_{ab}})$, $W_{ab} = \sum_h \gamma_{aa}(h)\gamma_{bb}(h)$ *($\leq B_{ab}$, with equality iff all products are nonnegative), above the nominal α/m when $W_{ab} > 1$. With $n_{\text{eff}} = n(1-\rho^2)^2/W_{ab}$ (W the delta-method Bartlett variance) the false-positive rate tends to α/m.*

Lean: not machine-checked (uses the multivariate Breuer–Major theorem).

The joint asymptotic normality used here follows from the multivariate Breuer–Major theorem (Arcones, 1994; Nourdin, Peccati, & Podolskij, 2011) (Appendix E). On real data n_{eff} is estimated by $n/\hat{B}_{ab}$, $\hat{B}_{ab} = \max(1, 1 + 2\sum_{h=1}^{K} \hat{\rho}_a(h)\hat{\rho}_b(h))$. If $K = K_n \to \infty$ with $K_n = o(n)$, $\hat{B}_{ab}$ is consistent, the plug-in test has asymptotic size at most α/m per pair, and the relatives set is recovered with probability at least $1-\alpha$ in the limit (Proposition 30). Our code uses a fixed K (20–100), which controls size only if the truncated sum captures the Bartlett sum: at AR coefficient 0.99 and $K = 50$ the size is 0.113 against 0.05, while on the B3 data K up to 300 changes none of 2,730 pair decisions (Appendix E). We did not run this check on the river data, which are the most persistent, or on study 3 and E1. For linear processes with non-Gaussian i.i.d. innovations the null variance, and hence the size results, are unchanged, because independent series share no fourth cumulant; for relatives the variance changes, which affects power but not the limits (Remark 32).

F2: saturation by a latent driver. Let $X_v = Y_v + b_v C$ with C a stationary unit-variance process independent of the VAR Y, and $\ell_v = b_v^2/\mathrm{Var} X_v$ the driver's share of variance.

Proposition 17 (saturation) *For summary non-relatives, $\rho_{uv} = \mathrm{sgn}(b_u b_v)\sqrt{\ell_u \ell_v} =: \kappa_{uv}$. If all $b_v \neq 0$, every pair is a population relative and the population rel-sortability is exactly $1/2$; $P(\hat{s}_n \neq \frac{1}{2}) \leq (d^2+3d)e^{-cn}$ with $c \approx \kappa_{\min}^2/(18(\sqrt{B}+\sqrt{D\gamma})^2)$. Asymptotically, a pair is detected with probability $> 1/2$ once $n > n_{1/2} = 3 + (z_c/\operatorname{atanh}\kappa)^2 \approx z_c^2/\ell^2$ (nominal test; for the effective test, the same bound on n_{eff}).*

Lean: prop21_corr_std, sOf_saturated, abs_sOf_sub_half_le, location of $n_{1/2}$: ncrit_eq_iff; the probability statements are not machine-checked.

$n_{1/2}$ does not depend on autocorrelation, which only sets the width of the transition. A driver with share $\ell = 0.1$ saturates at $n \approx 10^3$, and one with $\ell = 0.64$ at $n \approx 22$ (both for $z_c = 3.29$). A relatives count on a long record can recover the summary relatives only while $\max_{\text{non-rel}} |\kappa| < \tau_n < \min_{\text{rel}} |\rho|$, a window that closes at n of order $n_{1/2}$.

F3: conditioning on a proxy. A natural remedy is to condition on a measured proxy Z of the driver. It works only if the proxy is almost perfectly reliable.

Proposition 18 (conditioning) *Let $Z = \sqrt{r}\, C + \sqrt{1-r}\, e$ with reliability $r = \mathrm{corr}(Z, C)^2$, π_{uv} the partial correlation given Z, and ρ^Y the driver-free correlation. Then*

$$\pi_{uv} = \frac{\rho^Y_{uv}\sqrt{(1-\ell_u)(1-\ell_v)} + (1-r)\,\mathrm{sgn}(b_u b_v)\sqrt{\ell_u \ell_v}}{\sqrt{(1-r\ell_u)(1-r\ell_v)}}.$$

At $r = 1$ the conditional relatives are the summary relatives and Theorem 5 is restored. For non-relatives with $\ell_u, \ell_v \in (0,1)$ and $b_u b_v \neq 0$, $|\pi_{uv}|$ is strictly decreasing in r and vanishes only at $r = 1$; fewer than half of the non-relative pairs are declared only if $r > r_{\mathrm{crit}} = 1 - \frac{\tau'(1-\ell)}{\ell(1-\tau')}$, $\tau' = \tanh(z_c/\sqrt{n'-4})$ (equal shares). If the driver reaches nodes through different filters, a scalar proxy leaves a nonzero residual even at $r = 1$. The reliability is identified by $\rho_{uZ}\rho_{vZ}/\rho_{uv} = r$, the identity behind the three-indicator identification rule for one-factor models (Bollen, 1989, Ch. 7).

Lean: prop31_std, cor32, absPi_strictAntiOn, absPi_eq_zero_iff, rcrit_iff, thm35_bracket, pc_lipschitz, prop36_ar1_ne_zero, prop36_rank_one, prop38_ratio.

So spurious conditional relatives appear for every $r < 1$ once n is large; $r_{\mathrm{crit}} = 0.95$ for $\ell = 0.64$ and $\tau' = 0.082$. Formalisation found four missing nondegeneracy hypotheses in our earlier statements (e.g. $b_u b_v \neq 0$; Appendix G). The algebra of F2 and F3 is machine-checked; the probability statements are not. The non-asymptotic bounds are loose (factors 3–13 in n, Appendix F); the asymptotic statements matched every simulation to within 0.03 and are the ones used in Section 8.

7 Machine-Checked Proofs

The formalisation (Lean 4, de Moura & Ullrich, 2021, with Mathlib, The mathlib Community, 2020; 4,159 lines in 16 files, clean rebuild of the final state) certifies the

exact and stationary results of Section 3 (Theorems 1, 2 and 5, Corollaries 3 and 6, Proposition 4), including the Δ forms of the bound and the burn-in and the sharpness example of Remark 21, the loop-free exact edge value and the accuracy reading, the algebra of the real-data estimand $\tilde{s}$ (Appendix I), and the deterministic algebra of F1–F3. It does *not* certify the ER limits, the loop-free limit and periodicity, the lag-weighted extension, the statements of Sections 4.2 and 4.3, or any probability bound or limit theorem. The axiom report lists 163 declarations, each depending only on `propext`, `Classical.choice` and `Quot.sound`, and the sources contain no `sorry`, `admit` or `axiom` (33 headline theorems, 53 graph lemmas, 65 algebraic facts, many elementary, and 12 examples). Relatives, s_ρ, G_T and the windows are defined from scratch following Reisach et al. (2026). No human has yet checked that each Lean statement matches its paper statement. Appendix G maps every statement to its Lean names and status.

7.1 Use of AI tools

Most of this project was carried out with AI coding agents (large language models, Claude-based) working under the author's direction; no AI system is an author. The paper proofs (including Appendices O.1 and U), all Lean code, all experiment and analysis code, the analysis plans, the simulated pre-submission reviews that motivated several revisions, and the manuscript text were generated by these agents. What is machine-checked: the 163 Lean declarations of Section 7 (Lean kernel, standard axioms only), which cover the exact and stationary results of Section 3 including the Δ forms of Theorems 2 and 5 and the sharpness of $B \geq \Delta$, the estimand algebra and the deterministic algebra of F1–F3. No human has checked that the Lean statements say what the paper statements say; the map in Appendix G is the agents' reading. What is checked only by computation: the enumeration of Appendices C and M, the ER numerics and the simulations of Appendix F. What no human has yet read line by line: the paper-only proofs, namely Appendix O.1, Proposition 14, Theorem 15, the probabilistic parts of Theorem 27 and Propositions 16 to 18 and 30, Proposition 31, Remark 32, the loop-free limit results of Appendix M, the lag-weighted extension (Appendix N), and Propositions 8 to 12. The author takes responsibility for all content.

8 Real-Data Evidence

Protocol and its limits. We ran four studies (P1–P9, then E1/E2), each with an analysis plan (predictions, thresholds, estimator, seeds, kill rules) committed to version control before any dependence statistic of its data was computed (timeline in Appendix J). The plans, analyses and results were produced by AI agents within about six hours of one day, commit times are set by the client and no external registry was used, so we call them *time-stamped analysis plans*, not pre-registrations. An earlier exploratory project in the same repository had already found unrelated CausalRivers stations about as correlated as river siblings, so our account of that failure is explanatory, not predictive; and the graph refinement used by E1 was motivated by study-3 observations on the same device (Appendix J).

Estimand. For an observed node set O of a known summary graph, the predicted value $\tilde{s}$ is the edge average of the scores of the counts $|\mathrm{rel}_S(v) \cap O|$: population lag-0 relatives among the observed nodes, as a perfect lag-0 dependence oracle would give them (Proposition 14, Definition 33). $\tilde{s} > 1/2$ forces an up-edge among the induced edges, but not conversely (Proposition 34, machine-checked). The estimator is the relatives count of Section 6 at $\alpha = 0.01$ with Bonferroni; orientation accuracy is the mean edge score of $\hat{\rho}$; CIs are cluster bootstraps. Lohse and Wahl (2025) measured var- and R^2-sortability on the same families; we know of no earlier test of rel-sortability on real data.

CausalRivers: a failed test, explained post hoc. River networks are in-forests, so Corollary 6 predicts $\tilde{s} > 1/2$ for subgraphs with confluences (mean 0.82). The estimated relatives sort carried no orientation information (P2, P3 fail; Table 4) because 98% of non-relative station pairs tested dependent. Read through F2 (post hoc), non-relative correlations $\kappa \approx 0.64$–0.74 give $n_{1/2} \approx 20$–27, while the records have $n = 377$–$3{,}652$ steps, so the study sat deep in the saturated regime (Figure 4, Appendix H), where Proposition 17 predicts $\hat{s} = 1/2$; regional precipitation acts as a latent driver. The F3 remedy fixed in plan 2 (conditioning on a leave-out regional mean) also failed (P4: 0.74), consistently with F3: the triad estimate of the proxy's reliability, $\hat{r} \approx 0.62$, is below the required $r_{\mathrm{crit}} \approx 0.87$, and the in-sample plug-in prediction of the dependent fraction, 0.743, matches the observed 0.741.

Causal Chambers: designed independent sources. In the wind-tunnel random-walk runs (Gamella, Peters, & Bühlmann, 2025) five actuators are independent random walks, hence separate source SCCs, and 25 of 26 edges are predicted strictly sorted. Saturation is absent (P7, 0.039) and the relatives count orients confluence edges above chance (P8, 0.656), but the graded agreement P9 missed its threshold ($0.267 < 0.3$), so by the plan the study is *not* a positive demonstration. The reason is F1: 16 of 26 edges are essentially undetectable at lag 0 at $n \approx 10^3$. The light-tunnel replication passes all three criteria, and F-theory predictions from the first half of each unit match the second half to within 0.02 (Table 5, Appendix H). The wind-tunnel graph is bipartite, so the prediction coincides with "sensors have more relatives than actuators".

E1 and B3: the theorem against naive rules. Plan 4 tested the edge values of Corollary 6 against eight naive predictors (sensor indicator, ancestor count, in-degree, depth, "every edge 1", ...) on the light-tunnel series `citris_1` (10 units). On the primary graph G_{lin} it passed 8/8 ($\Delta = -0.50$ to -0.53 in absolute loss, 10/10 units, Holm $p = 0.0078$). This is weaker than it looks (Appendix K): G_{lin} removes six real edges that carry no linear dependence under an ideal-sensor assumption, and on the published graph the sensor indicator wins on strictly predicted edges; each weak component of G_{lin} has a single source, so the prediction coincides with the rule "tie iff same weak component, else compare component sizes"; and most of the separation comes from ties. B3 (Appendix T) therefore used held-out light-tunnel data whose components have several independent sources (30 units of three AR(1)-driven series; 20 units of an actuator collider), with constant 1/2 and the weak-component rule as competitors. On the 2,361 instances where the theorem and the weak-component rule disagree, the counts were sorted as predicted in 99.8% ($\Delta = -0.50$ against both

Table 4 Tests fixed in the time-stamped analysis plans. Pass/fail is by the point rule fixed in the plan. Brackets: 95% cluster-bootstrap CI. "non-rel. dep.": fraction of graph non-relative pairs declared dependent. "Acc.": orientation accuracy on edges predicted to be strictly sorted.

study (units)	test (threshold)	result (12 h / 24 h)	verdict
CausalRivers, marginal (120 subgraphs)	P1: confluence subgraphs have $\tilde{s} > 1/2$	112 of 114	fails as worded[a]
	P2: Acc. > 0.5, CI excl. 0.5	$0.501\,[0.492, 0.509]$ / $0.498\,[0.489, 0.507]$	**fail**
	P3: Spearman$(\tilde{s}, \text{acc.}) \geq 0.3$	-0.018 / 0.018	**fail**
	(non-rel. dep., nominal test)	0.984 / 0.978	kill rule triggered
CausalRivers, conditioned on regional proxy (120 subgraphs)	P4: non-rel. dep. < 0.5 given proxy	0.741 / 0.735	**fail**
	P5: Acc. > 0.5, CI excl. 0.5	$0.539\,[0.513, 0.565]$ / $0.540\,[0.514, 0.567]$	pass, not interp.[b]
	P6: Spearman ≥ 0.3	-0.113 / -0.121	**fail**
Causal Chambers, wind tunnel (15 runs, primary)	P7: non-rel. dep. < 0.5	$0.039\,[0.031, 0.047]$	pass
	P8: Acc. > 0.5, CI excl. 0.5	$0.656\,[0.637, 0.675]$	pass
	P9: Spearman ≥ 0.3 (300 subgraphs)	$0.267\,[0.163, 0.368]$	**fail**
light tunnel (10 segments, replication)	R7 / R8 / R9 (same thresholds)	0.000 / 1.000 / $0.779\,[0.696, 0.864]$	pass (replication)[c]
E1, light tunnel (10 units, G_{lin})	E1a: non-rel. dep. < 0.5; det. ≥ 0.8	0.000; $0.969\,[0.945, 0.992]$	pass
	E1b: $\Delta \leq -0.10$, Holm $p \leq 0.05$, all 8	$\Delta = -0.502$ to -0.533 (10/10)	pass (8/8)[d]
	E1c: as E1b, detected edges	$\Delta = -0.500$	pass (8/8)
E2, wind regimes (64 units; D=1, lags L=30 vs. 0)	Q7: non-rel. dep. < 0.5	$0.003\,[0.001, 0.005]$	pass
	Q8: Acc. > 0.5, CI excl. 0.5	$0.553\,[0.527, 0.579]$	pass
	Q9: Spearman ≥ 0.3 (1,280 subgr.)	$0.243\,[0.193, 0.294]$	**fail**
	Q10: detection gain, CI lower bound > 0	$+0.069\,[0.060, 0.079]$	pass
	E2 success (Q7–Q10 all)		**not met**

[a] The two exceptions are split-and-rejoin channels with a single source, for which the theorem itself predicts 1/2. [b] Kill rule (P4 failed at both resolutions): P5 is reported, not interpreted. [c] Fixed in the plan as not counting towards a positive demonstration (i.i.d. inputs, weak time-series aspect). [d] G_{lin} is the primary graph fixed in plan 4. It omits six real causal edges that, under a symmetry lemma and an assumption of ideal polariser sensors, carry no linear dependence. On the published graph the theorem still beats all eight overall, but not on its strictly ordered edges, and on G_{lin} the prediction equals a weak-component-size rule (Section 8, Appendix K).

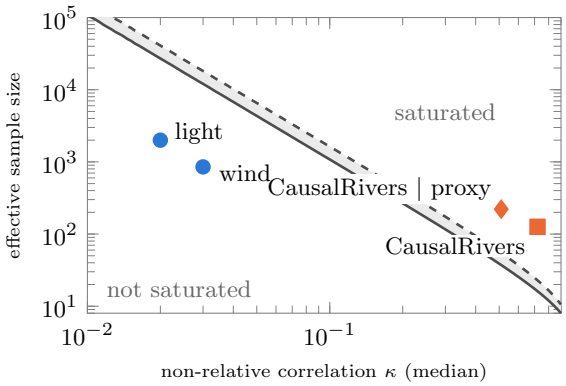


Fig. 4 Saturation regime of Proposition 17. Curves: onset $n_{1/2}$ for $z_c = 3.29$ ($d = 5$, solid) and 4.04 ($d = 20$, dashed); above them a non-relative pair is detected with probability $> 1/2$. Points: medians from the real-data studies (CausalRivers unconditioned and given the regional proxy, κ replaced by $|\pi|$; Causal Chambers wind and light tunnel). Values are in Table 4 and Appendix H.

Table 5 Out-of-sample F-theory check on the Causal Chambers: predictions from the first half of each unit, observed rates on the second half. Accuracy is predicted by Monte Carlo of independent pair decisions.

	wind tunnel		light tunnel	
	pred.	obs.	pred.	obs.
non-relatives declared (F2)	0.031	0.028	0.006	0.000
relatives detected (F1)	0.360	0.343	0.679	0.681
orientation accuracy	0.630	0.620	0.999	1.000

tie rules, 30/30 units); in the collider, R→G was sorted whenever B was observed (264/264) and tied otherwise (123/123). On these instances the theorem predicts strict order, so E1's naive sort rules agree with it there and are separated only on the 1,251 predicted ties whose own edge is detected ($\Delta = -0.43$, 30/30 units). The test was low-risk (same device; colour edges known to be detectable; results 3–4 minutes after the plan), and it failed where the graph is unfaithful: with an almost constant source the tie rule won ($\Delta = +0.12$, 0/10 units), and in one series the data contradicted G_{lin} (the design graph matched 0.96 against 0.04 where they differ).

E2: lagged relatives (Appendix L). A Bonferroni max-over-lags cross-correlation test (Haugh, 1976) on 64 held-out wind-tunnel regimes raised detection from 0.328 to 0.398 without more false positives, but failed the graded criterion ($0.243 < 0.3$), so E2 does not succeed, and neither did the lag-0 estimator on these regimes.

Data selection, preprocessing and secondary analyses are in Appendix H.

9 Discussion and Limitations

For benchmark design. The finite time-unrolled generator is rel-sortable above 1/2 for every summary graph that is not a union of strongly connected components (Corollary 3), strong connectivity alone does not help at the short horizons proposed for benchmarks (Section 4), and part of the excess comes only from the start at $t = 1$. We recommend three things. (a) Start trajectories from the stationary distribution (an exact Lyapunov or Cholesky start, as in Herman et al., 2025), or use a burn-in that is checked for distributional mixing. A burn-in of Δ slices ($\Delta\tau_{\max}$ with lags up to $\tau_{\max}$) removes the structural artefact and cannot be shorter in general (Theorem 5 and Remark 21), but it does not remove the distributional transient, which decays with the spectral radius: B1 needed 100 steps to pass its mixing check. (b) Standardise columns, since the fixed start also inflates var-sortability, and the raw-data gains of DAGMA, NOTEARS and GOLEM largely disappear on standardised data. (c) Score the window without first-slice pairs or as an ADMG: a latent burn-in makes the last burn-in slice a set of latent common causes of the first window slice, so a labels-using reference that has normalised SHD 0.044 under the fixed start has 0.273 after a burn-in, but 0.041 against 0.048 without first-slice pairs. An algorithm that sees time stamps orients every cross-time edge for free, so the artefact matters for methods that do not use time order. Random summary graphs are sortable only near criticality. Removing the time labels does not hide the time order in the population under the fixed start (Section 4.3), and for a method that has the labels the start carries no extra information (Proposition 8).

For real data. Population rel-sortability above 1/2 does not imply that a relatives count can be estimated. Two conditions are needed: no latent common driver with share ℓ such that $n \gtrsim z_c^2/\ell^2$ (or a near-perfect, rank-one proxy), and relatives detectable at lag 0 at the available n_{eff}. Diagnostic: compare n_{eff} with $3 + (z_c/\operatorname{atanh}\hat{\kappa})^2$, where $\hat{\kappa}$ is the typical correlation of pairs believed unrelated; if n_{eff} is far above it, the count is saturated whatever the graph.

Limitations. The closed forms assume self-loops at every node and unit lags (Appendices M and N). The ER limits are paper proofs not yet checked by a human probabilist; they are iterated limits ($T \to \infty$ at fixed p, then $p \to \infty$; the stationary value needs no T-limit), with no rates and no result for fixed T. Section 5 covers linear SVARs only. The finite-sample bounds are loose; Proposition 14 and the probabilistic parts of F1–F3 are proved on paper only, for Gaussian processes (the non-Gaussian CLT step is cited, not re-proved), and our code's fixed K controls size only away from a unit root. The analysis plans are internally time-stamped and agent-executed (push order cannot show that nothing was computed before a plan), and the CausalRivers plan was not blind to an earlier analysis of the same data. The positive real-data evidence comes from one device per chamber; E1 coincides with a weak-component-size rule on its primary graph and is carried mainly by ties; B3 separates the theorem from that rule but on the same light tunnel, with no external registration. The graded criteria P2, P3, P4, P6, P9, Q9 and one B3 secondary set failed. The benchmark experiments are simulations at $n = 1000$; the learner extension uses one linear-Gaussian parametrisation, $T \leq 5$, default hyperparameters and GOLEM at $p = 5$ only. Our

theory does not predict the ranking changes among competitive methods; their attribution to var-sortability is a hypothesis without a variance-matched control, and the plan's prediction of no rank change failed. The detectability calculation and the reanalyses with graph-cluster inference are post hoc, and the former does not isolate the structural mechanism. Exposure of deployed generators to the variance artefact was not measured (no mixing times). The identifiability statements of Section 4.3 are paper proofs that assume faithfulness, which we have not shown to hold for almost all tied parameters; Conjecture 13 is open; and they give no finite-sample guarantee and no algorithm.

Supplementary information. Code, a pinned environment (`requirements.txt`; Lean 4.35.0, Mathlib v4.35.0), dataset sources with SHA-256 checksums (`DATA.md`), the analysis plans, run logs and a reproduction script (`reproduce.sh`, with stages for the Lean build, the graph checks, the simulations including the benchmark experiments and the learner extension, the post hoc re-analyses, and the real-data studies; `REPRODUCE.md`) accompany the article.

Acknowledgements. [PLACEHOLDER: acknowledgements, to be completed by the author, or remove]

Statements and Declarations

- **Funding.** The author received no funding for this work.
- **Competing interests. [PLACEHOLDER: competing-interests statement, to be confirmed by the author]**
- **Ethics approval and consent to participate.** Not applicable. The study uses simulations and public sensor data sets and involves no human participants, human data or animals.
- **Consent for publication.** Not applicable.
- **Data availability.** No new data were collected. The real-data studies use the public CausalRivers benchmark (Stein, Shadaydeh, Blunk, Penzel, & Denzler, 2025) and the public Causal Chambers data sets (Gamella et al., 2025) (CC BY 4.0); sources, versions and SHA-256 checksums are listed in `DATA.md` of the supplementary repository, and the raw data are not redistributed. All simulated data are regenerated from fixed seeds by the reproduction script.
- **Code availability.** All code, including the Lean development, is in the supplementary repository **[PLACEHOLDER: public repository URL or archive DOI, to be completed by the author]**.
- **Materials availability.** Not applicable.
- **Author contributions.** Q.-V. Dang is the sole author: he conceived and directed the study, supervised the AI agents described in Section 7.1, and takes responsibility for all content.
- **Use of AI tools.** See Section 7.1.

Appendix A Overview and status labels

Appendix B gives full proofs of Section 3 (including Remark 21), Appendix C the exhaustive enumeration check, Appendix D the proofs of Section 5, Appendix E the proofs of Section 6 (including the plug-in effective sample size and the scope of the recovery bound), Appendix F the simulation checks of F1–F3, Appendix G the map from statements to Lean names, Appendix H data selection and preprocessing for the real-data studies, Appendix I the real-data estimand $\tilde{s}$, Appendix J the analysis-plan timeline for all studies and experiments, prior exposure and deviations, Appendices K and L the experiments E1 and E2 in full, Appendices M to P the loop-free graphs, lagged unrollings, the random summary graphs (with the full proof of Proposition 7 in Appendix O.1) and the accuracy interpretation, Appendices Q to T the benchmark experiment B1, its replication B1-NL, the extension to competitive learners with the post hoc re-analysis, detectability calculation and generator survey, and the discriminating test B3, and Appendix U the proofs of Sections 4.2 and 4.3.

Status labels. *Machine-checked*: a Lean theorem with no `sorry` and only the standard axioms. *Proved*: a complete paper proof given here. *Asymptotic*: a limit statement proved here from the multivariate Breuer–Major theorem (Arcones, 1994; Nourdin et al., 2011) (cited, not re-proved). *Sketched*: a derivation with one analytic step left out, stated as such (used only for statements outside the main claims: the windowed loop-free stationary value, the layered-DAG stationary value and the constants of the lagged recovery bound). *Enumerated*: checked by exact computation on many instances, not proved.

Appendix B Proofs for Section 3

Definition of s_ρ, verbatim. Reisach et al. (2026, Sec. 2.2) (arXiv v1) write: "For a graph $G = (V, E)$ with $|E| \geq 1$ we define sortability by the number of relatives, or rel-sortability for short, as $s_\rho(G) := \frac{\sum_{(x,y)\in E} s_\rho(x,y)}{|E|}$ with $s_\rho(x, y) := 0$ if $\rho(x) > \rho(y)$, $\frac{1}{2}$ if $\rho(x) = \rho(y)$, 1 if $\rho(x) < \rho(y)$, for $(x, y) \in E$." (their Eq. (4); the cases are typeset as a brace there), and add: "Note that our definition differs from the sortabilities in these works in that we count edges instead of paths." Our s_ρ is this edge average; in G_T the edges are the copies $(i(t), j(t+1))$.

Throughout, $G_S = (V, E_S)$ has a self-loop at every node, $p = |V| \geq 2$, and $d(u, v)$ is as in Section 2. We use two facts of Reisach et al. (2026), proved there for DAGs: (Thm. 1) if $x \in \mathrm{anc}(y)$ then $\mathrm{rel}(x) \subseteq \mathrm{rel}(y)$; (Lemma 1, eq. (10)) if $x \in \mathrm{anc}(y)$ then $\mathrm{rel}(x) \subsetneq \mathrm{rel}(y)$ iff $\mathrm{roots}(y) \setminus \mathrm{roots}(x) \neq \emptyset$, where $\mathrm{roots}(y) = \{x \in \mathrm{anc}(y) : \mathrm{anc}(x) = \{x\}\}$. Both are also machine-checked independently (relSet_subset_of_edge, and the direct characterisation isRel_unrolled_iff).

Lemma 19 (ancestors in G_T) *$u(s) \in \mathrm{anc}(v(t))$ iff $s \leq t$ and $d(u, v) \leq t - s$.*
Lean: unrolled_reach_iff, unrolled_reach_iff_dist.

Proof Every edge of G_T increases time by one, so a directed path from $u(s)$ to $v(t)$ has $t-s$ edges, and its projection to V is a walk of length $t-s$ from u to v in G_{S} (self-loops allowed as steps). Such a walk exists iff $d(u,v) \le t-s$: a walk contains a path, so its length is at least $d(u,v)$; conversely, pad a shortest path with $t-s-d(u,v)$ self-loop steps at u. Every such walk lifts to a directed path in G_T. □

Lemma 20 (roots) $\mathrm{roots}(v(t)) = \{u(1) : d(u,v) \le t-1\}$, *and* $\mathrm{desc}(u(1)) = \{w(s) : d(u,w) \le s-1\}$.

Lean: roots_unrolled_iff, isRoot_unrolled_iff, desc_root_unrolled_iff.

Proof A node $u(s)$ with $s \ge 2$ has the parent $u(s-1)$ through the self-loop, so it is not a root; $u(1)$ has no parent. Apply Lemma 19 with $s=1$, in both directions. □

Proof of Theorem 1 $i(t)$ is a parent of $j(t+1)$, so $\mathrm{rel}(i(t)) \subseteq \mathrm{rel}(j(t+1))$ and $s_\rho(e) \in \{1/2, 1\}$, with value 1 iff the inclusion is strict, iff $\mathrm{roots}(j(t+1)) \setminus \mathrm{roots}(i(t)) \neq \emptyset$. By Lemma 20 this set is $\{u(1) : d(u,j) \le t,\ d(u,i) > t-1\}$, and since d is integer-valued or ∞, $d(u,i) > t-1$ iff $d(u,i) \ge t$. □

Proof of Proposition 4 (and the general count) Since every ancestor of y descends from a root of y that is itself an ancestor, $\mathrm{rel}(y) = \mathrm{desc}(\mathrm{roots}(y))$. By Lemma 20, $w(s) \in \mathrm{rel}(v(t))$ iff some u has $d(u,v) \le t-1$ and $d(u,w) \le s-1$, i.e. iff $s \ge 1 + m_t(v,w)$ with $m_t(v,w) = \min\{d(u,w) : d(u,v) \le t-1\}$. Counting $s \in \{1,\dots,T\}$ gives $\rho(v(t)) = \sum_w \max(T - m_t(v,w), 0)$ (rho_unrolled_eq). For $t \ge p$ every finite $d(u,v)$ is at most $p-1 \le t-1$, so the admissible u are $\mathrm{anc}_S(v)$ and $m_t(v,w) = d(\mathrm{anc}_S(v), w)$, finite iff $w \in \mathrm{rel}_S(v)$ and then at most $p-1 < t \le T$. (The Lean statement rho_unrolled_large_t needs only $t \ge p$; an earlier hypothesis $T \ge p$ is redundant.) □

Proof of Theorem 2 Crossing edge $(i,j) \in E_\times$: j does not reach i, so $d(j,i) = \infty$, and $u = j$ gives $d(j,j) = 0 \le t \le \infty$ for every t: all $T-1$ copies score 1. Edge in $E_\circ$: i and j reach each other (or $i = j$), so for every u, $d(u,i) < \infty$ iff $d(u,j) < \infty$. If the condition of Theorem 1 held with $d(u,i) = \infty$ then $d(u,j) = \infty > t$, a contradiction; hence it forces $t \le d(u,i) \le \Delta$, and at most $N_T(i,j) \le \min(\Delta, T-1)$ copies score 1 ($\Delta \le p-1$, maxFinDist_le_card_sub_one). Summing over the $(T-1)|E_{\mathrm{S}}|$ edges,

$$s_\rho(G_T) = \tfrac{1}{2} + \frac{1}{2(T-1)|E_{\mathrm{S}}|}\Big[(T-1)|E_\times| + \sum_{(i,j)\in E_\circ} N_T(i,j)\Big],$$

which gives both bounds; let $T \to \infty$. Lean: sRho_unrolled_bounds_maxFinDist (the Δ form), with the $p-1$ form sRho_unrolled_bounds as a corollary. □

Proof of Corollary 3 $E_\times = \emptyset$ iff no edge joins two SCCs. A weakly connected component made of two or more SCCs is connected by undirected edges, one of which joins two different SCCs; conversely a strongly connected component has no crossing edge. The uniform bound is Theorem 2 with $|E_\times| \ge 1$. For the DAG case, every SCC is a singleton, $E_\times$ consists of the m non-loop edges, and $|E_{\mathrm{S}}| = m + p$. □

Proof of Theorem 5 In G_{B+T}, by Lemma 20 shifted by B, $w(s) \in \mathrm{rel}(v(t))$ iff some u has $d(u,v) \le t-1$ and $d(u,w) \le s-1$. For window nodes $t-1, s-1 \ge B \ge \Delta$, and every finite distance is at most Δ, so the condition is $u \in \mathrm{anc}_S(v) \cap \mathrm{anc}_S(w)$, independent of s,t. Hence $\mathrm{rel}_W(v(t)) = \{w(s) \in W : w \in \mathrm{rel}_S(v)\}$, of size $T|\mathrm{rel}_S(v)|$. On $V \times \mathbb{Z}$, Lemma 19 holds verbatim and the common ancestor can be taken arbitrarily early, giving the same set. Each copy of (i,j) therefore compares $|\mathrm{rel}_S(i)|$ with $|\mathrm{rel}_S(j)|$; as $i \in \mathrm{anc}_S(j)$, $\mathrm{rel}_S(i) \subseteq \mathrm{rel}_S(j)$, so the score is 1 iff $\mathrm{rel}_S(i) \ne \mathrm{rel}_S(j)$, and 1/2 for self-loops. Source SCCs: $\mathrm{rel}_S(j) \setminus \mathrm{rel}_S(i) \ne \emptyset$ iff $\mathrm{anc}_S(j) \not\subseteq \mathrm{rel}_S(i)$ (if $z \in \mathrm{rel}_S(i)$ then $\mathrm{desc}_S(z) \subseteq \mathrm{rel}_S(i)$). Every node of $\mathrm{anc}_S(j)$ is reached from some $\sigma \in \mathrm{Src}(j)$, and $\mathrm{desc}_S(\sigma) \subseteq \mathrm{rel}_S(i)$ whenever $\sigma \subseteq \mathrm{rel}_S(i)$. A source SCC meets $\mathrm{rel}_S(i)$ iff it reaches i (its only ancestors are its own nodes), and then it is contained in $\mathrm{rel}_S(i)$. So the condition is $\mathrm{Src}(j) \not\subseteq \mathrm{Src}(i)$, which, as $\mathrm{Src}(i) \subseteq \mathrm{Src}(j)$, is $\mathrm{Src}(i) \ne \mathrm{Src}(j)$. Lean: rhoBurn_eq_maxFinDist, sBurn_eq_maxFinDist (with $B \ge \Delta$), and the $B \ge p-1$ forms rhoBurn_eq, sBurn_eq as corollaries. □

Remark 21 ($B \ge \Delta$ cannot be lowered) For the single edge $0 \to 1$ with self-loops, $\Delta = 1$; with $B = 1$ the window value is the stationary value 1/2, and with $B = \Delta - 1 = 0$ it is at least 2/3 for every $T \ge 2$ (edgeLoops_burn_sharp). Enumeration agrees: on 726 random summary graphs $B = \Delta$ always gave the stationary value and $B = \Delta - 1$ did not in 445 (Appendix Q).

Proof of Corollary 6 If each weak component C has a unique source SCC σ_C, then $\mathrm{Src}(v) = \{\sigma_C\}$ on C and $E_\uparrow = \emptyset$. Conversely, if $E_\uparrow = \emptyset$, Src is constant along edges and hence on each weak component; a node v of a source SCC σ has $\mathrm{Src}(v) = \{\sigma\}$, so there is only one. Out-forests (at most one parent besides itself; acyclicity is not needed, stat_half_of_outForest) and strongly connected graphs (stat_half_of_strong) have one source SCC per component. In an in-forest (acyclic apart from self-loops, out-degree at most one), if $u \in \mathrm{anc}(v)$ then $\mathrm{desc}(u)$ is the unique downstream path through v, so $\mathrm{rel}(v) = \mathrm{anc}(v) \cup \mathrm{desc}(v)$ (inForest_isRel_iff); the source SCCs are the headwaters, and $\mathrm{Src}(j) = \mathrm{Src}(i) \cup \bigcup_k \mathrm{Src}(k)$ over the other parents k of j, with disjoint headwater sets on different branches, so $\mathrm{Src}(j) \ne \mathrm{Src}(i)$ iff j is a confluence (inForest_up_iff). Out-degree at most one alone is not enough: for $a \rightleftarrows b$ with $c \to a$, a has two parents but the only source SCC is $\{c\}$. □

Remark 22 (self-loops) Without self-loops the lemmas fail: for the single edge $0 \to 1$ without loops, $s_\rho(G_T) = 1/2$ for every $T \ge 2$ (edgeNoLoops_sRho); with loops the limit is 2/3 (edgeLoops_limit).

Appendix C Exhaustive Enumeration Check

Before formalisation, every statement of Section 3 was checked by brute force (bitset relatives with the definitions of Reisach et al., 2026) on all $4 + 64 + 4096$ summary graphs with $p = 2, 3, 4$ (all $T \le 9$) and on 10,000 random graphs ($p = 2$–7; Erdős–Rényi, DAG, in-forest, out-forest, chain with back edge and block-SCC families; three values of $T \le 14$ each, fixed seed). In total 46,724 (graph, T) cases and 1,895,609 edges were checked with zero failures of Theorem 1, Proposition 4, the bounds of Theorem 2, Theorem 5 (with $B = p-1$ and $B = 2p+1$) and the source-SCC characterisation. The conjecture fixed in plan 1 ("an edge is strictly sorted in the limit

iff $|\mathrm{rel}_S(j)| > |\mathrm{rel}_S(i)|$") disagreed with the exact G_T limit in 9,603 random graphs and never with the stationary value; the conjecture is Theorem 5, not the limit of G_T.

Appendix D Proofs for Section 5

Proof of Proposition 14 By the MA(∞) representation $X(t) = \sum_k \Psi_k \varepsilon(t-k)$, and because Ψ_k is supported on the reachability relation, $X_i(t)$ is a measurable function of $\{\varepsilon_k(s) : k \in \mathrm{anc}_S(i)\}$. Disjoint ancestor sets mean independent innovation sets, so the processes X_i and X_j are independent, and so are any per-node transforms. For the generic part, each covariance entry is real-analytic in (A, Σ) on the connected open set of stable parameters with the given support, hence either identically zero or nonzero almost everywhere. With all coefficients positive, including the self-loop at a common ancestor k, all Ψ entries are nonnegative and $\Psi_{ik}(m), \Psi_{jk}(m) > 0$ for all large m (self-loops pad the walks), so $\Gamma(0)_{ij} > 0$; the same holds for $\Gamma(h)$, $h \geq 0$, and therefore for the Fejér sums and S_{LR}. "Generic" does not mean "large": with little autocorrelation, $\Gamma(0)_{ij}$ along a lagged edge can be tiny (down to 10^{-14} in our exact computations; exactly 0 without autocorrelation). □

Proof of Theorem 15 (a) $D^2\mathrm{Cov}(\bar{Y}) = \sum_{u,w<D} \Gamma(u-w) = D\Gamma(0) + \sum_{h=1}^{D-1}(D-h)(\Gamma(h)+\Gamma(-h))$ (the counting identity is fejer_identity). The stable VAR has $\|\Gamma(h)\| \leq Cr^{|h|}$, $r < 1$, so $\sum_h |h|\|\Gamma(h)\| < \infty$ and $S_{LR} = \sum_h \Gamma(h) = \Psi(1)\Sigma\Psi(1)^\top$ with $\Psi(1) = (I - A(1))^{-1}$. Then $S_{LR} - D\,\mathrm{Cov}(\bar{Y}) = \sum_{|h|\geq D} \Gamma(h) + \sum_{|h|<D} \frac{|h|}{D}\Gamma(h)$, and $1 \leq |h|/D$ in the first sum. (b) Write $I - A(1) = (I - \Lambda) - N$ with Λ the diagonal and N the off-diagonal part; then $I - A(1) = (I - \Lambda)(I - \tilde{B})$ with $\tilde{B} = (I - \Lambda)^{-1}N$ strictly triangular in a topological order (longrun_factor, longrun_inv). Under acyclicity the AR polynomial factorises into univariate $\phi_j(z) = 1 - \sum_\tau (A_\tau)_{jj} z^\tau$; stability, $\phi_j(0) = 1$ and the intermediate value theorem give $\phi_j(1) = 1 - \lambda_j > 0$ (selfloop_sum_lt_one). (c) Stationarity. (d) For a linear SEM $X = BX + e$ with noise variances ω, $R_i^2 = 1 - 1/(\Sigma_{ii}(\Sigma^{-1})_{ii})$ and $(\Sigma^{-1})_{ii} = 1/\omega_i + \sum_{k\in\mathrm{ch}(i)} B_{ki}^2/\omega_k$. In an out-star with root 0 and leaves k, $1 - R_0^2 = 1/(1 + \sum_k r_k)$ and $1 - R_k^2 = 1/(1 + r_k)$ with $r_k = \omega_0 B_{k0}^2/\omega_k > 0$, so with ≥ 2 leaves the root has strictly larger R^2 than every leaf; the in-star is symmetric. Strict inequalities in the limit SEM of (b), plus (a) and continuity of $S \mapsto R^2$ at positive-definite S, give the claim for all large D. Under subsampling every contemporaneous statistic is a function of $\Gamma(0)$. □

Appendix E Proofs for Section 6

Notation. $\hat{C}_{ab} = \frac{1}{n}\sum_t (X_a(t) - \bar{X}_a)(X_b(t) - \bar{X}_b)$, $\hat{r}_{ab} = \hat{C}_{ab}/\sqrt{\hat{C}_{aa}\hat{C}_{bb}}$. Constants: B_{ab} as in the main text; $D_\gamma = \max_a 2\sum_h \gamma_{aa}(h)^2$; $\kappa_1 = \max_a \sum_h |\gamma_{aa}(h)|$; $\kappa_2 = \max_{a\neq b}(\sum_h |\gamma_{aa}(h)| + \sum_h |\gamma_{ab}(h)|)$. For i.i.d. sampling, $B_{ab} = 1 + \rho_{ab}^2$, $D_\gamma = 2$, $\kappa_1 = 1$.

Lemma 23 (Gaussian chaos tail) *Let $g \sim N(0, I_N)$, A symmetric, $S = g^\top Ag - \mathrm{tr}A$. For $x > 0$, $P(S \geq 2\|A\|_F\sqrt{x} + 2\|A\|_{op}x) \leq e^{-x}$, and the same for $-S$.*

Lean: algebraic steps phi_le, chernoff_logmgf_le, chernoff_choice, sqrt_two_v_x; the probability statement is not machine-checked.

Proof Diagonalise; S has the law of $\sum_i \lambda_i (g_i^2 - 1)$. For $2s\lambda < 1$, $\log E e^{s\lambda(g^2-1)} = \varphi(s\lambda)$ with $\varphi(u) = -\frac{1}{2}\log(1-2u) - u = \sum_{k\ge 2} (2u)^k/(2k) \le u^2/(1-2|u|)$ for $|u| < 1/2$. With $c = 2\|A\|_{op}$, $v = 2\|A\|_F^2$ and $0 \le s < 1/c$, $\log E e^{sS} \le vs^2/(2(1-cs))$. Markov's inequality with $t = \sqrt{2vx} + cx$, $q = \sqrt{2x/v}$ and $s = q/(1+cq)$ gives exponent exactly x. This is the Gaussian case of Hsu, Kakade, and Zhang (2012, Prop. 1), who derive it from Laurent and Massart (2000); the proof here is self-contained. □

Lemma 24 (exact variance) *With* $\tilde\gamma_{ab} = \frac{1}{n}\sum_t X_a(t)X_b(t) = g^\top A g$ *for* $A = C^{1/2} M C^{1/2}$, $M = \frac{1}{2n}\left(\begin{smallmatrix} 0 & I \\ I & 0 \end{smallmatrix}\right)$,

$$2n\|A\|_F^2 = \sum_{|h|<n} \Big(1 - \tfrac{|h|}{n}\Big)\big[\gamma_{aa}(h)\gamma_{bb}(h) + \gamma_{ab}(h)\gamma_{ab}(-h)\big] \le B_{ab},$$

$\|A\|_{op} \le \kappa_2/(2n)$*; on the diagonal* $2n\|A\|_F^2 \le D_\gamma$ *and* $\|A\|_{op} \le \kappa_1/n$*; and* $\bar X_a \sim N(0, \bar\sigma_a^2)$ *with* $\bar\sigma_a^2 \le \kappa_1/n$.

Lean: trace and Toeplitz algebra quad_M, trace_MC, lemma12_trace, lemma12_fejer, lemma12_le_B, lemma12_diag_le_D; the step $\|C^{1/2}MC^{1/2}\|_F^2 = \mathrm{tr}(MCMC)$ *and the operator-norm bounds are not machine-checked.*

Proof $\|A\|_F^2 = \mathrm{tr}(MCMC)$ and, with Toeplitz blocks $(C_{ab})_{st} = \gamma_{ab}(s-t)$, $4n^2\mathrm{tr}(MCMC) = 2\sum_{s,t}[\gamma_{aa}(s-t)\gamma_{bb}(s-t) + \gamma_{ab}(s-t)\gamma_{ab}(t-s)]$ (using that γ_{aa}, γ_{bb} are even); count pairs with $s-t = h$. $\|A\|_{op} \le \|C\|_{op}\|M\|_{op}$ and $\|C\|_{op}$ is at most the largest absolute row sum. □

Lemma 25 (deviation event) *Let* $\varepsilon_{ab} = \sqrt{2B_{ab}x/n} + \kappa_2 x/n + 2\kappa_1 y/n$ *and* $\varepsilon_d = \sqrt{2D_\gamma y/n} + 4\kappa_1 y/n$*. With probability at least* $1 - d(d-1)e^{-x} - 4de^{-y}$, $|\hat C_{ab} - \rho_{ab}| \le \varepsilon_{ab}$ *for all* $a \ne b$ *and* $|\hat C_{aa} - 1| \le \varepsilon_d$ *for all* a.

Lean: deterministic steps centred_cov_eq, centred_dev_offdiag, centred_dev_diag, lemma13_eps, lemma13_count.

Proof $\hat C_{ab} = \tilde\gamma_{ab} - \bar X_a \bar X_b$. Apply Lemmas 23 and 24 to each off-diagonal and diagonal $\tilde\gamma$, bound $|\bar X_a| \le \sqrt{2\kappa_1 y/n}$ by the Gaussian Chernoff bound, and take a union bound. □

Lemma 26 (correlation perturbation) *If* $a', b' \in [1-\varepsilon_d, 1+\varepsilon_d]$, $\varepsilon_d < 1$ *and* $|c' - \rho| \le \varepsilon$*, then* $|c'/\sqrt{a'b'} - \rho| \le (\varepsilon + |\rho|\varepsilon_d)/(1-\varepsilon_d)$.

Lean: corr_perturb (the hypothesis $|\rho| \le 1$ *of our earlier statement is not needed).*

Proof $|c'/\sqrt{a'b'} - \rho| \le |c' - \rho|/\sqrt{a'b'} + |\rho||1/\sqrt{a'b'} - 1|$, with $1/\sqrt{a'b'} \le 1/(1-\varepsilon_d)$ and $|1/\sqrt{a'b'} - 1| \le \varepsilon_d/(1-\varepsilon_d)$. □

Theorem 27 (recovery bound) *For* $x, y > 0$ *let* ε_{ab}, $\varepsilon_d < 1$ *be as in Lemma 25 and* $\delta_{ab} = (\varepsilon_{ab} + |\rho_{ab}|\varepsilon_d)/(1-\varepsilon_d)$*. If thresholds separate* ($\delta_{ab} \le \tau_{ab}$ *on non-relatives and* $|\rho_{ab}| - \delta_{ab} > \tau_{ab}$ *on relatives), the declared set equals the true one, and hence* $\hat s = s_\rho^{\mathrm{pop}}$*, with probability at*

least $1-d(d-1)e^{-x}-4de^{-y}$. *For a common threshold* $\tau<\rho_{\min}$ *the failure probability decays as* $\exp(-n\min(\tau,\rho_{\min}-\tau)^2/(2(\sqrt{B}+\rho_{\max}\sqrt{D_\gamma})^2))$.

Lean: deterministic core thm15_det, decide_eq_target, corr_perturb, sOf_cntDecl_congr; Chernoff and trace algebra chernoff_choice, phi_le, lemma12_trace, lemma12_le_B. The Gaussian tail bound itself is not machine-checked.

The proof combines a Gaussian-chaos tail bound (the Gaussian case of Hsu et al., 2012, Prop. 1; see also Laurent & Massart, 2000), an exact finite-n variance identity for $\hat{r}$, and a deterministic perturbation lemma.

Proof of Theorem 27 Set $\delta_{ab}=(\varepsilon_{ab}+|\rho_{ab}|\varepsilon_d)/(1-\varepsilon_d)$. On the event of Lemma 25, Lemma 26 gives $|\hat{r}_{ab}-\rho_{ab}|\le\delta_{ab}$, so non-relatives satisfy $|\hat{r}|\le\tau$ and relatives $|\hat{r}|>\tau$ (decide_eq_target, thm15_det). Equal declared sets give equal counts and hence $\hat{s}=s_\rho^{\mathrm{pop}}$ (sOf_cntDecl_congr). For a VAR observed as a long stationary series, s_ρ^{pop} is the stationary value of Theorem 5 on non-self-loop edges (Proposition 14). Explicit form: with $y=x+\log(4/(d-1))_+$ (explicit_y) and a common threshold $\tau<\rho_{\min}$, $P(\hat{R}\neq R)\le 2d(d-1)\exp\big(-n\min(\tau,\rho_{\min}-\tau)^2/(2(\sqrt{B}+\rho_{\max}\sqrt{D_\gamma})^2)(1+o(1))\big)$. □

The multivariate Breuer–Major theorem. We use it in the form stated by Nourdin et al. (2011, Thm. 1.1), who attribute the vector case to Arcones (1994): let $X=(X_k)_{k\in\mathbb{Z}}$ be a centred stationary Gaussian process in $\mathbb{R}^d$ with covariances $r^{(i,l)}(j)=E[X_1^{(i)}X_{1+j}^{(l)}]$, and $f:\mathbb{R}^d\to\mathbb{R}$ with $Ef(X_1)^2<\infty$ and Hermite rank $q\ge 1$ with respect to X (orthogonal to all polynomials of degree $<q$, not to some of degree q). If $\sum_{j\in\mathbb{Z}}|r^{(i,l)}(j)|^q<\infty$ for all i,l, then $n^{-1/2}\sum_{k\le n}(f(X_k)-Ef(X_k))\Rightarrow N(0,\sigma^2)$ with $\sigma^2=\mathrm{Var}f(X_1)+2\sum_{k\ge1}\mathrm{Cov}(f(X_1),f(X_{1+k}))\in[0,\infty)$.

Proof of Proposition 16 Take $d=2$, $X_k=(X_a(k),X_b(k))$ and, for $c\in\mathbb{R}^3$, $f_c(x)=c_1(x_a^2-1)+c_2(x_b^2-1)+c_3(x_ax_b-\rho_{ab})$. f_c is a centred polynomial of degree 2 without linear part, so it is orthogonal to every polynomial of degree ≤ 1 (odd Gaussian moments vanish); if it were orthogonal to every polynomial of degree 2 it would be orthogonal to itself, so either $f_c(X_1)=0$ a.s. (then the partial sums vanish) or its Hermite rank is 2. The condition $\sum_h|\gamma_{ab}(h)|^2<\infty$ holds because the covariances are absolutely summable and bounded. By the theorem and the Cramér–Wold device, $\sqrt{n}(\tilde{\gamma}_{aa}-1,\tilde{\gamma}_{bb}-1,\tilde{\gamma}_{ab}-\rho_{ab})\Rightarrow N(0,V)$, where, by Isserlis' theorem, $V(ab,cd)=\sum_h[\gamma_{ac}(h)\gamma_{bd}(h)+\gamma_{ad}(h)\gamma_{bc}(h)]$ (as in Lemma 24). Centring by the sample means changes each entry by $\bar{X}_a\bar{X}_b$, and $\sqrt{n}\bar{X}_a\bar{X}_b\to 0$ in probability since $E(\sqrt{n}\bar{X}_a)^2\le\kappa_1$ and $\bar{X}_b\to 0$; Slutsky's lemma applies. The delta method with gradient $(-\rho/2,-\rho/2,1)$ gives $\sqrt{n}(\hat{r}-\rho)\Rightarrow N(0,W)$, $W=V(ab,ab)-\rho[V(ab,aa)+V(ab,bb)]+\frac{\rho^2}{4}[V(aa,aa)+2V(aa,bb)+V(bb,bb)]$, which equals $\sum_h\gamma_{aa}(h)\gamma_{bb}(h)$ for an independent pair (this is B_{ab} when all products $\gamma_{aa}(h)\gamma_{bb}(h)$ are nonnegative, and at most B_{ab} in general). Then $\sqrt{n}\,\operatorname{atanh}\hat{r}\Rightarrow N(0,W_{ab})$ under the null and $P(|\operatorname{atanh}\hat{r}|>z_c/\sqrt{n-3})\to 2\bar{\Phi}(z_c/\sqrt{W_{ab}})$, which exceeds the nominal α/m in the limit whenever $W_{ab}>1$. □

Lemma 28 (sample autocovariances) *Let* X_a *be a standardised stationary Gaussian series with* $\sum_h|\gamma_{aa}(h)|\le\kappa_1$ *and* $\hat{\gamma}_a(h)=\frac{1}{n}\sum_{t\le n-h}(X_a(t+h)-\bar{X}_a)(X_a(t)-\bar{X}_a)$. *For* $0\le h<n$, $E(\hat{\gamma}_a(h)-\gamma_{aa}(h))^2\le(8\kappa_1+108\kappa_1^2)/n+2(h/n)^2\gamma_{aa}(h)^2$.

Proof For $\tilde\gamma(h) = \frac{1}{n}\sum_{t\le n-h} X(t+h)X(t)$, $E\tilde\gamma(h) = (1-h/n)\gamma(h)$ and, by Isserlis, $\operatorname{Var}\tilde\gamma(h) = n^{-2}\sum_{s,t\le n-h}[\gamma(s-t)^2 + \gamma(s-t+h)\gamma(s-t-h)] \le 2\kappa_1/n$, since $|\gamma| \le 1$ gives $\sum_k \gamma(k)^2 \le \kappa_1$ and $\sum_k |\gamma(k+h)\gamma(k-h)| \le \kappa_1$. The centring term is $-\bar X(S_1+S_2)/n + \frac{n-h}{n}\bar X^2$ with S_1, S_2 partial sums; $\bar X$, S_1/n, S_2/n are centred Gaussians with variance $\le \kappa_1/n$ and $E(UV)^2 \le 3\sigma^4$ for such pairs, so its second moment is $\le 27\kappa_1^2/n^2$. Combine with $(u+v+w)^2 \le 4u^2+4v^2+2w^2$ for $u+v$ and the bias $w = h\gamma(h)/n$. □

Lemma 29 (plug-in Bartlett factor) *Let X_a, X_b satisfy the assumptions of Section 6 (no condition on their cross-dependence), $\hat\rho_a(h) = \hat\gamma_a(h)/\hat\gamma_a(0)$, $\hat W_n = 1 + 2\sum_{h=1}^{K_n}\hat\rho_a(h)\hat\rho_b(h)$ and $W^\circ_{ab} = \sum_{h\in\mathbb{Z}}\rho_a(h)\rho_b(h)$. If $K_n \to \infty$ and $K_n/n \to 0$, then $|\hat W_n - W^\circ_{ab}| = O_P(n^{-1/2} + K_n/n) + 2\sum_{h>K_n}|\rho_a(h)\rho_b(h)| \to 0$; for fixed K, $\hat W \to 1 + 2\sum_{h\le K}\rho_a(h)\rho_b(h)$ in probability.*

Proof With $\Delta_a = \hat\rho_a - \rho_a$, on $\{\hat\gamma_a(0), \hat\gamma_b(0) \ge \frac12\}$ (probability $\to 1$), $|\Delta_a(h)| \le 2|\hat\gamma_a(h) - \gamma_{aa}(h)| + 2|\gamma_{aa}(h)||\hat\gamma_a(0) - 1|$. Write $\hat\rho_a\hat\rho_b - \rho_a\rho_b = \Delta_a\rho_b + \rho_a\Delta_b + \Delta_a\Delta_b$. By Lemma 28, $E\sum_{h\le K}|\hat\gamma_a(h) - \gamma_{aa}(h)||\rho_b(h)| \le \kappa_1\sqrt{c_0/n} + \sqrt{2}\kappa_1 K/n$, so the linear terms are $O_P(n^{-1/2} + K/n)$; by Cauchy–Schwarz and $E\sum_{h\le K}(\hat\gamma_a - \gamma_{aa})^2 \le Kc_0/n + 2\kappa_1(K/n)^2$, $\sum_{h\le K}|\Delta_a\Delta_b| = O_P(K/n)$; the tail is at most $\sum_{h>K}|\rho_a(h)| \to 0$. □

Only $K_n = o(n)$ is needed, because the target is a sum of products of two summable sequences; we do not rely on general HAC bandwidth conditions.

Proposition 30 (plug-in effective sample size) *Under the assumptions of Proposition 16, use the plug-in threshold $\tanh(z_c/\sqrt{n/\hat B_{ab} - 3})$ with $\hat B_{ab} = \max(1, \hat W_n)$, where $1 \le K_n = o(n)$. (i) For an independent pair with $W_{ab} > 0$ and $K_n \to \infty$, the false-positive probability tends to $2\bar\Phi(z_c\sqrt{\max(1, W_{ab})/W_{ab}})$, which is α/m if $W_{ab} \ge 1$ and smaller otherwise. For fixed K it tends to $2\bar\Phi(z_c\sqrt{\max(1, W^\circ_{K,ab})/W_{ab}})$, $W^\circ_{K,ab} = 1 + 2\sum_{h\le K}\rho_a(h)\rho_b(h)$. (ii) Every pair with $\rho_{ab} \ne 0$ is declared with probability $\to 1$. (iii) If the non-relatives are independent pairs (summary non-relatives of a Gaussian VAR) and every relative has $\rho_{ab} \ne 0$, then $\liminf_n P(\hat R = R) \ge 1 - |N|\alpha/m \ge 1 - \alpha$, and the same holds for $P(\hat s = s^{\mathrm{pop}}_\rho)$, provided $K_n \to \infty$.*

Lean: *not machine-checked (uses the multivariate Breuer–Major theorem through Proposition 16).*

Proof (i) $T = |\sqrt{n}\,\operatorname{atanh}\hat r|\,(1/\hat B - 3/n)^{1/2}$. By Proposition 16, $\sqrt{n}\,\operatorname{atanh}\hat r \Rightarrow N(0, W)$, and $\hat B \to b = \max(1, W)$ (or $\max(1, W^\circ_K)$) in probability by Lemma 29. By Slutsky, $T \Rightarrow |N(0, W/b)|$, whose law has no atom at z_c. (ii) By Lemma 24, $\hat r_{ab} \to \rho_{ab}$ in probability. By Cauchy–Schwarz $|\hat\rho_a(h)| \le 1$, so $\hat B \le 1 + 2K_n = o(n)$ and $n/\hat B \to \infty$. (iii) Union bound over N with (i), and (ii) on R; equal declared sets give equal $\hat s$. □

Proposition 31 (scope of Theorem 27) *Let the non-relatives $N \ne \emptyset$ have deterministic thresholds with $n\tau^2_{n,ab} \to t^2_{ab} \in (0, \infty)$: $t^2 = z_c^2$ for the nominal Fisher-z test, and $t^2 = z_c^2 b_{ab}$ for the effective one with $n' = n/b_{ab}$. Then the smallest failure probability certified by Theorem 27 tends to $d(d-1)\exp(-\min_N t^2_{ab}/(2B_{ab})) > 0$ when the relatives have*

$|\rho| \geq \rho_{\min} > 0$ *(and is at least this in general). For i.i.d. data and the nominal test this is* $d(d-1)e^{-z_c^2/2} \geq \sqrt{2\pi}\, z_c \alpha$ *(Mills), e.g.* 0.089 *for* $d = 5$, $\alpha = 0.01$. *If instead* $\alpha_n \to 0$ *with* $\log(1/\alpha_n) = o(n)$, *Theorem 27 gives* $P(\hat{R} \neq R) \leq d(d-1)\exp(-(1-\eta)z_{c,n}^2 r/2) + 4d/n \to 0$ *for any* $\eta > 0$ *and large* n, *with* $r = \min_N (n/n'_{ab})/B_{ab}$.

Proof For $\rho_{ab} = 0$ the separation condition $\delta_{ab} \leq \tau_{n,ab}$ and $\varepsilon_{ab} \geq \sqrt{2B_{ab}x/n}$ force $x \leq n\tau_{n,ab}^2/(2B_{ab})$, which gives the lower bound. For attainment, take $x_n = (1-\eta)\min_N n\tau_{n,ab}^2/(2B_{ab})$ and $y_n = \log n$. Then all lower-order terms are $o(n^{-1/2})$ and $\varepsilon_d \to 0$, so the conditions hold for large n on N, and on the relatives because $\tau_n \to 0$. The vanishing-level statement is the same computation with $z_{c,n} \to \infty$. □

Remark 32 (non-Gaussian innovations) Let $X(t) = \sum_j \Psi_j \varepsilon(t-j)$ with $\sum_j \|\Psi_j\| < \infty$ and i.i.d. innovations with independent unit-variance coordinates and fourth cumulants $\kappa_i = E\varepsilon_i^4 - 3$. Put $G_i(a,b) = \sum_j \Psi_j[a,i]\Psi_j[b,i]$. Then $n\,\mathrm{Cov}(\tilde{\gamma}_{ab}, \tilde{\gamma}_{cd}) \to V(ab,cd) + \sum_i \kappa_i G_i(a,b)G_i(c,d)$ (moment–cumulant formula and dominated convergence), so $W_{ab}^{\mathrm{ng}} = W_{ab} + \sum_i \kappa_i g_i^2$ with $g_i = G_i(a,b) - \frac{\rho_{ab}}{2}(G_i(a,a)+G_i(b,b))$. Independent series share no innovation, so $g_i = 0$: the null variance $\sum_h \rho_a(h)\rho_b(h)$, and hence Propositions 16 and 30, are unchanged, given the CLT for linear processes, which we do not re-prove. For relatives, W changes by $\sum_i \kappa_i g_i^2$ (of either sign, since $\kappa_i \geq -2$). This affects power and transition widths, not the population statements or the location of $n_{1/2}$. The non-asymptotic bounds use the Gaussian chaos tail. For sub-Gaussian innovations the Hanson–Wright inequality (Rudelson & Vershynin, 2013, Thm. 1.1) has the same form with unspecified constants.

Fixed K in practice (simulation and data check). Independent AR(1) pairs, 4,000 replications (2,000 at $n = 64{,}000$): at AR coefficient 0.9 and 0.97 the plug-in test with $K = 50$ has size 0.050 and 0.056 (level 0.05; limits 0.05 and 0.0556), but at 0.99 it has 0.113 (limit 0.118), while $K_n = n^{0.6}$ gives 0.0495 and the nominal test 0.853; for the opposite-sign pair the floor makes the test conservative (size 0). On the B3 primary data (30 segments of 3,000 rows, 2,730 unit-pairs) the declared status of no pair changes when $K = 50$ is replaced by 55, 122 or 300 (post hoc). This was not checked for study 3, E1 or the river data. Scripts: `finite2/`.

F2 and F3. The statements are Propositions 17 and 18 in Section 6.

Proof of Proposition 17 $\mathrm{Cov}(X_u, X_v) = \gamma_{uv}^Y(0) + b_u b_v$ and $\gamma_{uv}^Y = 0$ for non-relatives (Proposition 14); divide by the standard deviations (prop21_corr_std). For relatives the sum is nonzero except on a Lebesgue-null set. All counts equal d, so every edge scores 1/2 (sOf_saturated). The rate: apply Theorem 27 with target "all pairs" and $x = y = cn$, c small enough that $\delta \leq \kappa_{\min}/3$ and $\tau_n \leq \kappa_{\min}/3$ eventually; $|\hat{s} - 1/2| \leq 1/2$ always (abs_sOf_sub_half_le). The onset: by the asymptotic normality of $\operatorname{atanh} \hat{r}$, the declaration probability crosses 1/2 where $\tanh(z_c/\sqrt{n-3}) = |\kappa|$, i.e. at $n_{1/2}$ (ncrit_eq_iff). □

Proof of Proposition 18 $\mathrm{Cov}(X_u, Z) = b_u\sqrt{r}$ and $\mathrm{Var}Z = 1$; the partial covariance is $\gamma_{uv}^Y + (1-r)b_u b_v$ and the partial variance $\mathrm{Var}Y_u + (1-r)b_u^2$; divide and standardise (prop31, prop31_std; positive $\mathrm{Var}Y_u$ is required). At $r = 1$, $\pi = \rho^Y$ (cor32). For non-relatives $|\pi_{uv}| = \sqrt{\ell_u \ell_v}\sqrt{f_u f_v}$ with $f_w(r) = (1-r)/(1-r\ell_w)$ and $f'_w(r) = (\ell_w - 1)/(1-r\ell_w)^2 < 0$ (absPi

_strictAntiOn, absPi_eq_zero_iff; both need $b_u b_v \neq 0$). Bounds $(1-r)\sqrt{\ell_u \ell_v} \leq |\pi| \leq (1-r)\sqrt{\ell_u \ell_v}/(1-\ell_{\max})$ (absPi_lower, absPi_upper). r_{crit} solves $(1-r)\ell/(1-r\ell) = \tau'$ (rcrit_iff, rcrit_example). The finite-sample bracket uses the Lipschitz bound $|g(a',b',c') - g(a,b,c)| \leq L(q)\max(|a'-a|, |b'-b|, |c'-c|)$ for $g(a,b,c) = (a-bc)/\sqrt{(1-b^2)(1-c^2)}$, $|b|, |c| \leq q < 1$, $L(q) = 1/(1-q^2) + 4q/(1-q^2)^2$ (pc_lipschitz, thm35_bracket), combined with Lemma 25 for (X, Z). Multi-filter driver: for $D_v(t) = b_v C(t-k_v)$ with C an AR(1) with coefficient ϕ and $Z = C(t)$,

$$\pi_{uv} = \frac{b_u b_v (\phi^{|k_u - k_v|} - \phi^{k_u + k_v})}{\sqrt{(s_u^2 - b_u^2 \phi^{2k_u})(s_v^2 - b_v^2 \phi^{2k_v})}},$$

nonzero for $k_u, k_v > 0$, $\phi \in (0,1)$, $b_u b_v \neq 0$ (prop36_ar1, prop36_ar1_ne_zero); in general $\pi_{uv} \propto \text{Cov}(D_u - P_Z D_u, D_v - P_Z D_v)$ vanishes for all non-relative pairs iff the residual driver components are uncorrelated, which holds for a rank-one driver with $Z = F$ (prop36_pcov, prop36_zero_iff, prop36_rank_one). A proxy that contains summary relatives of u, v adds the terms $-c_Z(b_u q_v + b_v q_u) - q_u q_v$ to the numerator (prop37_pcov, prop37_bound). Triad identity: $\rho_{uZ}\rho_{vZ}/\rho_{uv} = r$ and, for $r > 0$, $\ell_u = \rho_{uZ}^2/r$ (prop38_ratio, prop38_loading). □

Appendix F Simulation Checks of F1–F3

All with fixed seeds; full tables are in the supplementary material. **F1** (5 processes: in-forests with $d = 5$ and $d = 10$, self-loop persistence $\lambda \in \{0.5, 0.9, 0.97\}$, and a near-white-noise case; $n = 200$–$20{,}000$; 1,000 replications per cell; nominal, oracle-effective and fixed thresholds). The bound was never violated in 105 cells. It is loose: with a fixed threshold $\rho_{\min}/2$, it reaches 0.9 at about 3–13 times the n at which the simulated probability does, and much later with Bonferroni thresholds, where the Chernoff step loses the Mills-ratio factor. The empirical $n\,\text{Var}(\hat{r})/W$ lies in $[0.93, 1.11]$. With the nominal test, $P(\hat{R} = R)$ plateaus at 0.94, 0.22, 0.03 and 0.27, as Proposition 16 predicts (predicted vs simulated false-positive rates, e.g. 0.33 vs 0.32–0.33 at $\lambda = 0.9$). **F2** ($\ell \in \{0.02, 0.1, 0.3, 0.64\}$): the asymptotic detection curve matches the simulated fraction of declared non-relatives within 0.02 for $n \geq 200$; predicted $n_{1/2} = 1{,}079$ for $\ell = 0.1$, simulated 0.479 at $n = 1{,}000$. **F3**: predicted $n_{\text{crit}} = 472$ ($\ell = 0.64$, $r = 0.9$) gives a simulated declared fraction 0.495 at $n = 471$; over 38 cells the asymptotic prediction is within 0.03. With travel-time lags $(0, 1, 2, 4, 8)$, $\phi = 0.9$, $r = 1$, the declared fraction still rises to 0.63 at $n = 10{,}000$, as the multi-filter part of Proposition 18 predicts.

Appendix G Lean Formalisation: Map and Findings

Lean 4.35.0 and Mathlib v4.35.0; 4,159 lines in 16 files. The build log (a clean rebuild of the final state, 9,064 jobs, including Observed.lean and the Δ refinements), the `#print axioms` output for each of the 163 declarations in the axiom report (all `[propext, Classical.choice, Quot.sound]`) and a grep showing no `sorry`/`admit`/`axiom` are in the supplementary material.

What is and is not certified. Certified: the exact and stationary results of Section 3, the loop-free exact edge value, the accuracy reading, the algebra of the estimand $\tilde{s}$ (Appendix I), and the deterministic and algebraic parts of Sections 5

and 6. Not certified: the ER limits (only the deterministic witness lemmas), the loop-free limit, periodicity and stationary value, the lag-weighted extension, and every probability bound, limit theorem and genericity statement.

Breakdown of the 163 declarations (counted from the Lean files): 33 headline theorems, i.e. Lean counterparts of numbered paper statements (29 for Section 3, the loop-free edge value and the accuracy reading, including the four Δ forms of Theorems 2 and 5 and Corollary 6; 4 for Proposition 34); 53 supporting graph lemmas (ancestry, roots, distance, the largest finite distance Δ (maxFinDist, 10 lemmas), source SCCs, 5 ER witness lemmas, 6 estimand lemmas); 65 algebraic identities and inequalities of the aggregation and finite-sample files (correlation perturbation, Chernoff algebra, trace and Toeplitz identities, partial correlation formulas, threshold locations; several are elementary, e.g. $|\hat{s} - \frac{1}{2}| \leq \frac{1}{2}$); 12 examples (10 concrete graphs, the sharpness example edgeLoops_burn_sharp, 1 numerical r_{crit} instance). No human has checked, line by line, that each Lean statement says what the corresponding paper statement says; the map below is the authors' (AI-generated) reading.

statement	Lean names	status
Lemmas 19 and 20	unrolled_reach_iff, unrolled_reach_iff_dist, roots_unrolled_iff, desc_root_unrolled_iff, isRel_unrolled_iff	checked
Theorem 1	edge_score, edge_score_dist, strictCond_iff_dist	checked
Proposition 4	rho_unrolled_eq, mT_le_iff, rho_unrolled_large_t	checked
Theorem 2	sRho_unrolled_exact, sRho_unrolled_bounds_maxFinDist (Δ form), sRho_unrolled_bounds ($p-1$ form), tendsto_sRho_unrolled, tendsto_sRho_GT; Δ: maxFinDist, dist_le_maxFinDist, exists_dist_eq_maxFinDist, maxFinDist_le_card_sub_one	checked
Corollary 3	tendsto_half_iff, crossEdges_eq_empty_tfae, sRho_unrolled_ge_of_cross, reisach_theorem4, limitValue_dag	checked
Theorem 5	rhoBurn_eq_maxFinDist, sBurn_eq_maxFinDist, relBurn_iff_maxFinDist, walkN_of_reach_maxFinDist ($B \geq \Delta$); rhoBurn_eq, sBurn_eq ($B \geq p-1$); rhoZ_eq, sZ_eq, sOf_window, relSet_eq_iff_src, mem_upEdges_iff	checked
Remark 21 ($B \geq \Delta$ cannot be lowered)	edgeLoops_burn_sharp, sBurn_zero, relBurn_zero_iff, maxFinDist_edgeLoops	checked
Corollary 6	sBurn_eq_half_iff_maxFinDist, sBurn_eq_half_iff, sBurn_eq_half_iff_common_anc, sZ_eq_half_iff, upEdges_eq_empty_tfae, stat_half_of_strong, stat_half_of_outForest, inForest_isRel_iff, inForest_up_iff, statValue_inForest, statValue_inForest_gt_iff	checked
Table 1	collider_limit, fork_limit, chain3_limit, collider_stat, fork_stat, chain3_stat	checked
self-loop remark	edgeLoops_limit, edgeNoLoops_sRho	checked
Theorem 36 (ancestors, relatives, edge value)	isRel_unrolled_iff_NL, reduce_ancestor, relSet_unrolled_eq_iff_NL, edge_score_NL	checked
Corollary 37	sRho_unrolled_eq_NL, sRho_unrolled_ge_half_NL	checked
Proposition 40 (witness $u = j$, fresh parent)	strictNL_of_closed_walk, strictNL_of_fresh_parent, sRho_unrolled_eq_one_NL	checked
loop-free examples (Table M6)	colliderNL_sRho, edgeLoopHead_sRho, edgeNoLoops_sRho	checked
eventual periodicity, limit, same-SCC edges, Theorem 42	–	paper / enumerated

statement	Lean names	status
Proposition 7	witness lemmas upEdge_witness, mem_upEdges_of_witness, mem_upEdges_of_disjoint_anc, upEdges_subset_confluenceNL, card_upEdges_le_confluenceNL	paper proof (App. O.1); lemmas checked
Proposition 43	–	paper only
Proposition 34 (a)–(d)	relObs_subset_of_edge, score_obsEdge, sTilde_eq, sTilde_ge_half, upObs_subset_upEdges, sTilde_gt_half_iff, exists_upEdge_of_sTilde_gt_half, sTilde_eq_half_of_no_up, mem_upObs_of_witness, upObs_univ	checked
Proposition 53 (oracle sort accuracy; optimality)	accRule_sortRule, accRule_sortRule_rho, accRule_le_sOf, accRule_le_sRho, edgeAcc_sortRule_strict, rho_not_gt_of_edge	checked
Proposition 14	–	paper only
Theorem 15(a)	fejer_identity (counting identity)	partial
Theorem 15(b)	longrun_factor, longrun_inv, selfloop_sum_lt_one	partial
Theorem 15(c,d)	–	paper only
Lemma 23	phi_le, chernoff_logmgf_le, chernoff_choice, sqrt_two_v_x	algebra only
Lemma 24	quad_M, trace_MC, lemma12_trace, lemma12_fejer, avg_toeplitz_le_tsum, lemma12_le_B, lemma12_diag, lemma12_diag_le_D	algebra only
Lemma 25	centred_cov_eq, centred_dev_offdiag, centred_dev_diag, lemma13_eps, lemma13_count	deterministic part
Lemma 26	corr_perturb	checked
Theorem 27	decide_eq_target, decide_eq_support, thm15_det, explicit_y, sOf_cntDecl_congr	deterministic part
Proposition 16	–	asymptotic (cited CLT)
Proposition 17	prop21_corr, prop21_corr_std, sOf_saturated, sOf_const, abs_sOf_sub_half_le, thm22_const, ncrit_iff, ncrit_eq_iff	deterministic part
Proposition 18	pcorr_eq_pcov, pcorr_eq_corr_resid, prop31, prop31_std, cor32, prop33, prop33_abs, absPi_lower, absPi_upper, absPi_le_kappa, absPi_lt_kappa, absPi_eq_zero_iff, absPi_strictAntiOn, absPi_equal, rcrit_iff, rcrit_eq_iff, rcrit_example, pc_lipschitz, thm35_bracket, prop36_pcov, prop36_zero_iff, prop36_rank_one, prop36_ar1_pcov, prop36_ar1_pvar, prop36_ar1, prop36_ar1_ne_zero, prop37_pcov, prop37_bound, prop38_ratio, prop38_loading	population part checked

Modelling choices. Summary graphs are relations on a finite type with a self-loop hypothesis; anc and desc are reflexive-transitive closures ("a path exists iff a walk exists" is the one convention adopted without proof); G_T with $T = n+1$ slices lives on $V \times \mathrm{Fin}(n+1)$; source SCCs are represented by their member nodes (equality of SCC sets as equality of node unions is elementary and not formalised). In the finite-sample files, zero-mean variables are vectors of an arbitrary real inner-product space and lag-0 covariance is the inner product; independence enters only as vanishing inner products.

Findings. No statement was false. Formalisation added four missing nondegeneracy hypotheses to our earlier finite-sample statements: (i) the strict claims of Proposition 18 need $b_u b_v \neq 0$; (ii) $\ell_u = \rho_{uZ}^2 / r$ needs $r > 0$; (iii) nonvanishing of the multi-filter residual needs $b_u b_v \neq 0$; (iv) the partial correlation needs positive partial variances ($\mathrm{Var} Y_u > 0$ suffices). It also showed that one hypothesis of Proposition 4 and one of Lemma 26 are redundant, and that the in-forest statement needs acyclicity, not only out-degree at most one.

Appendix H Real-Data Details

The account of the real-data studies, with Tables 4 and 5 and Figure 4, is in Section 8. This section gives data selection, preprocessing and secondary analyses.

CausalRivers (Stein et al., 2025). Study 1 (P1–P3): the benchmark's connected-subgraph sampler, seeds 7000–7029, East Germany and Bavaria, $d \in \{5, 10\}$, 120 subgraphs, 791 induced edges; block means at 12 h and 24 h; nominal Fisher-z. Study 2 (P4–P6): seeds 8000–8029, 120 subgraphs, 1,376 non-relative pairs; proxy Z is the mean of standardised stations of the region outside the subgraph and outside 2 undirected river hops (429–615 stations); primary test on partial correlations with a residual Bartlett n_{eff} ($K = 100$ steps at 12 h, 50 at 24 h); median n_{eff} 139/113 (raw) and 231/211 (residual). Before conditioning, 0.98 (nominal) and 0.87 (effective) of non-relative pairs tested dependent. Median proxy contamination by river relatives was 2.6% (maximum 72%), a case of the contaminated-proxy term in Proposition 18. Post hoc and without verdict: conditioning on $k = 1, 3, 5, 10$ principal components of the leave-out stations gave dependent fractions $0.745, 0.605, 0.576, 0.491$ (12 h), with accuracy 0.580 and Spearman 0.16 at $k = 10$.

Causal Chambers (Gamella et al., 2025). Candidate selection (recorded before analysis) compared Causal Chambers, CausalRivers with measured precipitation (rejected: the driver reaches each gauge through its own catchment filter, so by Proposition 18 a scalar proxy cannot remove it), a semi-synthetic cross-sectional manufacturing benchmark (rejected: not real, not time series) and fMRI with physiological regressors (rejected: no ground truth). Wind tunnel: runs `actuators_random_walk_2`–`16` (run 1 excluded before analysis: truncated); 16 random variables, 26 edges, 63 non-relative pairs per run; block means over 10 rows, first differences, standardisation, $n = 999$; effective-n Fisher-z with Bartlett $K = 20$. Light tunnel: `actuators_white`, 20 variables, 39 edges, ten segments of 2,000 steps, levels. Predicted values come from the package's published ground-truth graph with constant parameters removed. Secondary analyses (no verdicts): nominal n gives slightly more false positives (0.048), as Proposition 16 predicts; levels instead of increments gave 0.054, contradicting our own

secondary prediction that levels of random walks would produce many spurious dependences; without block means ($D = 1$) relative recovery drops to 0.18 and Spearman to 0.10. In the light tunnel, polariser angles act on two sensors through Malus' law, a nonlinearity with near-zero linear correlation; as stated in the plan, these relative pairs were never detected, without affecting orientation.

Appendix I The Real-Data Estimand $\tilde{s}$

The theorems of Section 3 concern s_ρ of an unrolled DAG. The real-data studies estimate something else: lag-0 dependences between the observed series of one long record. This section defines the population quantity they target and proves what it does and does not say.

Definition 33 (estimand) Let $G_{\mathrm{S}} = (V, E_{\mathrm{S}})$ be a summary graph (no loop assumption) and $O \subseteq V$ the observed nodes. For $v \in O$ let $\tilde{\rho}_O(v) = |\mathrm{rel}_S(v) \cap O|$ (relatives computed in the full graph G_{S}, so unobserved nodes act as latent common ancestors). Let $E_O = \{(i,j) \in E_{\mathrm{S}} : i, j \in O,\ i \neq j\}$ be the induced non-self-loop edges and, for $E_O \neq \emptyset$, $\tilde{s}(G_{\mathrm{S}}, O) = \frac{1}{|E_O|} \sum_{(i,j)\in E_O} s_\rho\big(\tilde{\rho}_O(i), \tilde{\rho}_O(j)\big)$, with the edge score of Section 2. Edges of the latent projection that are not edges of G_{S} (paths through unobserved nodes) are not scored.

Why this is the target. Under the SVAR model of Appendix D, Proposition 14 gives, for almost every parameter, $\Gamma(0)_{uv} \neq 0$ iff $\mathrm{anc}_S(u) \cap \mathrm{anc}_S(v) \neq \emptyset$ iff $v \in \mathrm{rel}_S(u)$, for every observation scheme. So a perfect lag-0 dependence oracle on the observed series declares exactly $\mathrm{rel}_S(v) \cap O$, and the relatives sort of the observed counts, averaged over E_O, is $\tilde{s}$. (Lag-0 faithfulness is the generic case of Proposition 14; it can fail for specific parameters, e.g. through Malus-law nonlinearities.) Relation to Theorem 5: with $O = V$, every window node of the stationary unrolled DAG has $\rho_W(v(t)) = T|\mathrm{rel}_S(v)|$, so $\tilde{s}(G_{\mathrm{S}}, V)$ is the stationary value computed over non-self-loop edges only.

Proposition 34 (sign relation) *Let $E_O \neq \emptyset$ and $E_O^{\uparrow} = \{(i,j) \in E_O : \mathrm{rel}_S(i) \cap O \neq \mathrm{rel}_S(j) \cap O\}$. (a) Every edge of E_O scores $1/2$ or 1, and $\tilde{s} = \frac{1}{2} + |E_O^{\uparrow}|/(2|E_O|) \geq \frac{1}{2}$. (b) $E_O^{\uparrow} \subseteq E_{\uparrow}$. Hence $\tilde{s} > \frac{1}{2}$ iff $E_O^{\uparrow} \neq \emptyset$, and then some induced edge is an up-edge of G_{S} ($\mathrm{Src}(i) \neq \mathrm{Src}(j)$); if no induced edge is an up-edge, $\tilde{s} = \frac{1}{2}$. (c) $(i,j) \in E_O$ belongs to $E_O^{\uparrow}$ iff some observed w lies in $\mathrm{rel}_S(j) \setminus \mathrm{rel}_S(i)$, i.e. shares an ancestor with j but none with i; in particular an observed other parent k of j with $\mathrm{anc}_S(k) \cap \mathrm{anc}_S(i) = \emptyset$ suffices. (d) With $O = V$, $E_V^{\uparrow} = E_{\uparrow} \setminus \{\text{self-loops}\}$, so $\tilde{s}(G_{\mathrm{S}}, V) - \frac{1}{2}$ and $s_\rho^W - \frac{1}{2}$ have the same sign. (e) The converse of (b) fails: for the collider $a \to c \leftarrow b$ with loops and $O = \{a, c\}$, the only induced edge (a, c) is an up-edge, but $\mathrm{rel}_S(a) \cap O = \mathrm{rel}_S(c) \cap O = \{a, c\}$, so $\tilde{s} = 1/2$.*

Lean: (a)–(d): relObs_subset_of_edge, score_obsEdge, sTilde_eq, sTilde_ge_half, upObs_subset_upEdges, sTilde_gt_half_iff, exists_upEdge_of_sTilde_gt_half, sTilde_eq_half_of_no_up, mem_upObs_of_witness, upObs_univ (file Observed.lean); (e) paper proof only (not machine-checked; written by the AI agents and not yet read by a human).

Proof (a) $i \in \mathrm{anc}_S(j)$ gives $\mathrm{rel}_S(i) \subseteq \mathrm{rel}_S(j)$ (Reisach et al., 2026, Thm. 1), hence the same after intersecting with O; nested sets have equal or strictly ordered sizes. (b) If the observed parts differ, the full sets differ, and by Theorem 5 that is $\mathrm{Src}(i) \neq \mathrm{Src}(j)$. (c) By (a) the observed sets are nested, so they differ iff the larger one has an observed element outside the smaller; $w \in \mathrm{rel}_S(j) \setminus \mathrm{rel}_S(i)$ means $\mathrm{anc}_S(w) \cap \mathrm{anc}_S(j) \neq \emptyset = \mathrm{anc}_S(w) \cap \mathrm{anc}_S(i)$; k itself is such a w. (d) is (b) with $O = V$ and $\mathrm{Src}(i) \neq \mathrm{Src}(j)$ iff $\mathrm{rel}_S(i) \neq \mathrm{rel}_S(j)$ (Theorem 5); $s_\rho^W > \frac{1}{2}$ iff $E_\uparrow \neq \emptyset$, and self-loops are never in $E_\uparrow$. (e) $\mathrm{rel}_S(a) = \{a, c\}$, $\mathrm{rel}_S(c) = \{a, b, c\}$; b is unobserved. □

What is and is not implied. For river subgraphs (in-forests) an induced edge (i, j) counts iff an observed station lies upstream of j on a branch other than i's (in an in-forest $\mathrm{rel}_S(v) = \mathrm{anc}_S(v) \cup \mathrm{desc}_S(v)$ and $\mathrm{desc}_S(j) \subseteq \mathrm{desc}_S(i)$), so a confluence in the induced subgraph whose other branch is observed is sufficient. $\tilde{s}$ is a statement about population lag-0 relatives of a stationary process; it is not s_ρ of the finite unrolled DAG, of the induced subgraph's unrolled DAG, or of the latent projection, and it says nothing about lagged dependence. A deviation of the estimate from $\tilde{s}$ is a failure of estimation (F1–F3) or of lag-0 faithfulness, not evidence against Theorems 1 and 2.

Appendix J Time-Stamped Analysis Plans: Process, Prior Exposure, Timeline, Deviations

Process. The four real-data analysis plans, all analyses and the results were produced by AI coding agents (Section 8, AI-use statement) within about six hours of one day; the later plans for B1, B1-NL, B3 and the learner extension are listed at the end of this section. Each plan was committed to version control before its study's data were loaded for any dependence statistic in this project. Commit timestamps are set by the committing client; no third-party registry (e.g. OSF) was used. The hosting service's push records are given below for the later plans and can be supplied for the others. The gap between plan and results was short: about 3 min 41 s for plan 3 and 3 min 36 s for plan 4 (each including a run of about two minutes). We therefore call these *internally time-stamped analysis plans*. The commits are in the history of the supplementary repository.

Prior exposure to CausalRivers (disclosure). One day before plan 1, a separate exploratory project in the same repository (a test of contemporaneous-edge discovery on aggregated CausalRivers; commits dated 2026-11-03) had analysed CausalRivers with the same regions (East Germany, Bavaria), at resolutions 6 h, 12 h and 24 h (the studies here use 12 h and 24 h), and with the same data loaders and subgraph sampler that the plan-1 code reuses (with new seeds). It ran correlation, VAR, PCMCI+, VAR-LiNGAM and DYNOTEARS baselines and recorded a hidden-driver diagnostic: at 24 h the VAR(1) residual correlation among sibling stations was 0.51, the same as among unrelated non-adjacent stations in the same subgraph (0.51), and below adjacent pairs (0.65); it concluded that shared regional forcing is large and contemporaneous dependence is not specific to the river edges. This was known to the agents (and recorded in the repository) before plan 1 was written. Plan 1 nevertheless predicted that the relatives sort would orient river edges above chance; it failed.

Because the strong shared dependence had been observed beforehand, the F2 saturation account of that failure (and the F3 account of study 2) is *explanatory*, not a prediction made blind to the data. The Causal Chambers data and the E1/E2 held-out sets had not been analysed before their plans.

Times below are commit timestamps relative to the first plan commit (T_0).

time	event
$T_0 + 0{:}00$	plan 1 committed: targets T1–T3, predictions P1–P3, kill rule
$T_0 + 0{:}14$	theory of Sections 3 and 5, enumeration checks, operationalisation of P1–P3 (before data)
$T_0 + 0{:}17$	study 1 results (P2, P3 fail; kill rule triggered)
$T_0 + 1{:}05$	plan 2 committed: F1–F3 targets, Lean target, predictions P4–P6, kill rule
$T_0 + 1{:}18$–$1{:}31$	Lean formalisation of Section 3 and algebraic parts of Section 5
$T_0 + 1{:}38$	finite-sample theory, simulations, operationalisation of P4–P6 (before data)
$T_0 + 1{:}44$	study 2 results (P4, P6 fail; kill rule triggered)
$T_0 + 4{:}58$	plan 3 committed: candidate scoping, graph-only predictions, P7–P9 (before any dependence statistic)
$T_0 + 5{:}02$	study 3 results (P7, P8 pass; P9 fails; replication passes)
$T_0 + 5{:}14$–$5{:}24$	Lean formalisation of the finite-sample lemmas; errata
$T_0 + 5{:}45$	manuscript draft committed (with placeholders for E1/E2)
$T_0 + 5{:}50$	plan 4 committed: E1 (test of the theorem against eight naive predictors), E2 (lagged relatives), theory addendum A1–A3 and B1–B3, graph-only predictions, design check; development runs on the study-3 data only; no dependence statistic of any held-out dataset computed
$T_0 + 5{:}54$	E1/E2 results committed as planned (E1 passes 8/8 on the primary graph; Q7, Q8, Q10 pass, Q9 fails); out-of-sample F checks; run logs

Deviations (all recorded at the time). (1) Plan 1 said "theorem-predicted s_ρ" without naming the version; the stationary version was fixed in writing before data were loaded (the finite version would predict $> 1/2$ for every subgraph, including chains). (2) Self-loops are excluded from the predicted average because the estimator cannot orient them; this does not change which side of $1/2$ the prediction is on. (3) Only 6 of 120 study-1 subgraphs lacked a confluence; the plan set no minimum size for that group, and its trivial pass is not interpreted. (4) Plan 2 did not say which test defines "dependent" in P4; the effective-n test was fixed as primary before data, and the nominal test fails P4 as well. (5) In study 3, a planned triad-reliability check was not applicable (the measured driver reaches only four mutually related barometers) and is not interpreted. (6) The P9 failure is close to the threshold and its CI contains 0.3; as fixed in the plan, the point estimate decides. (7) Plan 4 (E1/E2): no threshold, unit, seed, K, L, D, graph or node set changed after the results were seen. A stray cache file committed with plan 4 was removed in the results commit (no content change). The F-check tolerance of ± 0.03 is the one fixed in plan 4; the citris "predicted accuracy" is undefined (no full-set edge of G_{lin} is predicted 1). The constant-1/2

comparison and the weak-component-size comparison are post hoc. The choice of G_{lin} as the primary graph was fixed in plan 4, together with the secondary analysis on the published graph G_D, which is the one on which the theorem fails against the sensor indicator on strictly ordered edges.

Later plans (benchmark experiments and B3). Times in the table are commit times (UTC, set by the committing client). For three plans we also retrieved the hosting service's own push records (its activity log, UTC, 09 Nov): B1-NL plan 04:10:47; B3 plan 04:21:12 and B3 results 04:24:37; learner-extension plan 08:20:31 and results 12:44:27. Each is 2 to 28 seconds after the corresponding commit time. We did not retrieve push records for the B1 plan or the four earlier plans. Push records order pushes, not computations: a plan pushed three minutes before its results, with a run of about two minutes in between, is consistent with the stated process but cannot exclude that something was computed before the plan was written. Only registration with a third party before data access could, and no plan was posted to an external registry.

time (2026)	event
08 Nov 22:31	B1 plan committed (generators, methods, hypotheses H1–H3, decision rules)
08 Nov 22:57	B1 code; deviations D1 (burn-in 50→100 after the validation mixing check) and D2 recorded before the test run
09 Nov 01:26	B1 results (test run redone once after a container restart, D3)
09 Nov 04:10	B1-NL plan committed
09 Nov 04:20	B3 plan committed and pushed before any held-out byte was read
09 Nov 04:24	B3 results
09 Nov 04:34	B1-NL analysis code; validation notes N1, N2
09 Nov 06:49	B1-NL results
09 Nov 08:20	learner-extension plan (Appendix S) committed and pushed before any score
09 Nov 08:34	post hoc re-analysis of B1/B1-NL and detectability oracle (no new data)
09 Nov 12:44	learner-extension results (one run, 08:28–12:41)

Appendix K Experiment E1: Discriminating Test Against Naive Predictors

Status and provenance. Specified in plan 4 (Appendix J); the plan was committed before any dependence statistic of any held-out dataset was computed. Done before it: file hashes, column and row counts, regeneration of the `citris_1` input protocol from its published generator (it matches the recorded inputs exactly), graph-only predictions, and development runs on the study-3 data only. All numbers below come from the committed result files of the experiment folder (`results4.json`, `summary4.txt`); none is estimated or rounded beyond the printed digits.

Why `citris_1`. Every published Causal Chambers graph is bipartite except the pressure-control configuration, whose only experiment varies no independent input,

so we rejected it. The light-tunnel `citris_1` series (100,000 steps about 1.8 s apart) is generated by a time-series structural model among the actuators: red is a reflected random walk, green moves halfway towards red and blue a quarter of the way towards green (each plus a uniform step), pol_1 is a random walk and pol_2 moves towards or away from pol_1 according to sign(blue − red). With the sensors this is a chain of length 3, red → green → blue → sensors. The secondary data are snapshot (i.i.d.) structural models among the actuators (`buchholz_1_obs`; `scm_{1,2,4,5}_reference`, including a chain and a collider) and the wind-tunnel regime-jump runs (bipartite; disagreements arise only from latent-node ties).

Linear-zero lemma (the refinement). Write $\mathcal{C}$ for the colour block (red, green, blue, ir_1, vis_1, ir_2, vis_2, ir_3, vis_3, current) and $\mathcal{Q}$ for the polariser block (pol_1, pol_2, angle_1, angle_2). *B1 (proved for the actuators).* Conditionally on the flag and the colour paths, the law of the polariser paths is invariant under negation, because every update (uniform draw on a symmetric interval, move towards or away with a sign fixed by the colours, reflection at ±90, rounding to 0.1) commutes with negation. *B2 (assumed).* The first-two-layer sensors depend only on the colours and independent noise; ir_3 and vis_3 depend on the polarisers only through an even function of $\theta_1 - \theta_2$ (Malus' law with aligned zero marks); each angle sensor is affine in its polariser plus independent noise. This is a physical assumption that the design documents do not let us check. *B3.* Under B1 and B2 every variable of $\mathcal{C}$ has zero cross-covariance, at every pair of times, with every variable of $\mathcal{Q}$ (a product of an even and an odd function of the polariser path has conditional mean zero). The six real edges red → pol_2, blue → pol_2 and pol_k → ir_3, vis_3 therefore carry no linear dependence. The relatives that a correlation-based estimator targets are those of the *linear-dependence graph* G_{lin} (the design graph G_D without these six edges), which has two weak components, each with a single source (red; pol_1). By Corollary 6, every edge inside a block has predicted value 1/2 for every node set; the six cross-block edges are compared by block size: red → pol_2 and blue → pol_2 are predicted 0 (anti-sorted) and pol_k → ir_3, vis_3 are predicted 1 for the full set. Faithfulness inside the blocks is assumed, as in Proposition 14. A zero-mark offset or a non-affine angle sensor would break B2 and move the prediction towards G_D.

Metric and tests. For each edge instance (the full node set plus 20 subgraphs per unit) the observed value is $o \in \{0, \frac{1}{2}, 1\}$ from the estimated counts, and the loss is $|o-p|$ for the predicted value p. For each competitor take the instances where it disagrees with the theorem; Δ = loss(theory) − loss(competitor), pooled, with a unit-level exact sign-flip test (1,024 flips for 10 units) and Holm correction over eight competitors, and a unit-bootstrap CI. E1a (prerequisites): non-relatives declared < 0.5 and relatives detected ≥ 0.8. E1b: $\Delta \leq -0.10$ and Holm $p \leq 0.05$ for all eight. E1c: the same restricted to instances whose own pair is declared dependent ($\Delta < 0$). Separation was to be claimed iff E1a, E1b and E1c all pass on `citris_1`. Kill rules: K1 (E1a fails: no claim), K2 (a competitor is significantly better: no claim against it), K3 (E1b passes and E1c fails: attributed to power-induced ties). None was triggered on the primary analysis.

Observed values given the theorem's prediction: predicted 1/2 (1,365 instances) observed exactly 1/2 in 99.2%; predicted 1 (187 instances, polariser → ir_3, vis_3) observed 1 in 99.5%; predicted 0 (110 instances, red and blue → pol_2) observed 0 in

Table K1 E1 primary, `citris_1`, graph G_{lin}, lag 0, 1,662 instances. Δ [95% CI]; all rows: 10/10 units, Holm $p = 0.0078$. "Exact": share of instances whose predicted value equals the observed one. "Spearman": agreement of the mean predicted edge value with observed accuracy over 200 subgraphs.

predictor	Δ [CI]	Δ, detected edges	exact	Spearman
theorem	—	—	0.992	0.962
sensor indicator	-0.502 $[-0.507, -0.496]$	-0.500	0.203	0.408
ancestors, full graph	-0.515 $[-0.519, -0.510]$	-0.500	0.147	0.334
ancestors, subgraph	-0.509 $[-0.513, -0.505]$	-0.500	0.120	0.121
in-degree, full graph	-0.515 $[-0.520, -0.511]$	-0.500	0.180	0.349
in-degree, subgraph	-0.510 $[-0.514, -0.505]$	-0.500	0.143	0.105
depth, full graph	-0.515 $[-0.519, -0.510]$	-0.500	0.147	0.334
depth, subgraph	-0.510 $[-0.514, -0.505]$	-0.500	0.114	0.113
every edge 1	-0.533 $[-0.536, -0.530]$	-0.500	0.115	undefined

99.1%. The detected-edge subset has 1,338 instances, all observed exactly 1/2. Prerequisites: non-relatives declared 0.000 [0, 0] (40 pairs per unit), relatives detected 0.969 [0.945, 0.992] (51 pairs per unit).

Strict subset and the power safeguard. On the 297 instances where the theorem predicts 0 or 1 it beats all eight ($\Delta = -0.539$ for the sensor indicator, -0.948 for ancestors and depth on the full graph, -0.991 for "every edge 1"; Holm $p = 0.0078$, 10/10 units). There, an estimator without power would favour the competitors (the "all counts tied" baseline is +0.443 to 0), so this part cannot come from power-induced ties.

Published graph G_D (secondary). Applying the theorem to the unmodified design graph, relatives detection is 0.659 [0.643, 0.675] (75 relative pairs per unit) and non-relatives declared 0.000. Overall the theorem still beats all eight ($\Delta = -0.323$ for the sensor indicator; -0.424 for ancestors, depth and "every edge 1"; -0.420 and -0.417 for in-degree; 10/10 units, Holm $p = 0.0078$), and its exact-match rate is 0.714. But on the 442 edges where G_D predicts strict order (1), the theorem matches 0.222 and the sensor indicator 0.319; the sensor indicator is significantly better ($\Delta = +0.453$ [0.410, 0.492], theorem better in 0/10 units, reverse Holm $p = 0.002$). A constant-1/2 rule (post hoc) matches 0.815 of all instances, against 0.714 for the theorem on G_D and 0.992 on G_{lin}; mean $|o - 1/2| = 0.092$ against 0.166 for the theorem on G_D. The Spearman statistic is 0.341 (theorem on G_D) against 0.408 (sensor indicator). This is why the claim is restricted to the linearly dependent relatives.

Lagged estimator on `citris_1` (secondary). With $L = 5$ and G_{lin}: detection 0.976 [0.953, 0.992], non-relatives declared 0.000, 8/8 passed ($\Delta = -0.505$ to -0.535), theorem exact match 0.998, Spearman 0.996.

On the snapshot sets, relatives detected: 0.933 (`buchholz_1_obs`), 1.000 (`scm_1`), 0.965 (`scm_2`), 1.000 (`scm_4`, `scm_5`); non-relatives declared 0.000 wherever non-relative pairs exist (`scm_4` has none). The chain (`scm_4`) gives 1/2 on all 2,267 instances (observed exactly 1/2 every time); the collider gives 1 on edges into the collider and its sensors and 1/2 elsewhere (all 2,238 instances exact). Pooled, the theorem's strict predictions number 2,768 (observed value 1 on 2,403, 1/2 on 349, 0 on 16: exact 0.868);

Table K2 E1 on i.i.d. structural-model sets (no verdict; secondary). Competitors' exact-match range over the eight predictors. Units: `scm_1` 5, the others 10 each (45 pooled).

set	instances	theorem exact	competitors exact	Δ range	result
`buchholz_1_obs`	1,712	0.791	0.328–0.375	-0.499 to -0.496	pass 8/8
`scm_1_reference`	867	1.000	0.314–0.392	-0.500	not met[†]
`scm_2_reference`	1,650	0.812	0.324–0.408	-0.443 to -0.426	pass 8/8
`scm_4_reference` (chain)	2,267	1.000	0.000–0.088	-0.500	pass 8/8
`scm_5_reference` (collider)	2,238	1.000	0.464–0.495	-0.500	pass 8/8
pooled (45 units)	8,734	0.923	0.283–0.322	-0.489 to -0.487	pass 8/8, Holm $p = 8 \times 10^{-5}$

[†]Five units: the smallest attainable Holm-adjusted p is $8/32 = 0.25$.

on them only the sensor indicator disagrees often (390 instances, $\Delta = -0.5$, 35/35 units) and the other competitors coincide with the theorem. The pooled Spearman is 0.913 for the theorem and -0.138 for the sensor indicator. Post hoc, a constant-1/2 rule matches 0.688 of the pooled instances against 0.923.

Wind-tunnel regimes (tertiary, 64 units, no verdict). The same test passes against all eight in each configuration. $D = 10$, $L = 0$: non-relatives declared 0.014, relatives detected 0.592, $\Delta = -0.203$ $[-0.227, -0.179]$ in 63/64 units, exact match 0.539 against 0.432; detected-edge subset $\Delta = -0.169$. $D = 1$, $L = 30$: 0.003, 0.398, $\Delta = -0.225$ $[-0.250, -0.197]$ in 62/64 units, exact 0.443 against 0.353; detected-edge subset $\Delta = -0.134$. $D = 1$, $L = 0$: $\Delta = -0.218$ (64/64 units), exact 0.411 against 0.326; $D = 10$, $L = 3$: $\Delta = -0.218$, exact 0.533 against 0.411. Detection of 0.33–0.59 means that part of the advantage is power-induced, as plan 4 stated in advance. Post hoc, the constant-1/2 rule has mean absolute error 0.311 against 0.363 for the theorem at $D = 1$, $L = 30$ (and 0.313 against 0.294 at $D = 10$, $L = 0$), and matches 0.378 (0.375) of the instances against 0.443 (0.539).

A trivial rule that E1 did not test (post hoc). Each weak component of G_{lin} has a single source (red; pol_1). By the proof of Corollary 6, every node of such a component is reached from that source, so its relatives are the whole component, and for every node set the predicted counts are the numbers of observed nodes in the endpoints' weak components. The theorem's prediction therefore coincides on all 1,662 instances with the rule "1/2 iff both endpoints lie in the same weak component of G_{lin}, otherwise compare the observed sizes of the two components". That rule was not among the eight competitors, so E1 cannot separate the theory from it; the only data in this appendix where the source structure matters beyond component membership are the snapshot sets with several sources per component, notably the collider `scm_5`.

What E1 does and does not show. It shows that, on one multi-layer light-tunnel series, the estimated relatives counts are tied along the whole chain red $\to$ green $\to$ blue $\to$ sensors and between the polariser block and the angle sensors, and strictly ordered (including against the causal direction) on red, blue $\to \text{pol}_2$ — and that the source-component rule of Theorem 5 and Corollary 6 predicts this while eight naive rules do not. It does not show that the theorem predicts relatives under the published graph; the faithfulness of the graph to linear dependence fails here by design. Most of the separation is in the ties, which naive rules cannot express, and on G_{lin} the prediction equals the weak-component-size rule. The evidence is one device in a favourable regime, and the wind tunnel, where detection is limited by F1, gives weaker support.

Appendix L Experiment E2: Lagged Relatives

Theory (plan 4, addendum A). For $L \in \{0, 1, \dots\}$ let the lag-L relatives of u be u and every v with $\gamma_{uv}(h) \neq 0$ for some $|h| \leq L$. *A1 (proved).* Under the independent-innovations assumption behind Proposition 14, $\text{rel}_0 \subseteq \text{rel}_L \subseteq \text{rel}_\infty \subseteq \text{rel}_S$; for a VAR model all are equal for almost every parameter value; and Theorem 5, Corollary 6 hold verbatim with "relatives" read as rel_L under lag-L faithfulness. Lag-0 faithfulness implies lag-L faithfulness; the converse fails (a white-noise source s with

an edge $s \to j$ and no self-loop at s has $\gamma_{sj}(0) = 0 \neq \gamma_{js}(1)$). *Estimator.* This is a Bonferroni max-over-lags variant of the classical cross-correlation tests of independence between two series (Haugh, 1976 cross-correlates prewhitened residuals and uses a portmanteau statistic; we use effective-n Fisher-z statistics without prewhitening). Declare u, v relatives iff $\max_{|h|\le L} |\operatorname{atanh} \hat{r}_{uv}(h)| \sqrt{\hat{n}_{\text{eff},uv}(h) - 3} > z_c(L)$, with $\hat{n}_{\text{eff},uv}(h) = (n - |h|)/\hat{B}_{uv}$ and $z_c(L) = \Phi^{-1}(1 - \alpha/(2m(2L+1)))$, a Bonferroni correction over the m pairs and the $2L+1$ lags; $L = 0$ is exactly the study-3 test. *A2 (proved for finite n).* For independent stationary Gaussian a, b the null variance of the lag-h cross-covariance has the same bound at every h, so one Bartlett factor serves all lags. *A3 (deterministic part proved, constants sketched).* The recovery bound of F1 holds with $\rho_{\min}$ replaced by $\rho^*_{\min,L} = \min_{\text{relative pairs}} \max_{|h|\le L} |\rho_{ab}(h)| \ge \rho_{\min}$ (the gain) and a factor $2L + 1$ in the failure probability and the Bonferroni count (the cost): for 105 pairs at $\alpha = 0.01$, z_c rises from 3.90 ($L = 0$) to 4.35 ($L = 3$) and 4.80 ($L = 30$). A lagged correlation must therefore exceed the lag-0 correlation by about 1.12 ($L = 3$) or 1.23 ($L = 30$) times to help.

Design. Held-out data: the wind-tunnel regime-jump runs (`regime_jumps_single`, `regime_jumps_multi`), $2 \times 32 = 64$ regimes of 10,000 steps, in the same archive as the study-3 data but never analysed in study 3. Load_in, load_out, pot_1, pot_2 follow independent mirrored random walks within a regime; the hatch and sensor parameters are constant within a regime. The first 100 rows of each regime are dropped. Graph: 20 edges without the hatch (19 predicted 1, $\text{pot}_1 \to \text{signal}_1$ predicted 1/2); 54 non-relative and 51 relative pairs; 1,280 subgraphs (10 of size 5 and 10 of size 8 per unit). Primary: $D = 1$ (first differences), $L = 30$ (about 9 s) against $L = 0$; secondary: block means with $D = 10$ and $L = 3$ against $L = 0$. Success required Q7–Q10 all to pass at the primary configuration. Kill rules: Q7 fails (saturation) or Q10 fails (lags do not help detection, and the lagged estimator is not recommended). Plan 4 states that E2 is not a re-analysis of P9 and that P9's verdict is final.

Detection of the 51 relative pairs: $D = 1$: 0.328 [0.312, 0.345] at lag 0 and 0.398 [0.383, 0.412] with $L = 30$; $D = 10$: 0.592 [0.584, 0.599] and 0.585 [0.576, 0.592] with $L = 3$ (lags reduce detection: the Bonferroni cost exceeds the gain). On `citris_1` with $L = 5$ against $L = 0$: 0.976 against 0.969, difference +0.008 [0, 0.024], so Q10 fails there; detection is already nearly saturated at lag 0. **Verdict.** Q9 fails at the primary configuration, so E2 does not succeed; no kill rule is triggered (Q7 and Q10 pass). Lags raise detection at fine resolution without adding false positives, as A1 and A3 predict. The study-3 estimator (lag 0) rerun on these held-out regimes fails the graded criterion again (0.202 at $D = 10$, 0.120 at $D = 1$): the P9-type failure repeats on held-out data. P9's original verdict (0.267, failed) stands; E2 is not presented as rescuing it. The development check of plan 4 showed that, on the study-3 data with the same estimator, drawing different subgraph seeds moves the P9 statistic from 0.267 to 0.338. We report this as variability of the statistic (about 0.07 from the subgraph draw alone); it is a development run, not a verdict, and it does not change the P9 outcome.

Lags recover the pressure edges at fine resolution (load_in $\to$ upwind pressure 0.562 $\to$ 0.984, load_out $\to$ upwind pressure 0.016 $\to$ 0.812) but lose the pot edges to the larger threshold (0.359 $\to$ 0.156 and 0.266 $\to$ 0.109); every microphone edge, $\text{pot}_2 \to \text{signal}_2$ and the cross-current edges stay undetected. Even at its best ($D = 10$,

Table L3 E2, 64 held-out regimes, 95% unit-bootstrap CIs. Q8: accuracy on the 1,216 instances predicted 1. Q10: relatives detected, lagged minus lag 0 (non-relatives declared difference in the last column).

configuration	Q7	Q8	Q9	Q10	non-rel. diff.
primary D=1, L=30	0.003 [0.001, 0.005] pass	0.553 [0.527, 0.579] pass	0.243 [0.193, 0.294] **fail**	+0.069 [0.060, 0.079] pass	+0.001 [−0.001, 0.002]
secondary D=10, L=3	0.011 [0.008, 0.014]	0.711 [0.692, 0.729]	0.202 [0.152, 0.253] fail	−0.007 [−0.012, −0.003] fail	−0.002 [−0.004, −0.0003]
lag 0, D=10 (replication of study 3)	0.014 [0.010, 0.018]	0.742 [0.722, 0.760]	0.202 [0.145, 0.259] fail	—	—
lag 0, D=1	0.002 [0.001, 0.004]	0.552 [0.532, 0.570]	0.120 [0.066, 0.176] fail	—	—

Table L4 Detection rate per edge over 64 regimes: share of regimes in which the pair was declared dependent.

edge	D=1, L=0	D=1, L=30	D=10, L=0
load_in → rpm_in	0.859	0.875	1.000
load_in → rpm_out	0.000	0.172	0.094
load_in → current_in	1.000	1.000	1.000
load_in → current_out	0.000	0.000	0.047
load_out → rpm_in	0.000	0.016	0.000
load_out → rpm_out	0.844	0.969	1.000
load_out → current_out	1.000	1.000	1.000
load_out → current_in	0.000	0.000	0.000
load_in → pressure_intake	0.000	0.094	0.953
load_out → pressure_intake	0.000	0.031	0.906
load_in → pressure_upwind	0.562	0.984	1.000
load_out → pressure_upwind	0.016	0.812	1.000
load_in → pressure_downwind	0.734	0.953	1.000
load_out → pressure_downwind	0.750	1.000	1.000
pot_1 → signal_1	0.359	0.156	0.969
pot_1 → signal_2	0.266	0.109	0.969
pot_2 → signal_2	0.000	0.000	0.266
pot_1, load_in, load_out → mic	0.000	0.000	0.016 (pot_1), 0.000

lag 0) wind-tunnel detection (0.592) is far below that of the light tunnel, so lags are not the main cause of the F1 shortfall.

All six checks pass. With lags the observed detection sits at or slightly below $\bar{p}_{\max}$, consistent with strongly correlated lag statistics; the independence formula overstates both detection and false positives, as the theory said it would. For `citris_1` the predicted accuracy is undefined because no full-set edge of G_{lin} is predicted 1 inside a block.

Appendix M Summary Graphs without Self-Loops

Here $G_{\text{S}} = (V, E_{\text{S}})$ is an arbitrary directed graph with $|E_{\text{S}}| \geq 1$; no loop assumption is made. B_n is the boolean n-th power of the adjacency matrix ($B_0 = I$), $\Lambda(u, v) = \{n : B_n[u, v] = 1\}$ the set of lengths of walks from u to v, and u is *fresh* if it has no incoming edge in E_{S} (a self-loop is an incoming edge). Status: Theorem 36 and Corollary 37 are machine-checked; the rest of this section is proved on paper and checked by enumeration, as marked.

Lemma 35 (ancestors and roots without loops) *$u(a) \in \text{anc}(v(t))$ iff $a \leq t$ and $t - a \in \Lambda(u, v)$. A node $u(a)$ is a root of G_T iff $a = 1$ or u is fresh.*

Lean: unrolled_reach_iff (already free of the loop hypothesis).

Proof Every edge of G_T raises time by one, so paths from $u(a)$ to $v(t)$ are the lifts of walks of length exactly $t - a$, and every walk lifts. Without loops one cannot pad. A node $u(a)$ with $a \geq 2$ has a parent iff some edge (w, u) exists. □

Table L5 Out-of-sample F-theory checks: the first half of each unit predicts the second half. Predicted detection is a bracket $[\bar{p}_{\max}, \bar{p}_{\mathrm{ind}}]$ when lags are used. Pass if the observed value lies within 0.03 of the bracket (criterion fixed in plan 4).

configuration	predicted detection	observed	non-rel. declared, pred. / obs.	accuracy, pred. / obs.
wind D=1, L=0	0.250	0.242	0.007 / 0.003	0.573 / 0.590
wind D=1, L=30	[0.344, 0.426]	0.332	[0.021, 0.074] / 0.003	0.504 / 0.506
wind D=10, L=0	0.548	0.533	0.014 / 0.008	0.649 / 0.632
wind D=10, L=3	[0.536, 0.554]	0.524	[0.016, 0.027] / 0.008	0.628 / 0.618
`citris_1`, L=0	0.935	0.945	0.007 / 0.000	undefined
`citris_1`, L=5	[0.940, 0.988]	0.945	[0.010, 0.045] / 0.000	undefined

Theorem 36 (edge value without loop assumption) *Let* $(i,j) \in E_S$ *and* $e_t = (i(t), j(t{+}1))$, $1 \le t < T$. *Then* $s_\rho(e_t) \in \{\frac{1}{2}, 1\}$ *independently of* T, *and* $s_\rho(e_t) = 1$ *iff* $\mathrm{Strict}_{ij}(t)$ *holds, where* $\mathrm{Strict}_{ij}(t)$ *is*

$$\exists u:\ t \in \Lambda(u,j),\ t{-}1 \notin \Lambda(u,i) \quad \text{or} \quad \exists \text{ fresh } u,\ \exists n \in \{1,\dots,t\}:\ n \in \Lambda(u,j),\ n{-}1 \notin \Lambda(u,i).$$

Lean: isRel_unrolled_iff_NL, reduce_ancestor, relSet_unrolled_eq_iff_NL, edge_score_NL.

Proof As in Theorem 1, $\mathrm{rel}(i(t)) \subseteq \mathrm{rel}(j(t{+}1))$, and the score is 1 iff $\mathrm{roots}(j(t{+}1)) \setminus \mathrm{roots}(i(t)) \neq \emptyset$ (Reisach et al., 2026, Thm. 1, Lemma 1). By Lemma 35 an element of the difference is $u(a)$ with ($a = 1$ or u fresh), $n = t + 1 - a \in \Lambda(u,j)$ and $u(a) \notin \mathrm{roots}(i(t))$. The case $a = t+1$ would make j fresh, impossible since j has the parent i; so $a \le t$ and, by the lemma again, $u(a)$ is not a root of $i(t)$'s ancestry iff $n - 1 = t - a \notin \Lambda(u,i)$. The branch $a = 1$ is $n = t$ with no condition on u; the other is a fresh u with $1 \le n \le t$. (The Lean proof avoids Lemma 1: the extra relative is $u(0)$ in the first case and a node of a fresh u in the second, and reduce_ancestor pushes any common ancestor back along parents to slice 0 or to a fresh node.) □

Corollary 37 *For every summary graph and* $T \ge 2$, $s_\rho(G_T) = \frac{1}{2} + \frac{1}{2(T-1)|E_S|} \sum_{(i,j)\in E_S} N_T(i,j)$ *with* $N_T(i,j) = \#\{t < T : \mathrm{Strict}_{ij}(t)\}$; *in particular* $s_\rho(G_T) \ge 1/2$. *With a loop at every node no node is fresh,* $\Lambda(u,j) = [d(u,j), \infty)$, *and* $\mathrm{Strict}_{ij}(t)$ *becomes* $\exists u : d(u,j) \le t \le d(u,i)$, *i.e. Theorem 1.*

Lean: sRho_unrolled_eq_NL, sRho_unrolled_ge_half_NL.

Exhaustive check of Theorem 36 against brute-force relatives: $p = 2, 3$ ($T = 2, \dots, 9$), all 65,535 graphs with $p = 4$ ($T = 3, 5, 8$), and random graphs with $p = 5, 6$; 0 mismatches in 120, 3,577, 196,605, 8,928 and 4,494 (graph, T) cases (enumerated, not part of the Lean proof).

Proposition 38 (limit exists and is rational; paper proof, enumerated) (B_n) *is eventually periodic: there are* $n_0, \pi \ge 1$ *with* $B_{n+\pi} = B_n$ *for* $n \ge n_0$. *Hence* $L = \lim_T s_\rho(G_T) = \frac{1}{2} + \frac{1}{2|E_S|} \sum_e \delta_e$ *with* $\delta_e = 1$ *if some fresh* u *has* $n \ge 1$, $n \in \Lambda(u,j)$, $n - 1 \notin \Lambda(u,i)$, *and otherwise* δ_e *the density, over one period of* $t > n_0$, *of* t *with some* u, $B_t[u,j] = 1$, $B_{t-1}[u,i] = 0$; *and* $|s_\rho(G_T) - L| \le (n_0 + \pi + 1)/(2(T-1))$.

Proof B_{n+1} is a function of B_n and there are finitely many boolean matrices. A fresh witness is monotone in t and has its first n at most $n_0 + 1 + \pi$ (the set $\Lambda(u,j) \setminus (\Lambda(u,i) + 1)$ is eventually periodic), so such an edge is strict for all $t \ge n_0 + 1 + \pi$. Without a witness, $\mathrm{Strict}_{ij}(t)$ depends for $t > n_0$ only on $t \bmod \pi$. Counting strict $t < T$ gives the error bound; average over edges. □

Checked on 1,115 graphs (all $15 + 511$ graphs with $p = 2, 3$, 390 random with $p = 4$, 199 random with $p = 5, 6$): the exact value at $T = 600$ is within the bound and $(L - \frac{1}{2})\, 2|E_S|\, \pi$ is an integer in every case.

Proposition 39 (edges inside an SCC; paper proof, enumerated) *Let $(i, j) \in E_S$ with j reaching i and no fresh witness. Then* $\mathrm{Strict}_{ij}(t)$ *fails for every $t \geq n_0 + 1$.*

Proof Fix u; put $f(n) = B_n[u, j]$, $h(n) = B_{n-1}[u, i]$. The edge gives $h(n) \leq f(n)$. If m is the length of a walk $j \to i$ and $c = m + 1$, then $f(n) \leq h(n + c)$. Take $k \geq 1$ with kc a multiple of π. For $n \geq n_0 + 1$, $h(n) \leq f(n) \leq h(n + c) \leq \cdots \leq h(n + kc) = h(n)$, so $f(n) = h(n)$ for every u. □

Proposition 40 (crossing edge into a cyclic node) *If (i, j) is crossing and j lies on a cycle with period g (gcd of the closed-walk lengths at j), then* $\mathrm{Strict}_{ij}(t)$ *holds for every $t \in \Lambda(j, j)$, which contains all large multiples of g, so $\delta_e \geq 1/g$, and $\delta_e = 1$ if $g = 1$. A fresh parent $u \neq i$ of j makes every layer strict.*
Lean: *strictNL_of_closed_walk, strictNL_of_fresh_parent, sRho_unrolled_eq_one_NL.*

Proof Take $u = j$: $t \in \Lambda(j, j)$ and $t - 1 \notin \Lambda(j, i) = \emptyset$. Closed-walk lengths at j form a numerical semigroup in $g\mathbb{Z}$, which contains all large multiples of g. □

Corollary 41 (loop-free versions of the loop-dependent statements) *(a) If every weakly connected component of G_S is strongly connected (any loop pattern, any periods), then $L = 1/2$ and $s_\rho(G_T) - \frac{1}{2} \leq n_0/(2(T - 1))$ (paper proof from Proposition 39; Reisach et al.'s Theorem 4 without loops). (b) The converse and the closed form are false without loops (Table M6). (c) If every crossing edge has a head on a cycle, then $L > 1/2$ iff $E_\times \neq \emptyset$.*

Table M6 Loop-free examples (L: finite limit, S: stationary). Lean: edgeNoLoops_sRho, colliderNL_sRho, edgeLoopHead_sRho; the others are computed values (example scripts in the supplement).

summary graph	without loops	with loops
single edge $0 \to 1$	$s_\rho(G_T) = 1/2$ for all T	$L = 2/3$, $S = 1/2$
chain $0 \to 1 \to 2$	$L = 1/2$	$L = 0.7$
2-cycle $\{0, 1\}$ plus $0 \to 2$	$L = 1/2$	loop at 2 only: $L = 5/8$
collider	$L = S = 1$	$L = S = 0.7$
edge, loop at head only	$L = S = 1$	–

For an acyclic loop-free graph, $L(u, v)$ is finite, $B_n = 0$ for $n \geq p$, and $L = \frac{1}{2} + \#\{e : \text{fresh witness}\}/(2|E_S|)$ (paper proof); for a loop-free in-forest an edge (i, j) has a fresh witness iff j has another parent, so $L = \frac{1}{2} + (m - n_1)/(2m)$ with $m = |E_S|$ and n_1 the number of nodes with one parent, which equals the stationary value below. On 362 random acyclic loop-free graphs $L = S$ in all cases, and the in-forest formula holds on 298 in-forests.

Theorem 42 (stationary value without loops; (a), (c) proved, (b) sketched) *Let G_{S} be arbitrary, $G_{\mathbb{Z}}$ the infinite-past unrolled DAG on $V \times \mathbb{Z}$, and $\Delta(v,w) = \{m - n : \exists u,\ n \in \Lambda(u,v),\ m \in \Lambda(u,w)\}$. Then $\mathrm{rel}_{G_{\mathbb{Z}}}(v(t)) = \{w(t+d) : d \in \Delta(v,w)\}$ (proved), the edge $(i(t), j(t{+}1))$ is strict iff $\Delta(j,w){+}1 \neq \Delta(i,w)$ for some w (proved; always $\Delta(i,w) \subseteq \Delta(j,w){+}1$), and $S^{\mathbb{Z}} = \frac{1}{2} + |E^{\mathbb{Z}}_{\uparrow}|/(2|E_{\mathrm{S}}|)$. (a) Outside $|d| < n_0 + \pi$, $\Delta(v,w)$ is π-periodic, so $E^{\mathbb{Z}}_{\uparrow}$ is detected with witnesses of $|d| \leq n_0 + 2\pi$ (proved). (b) In a finite window after a burn-in $B \geq 2n_0 + 3\pi + 1$ the windowed value equals $S^{\mathbb{Z}} + O(K_0/T)$, $K_0 = n_0 + 2\pi$ (*sketched*). (c) $S^{\mathbb{Z}} = 1/2$ whenever every weak component is strongly connected, whatever its periods (a node colouring* lab *with $\Delta(v,w) = \mathrm{lab}(w) - \mathrm{lab}(v) + g\mathbb{Z}$) (proved). With loops everywhere Δ is $\mathbb{Z}$ or empty and this is Theorem 5.*

Brute-force relatives in G_{B+T} ($B = 40$, $T = 90$, edges at distance ≥ 30 from the window ends) agree with $E^{\mathbb{Z}}_{\uparrow}$ on every edge of all $15 + 511$ graphs with $p = 2, 3$ and 248 random graphs with $p = 4$ (enumerated). Not formalised in Lean: periodicity, Proposition 39, Corollary 41(a), Theorem 42.

Appendix N Lagged Unrollings

Theorems 1, 2 and 5 and their Lean proofs are for VAR(1)-shaped unrollings: every summary edge acts at lag one. Here every non-loop summary edge $e = (i,j)$ carries a lag $\tau_e \in \{1, \ldots, \tau_{\max}\}$ (a pair may carry several lags, each a separate edge), self-loops act at lag one, and G_T has the edges $(i(t), j(t{+}\tau_e))$, $1 \leq t \leq T - \tau_e$. Let $d_\tau(u,v)$ be the smallest total lag of a path from u to v (0 if $u = v$, ∞ if none); $d_\tau(u,v) \leq d(u,v)\,\tau_{\max} \leq \Delta\tau_{\max}$ when finite. Status: paper proof, not machine-checked and not enumerated. Contemporaneous (lag-0) edges are excluded: they break the time-monotonicity used below.

Proposition 43 (lag-weighted version) *(a) $u(s) \in \mathrm{anc}(v(t))$ iff $d_\tau(u,v) \leq t - s$. (b) The copy $(i(t), j(t{+}\tau_e))$ scores 1 iff some u has $d_\tau(u,j) - \tau_e + 1 \leq t \leq d_\tau(u,i)$, and 1/2 otherwise. (c) Crossing edges score 1 in every copy; an edge inside an SCC has at most $\Delta\tau_{\max}$ strict copies, so $\lim_{T\to\infty} s_\rho(G_T) = \frac{1}{2} + |E_\times|/(2|E_{\mathrm{S}}|)$ with $|s_\rho(G_T) - L| = O(\Delta\tau_{\max}/T)$. (d) With a latent burn-in $B \geq \Delta\tau_{\max}$ (or an infinite past), Theorem 5 holds verbatim.*

Proof (a) Every edge raises time by its lag, so a path from $u(s)$ to $v(t)$ projects to a walk of total lag $t - s$; conversely a path of total lag $d_\tau(u,v)$ can be padded by lag-one self-loops at u. (b) As in Lemma 20, the roots of $v(t)$ are $\{u(1) : d_\tau(u,v) \leq t - 1\}$ (every $u(s)$, $s \geq 2$, has the parent $u(s-1)$), and $i(t)$ is a parent of $j(t{+}\tau_e)$, so the score is 1 iff $j(t{+}\tau_e)$ has a root $u(1)$ with $d_\tau(u,j) \leq t + \tau_e - 1$ that $i(t)$ lacks, $d_\tau(u,i) > t - 1$. (c) For a crossing edge take $u = j$: $d_\tau(j,j) = 0 \leq t + \tau_e - 1$ and $d_\tau(j,i) = \infty$. Inside an SCC, $d_\tau(u,i) < \infty$ iff $d_\tau(u,j) < \infty$, so a strict copy needs $t \leq d_\tau(u,i) \leq \Delta\tau_{\max}$. Each edge has $T - \tau_e$ copies; average. (d) By (a), $w(s) \in \mathrm{rel}(v(t))$ iff some u has $d_\tau(u,v) \leq t - 1$ and $d_\tau(u,w) \leq s - 1$; for window nodes $t - 1, s - 1 \geq B \geq \Delta\tau_{\max}$ this is $\mathrm{anc}_S(v) \cap \mathrm{anc}_S(w) \neq \emptyset$, and the proof of Theorem 5 applies. □

Appendix O Random Summary Graphs: Derivations and Checks

All graphs here are a non-loop random graph plus a self-loop at every node, so $|E_{\mathrm{S}}| = p + m$ with m the number of non-loop edges, and for acyclic graphs $|E_{\times}| = m$.

In-forests (proved identities). Let the non-loop part be an in-forest with m edges and n_1 nodes with exactly one parent. By Corollaries 3 and 6, $L = \frac{1}{2} + \frac{m}{2(m+p)}$, $S = \frac{1}{2} + \frac{m-n_1}{2(m+p)}$, $L - S = \frac{n_1}{2(m+p)}$ (the last checked to machine precision on 18 random forests). For a sequence of in-forests with parent-count distribution π and mean μ: $L \to \frac{1}{2} + \frac{\mu}{2(1+\mu)}$ and $S \to \frac{1}{2} + \frac{\mu-\pi_1}{2(1+\mu)}$. Exact computation at $p = 20{,}000$ (6 seeds each): uniform recursive in-tree ($\pi_1 = 1/4$, $S \to 11/16 = 0.6875$, measured 0.68739 ± 0.00026); preferential attachment (measured $\pi_1 = 0.1656$, $S = 0.70859 \pm 0.00028$); uniform random labelled tree rooted and oriented to the root (Poisson(1) parents, $\pi_1 \to e^{-1}$, $S \to \frac{1}{2} + (1 - e^{-1})/4 = 0.65803$, measured 0.65783 ± 0.00020). In all three $L = 0.74999$.

Directed ER: summary. $G(p, c/p)$, $m/p \to c$. The finite limit needs the fraction of non-loop edges inside an SCC ($\to x^2$, the two endpoints lie in the giant SCC iff the ancestor tree of i and the descendant tree of j are infinite); the stationary value needs the fraction of up-edges ($\to F_{\uparrow}$). Both are proved in Appendix O.1 below. Large c: $1 - x \sim e^{-c}$, $L - \frac{1}{2} \sim e^{-c}$, $S - \frac{1}{2} \sim e^{-c}/2$.

Layered DAG (sketch; L proved given edge concentration). M layers of width w, each node of layer $l + 1$ receiving an edge from each node of layer l with probability c/w; as $w \to \infty$ with $\mu = c(M - 1)/M$, $L \to \frac{1}{2} + \frac{\mu}{2(1+\mu)}$ and $S \to \frac{1}{2} + \frac{\mu(1-e^{-c})}{2(1+\mu)}$. For fixed w and deep M the stationary value falls below this (observed 0.7824 against 0.7878 at $w = 50$, $M = 400$, $c = 2$); this correction is not derived.

Table O7 Exact computation (means ± s.d. over seeds; SCCs by scipy and source sets by bitsets) against the limits of Proposition 7, proved in Appendix O.1.

c	p	L exact	L formula	S exact	S formula
0.5	20000	0.6666 ± 0.0011	0.6667	0.5658 ± 0.0014	0.5656
1.0	20000	0.7486 ± 0.0008	0.7500	0.6557 ± 0.0013	0.6580
1.5	50000	0.6975 ± 0.0020	0.6981	0.6159 ± 0.0018	0.6163
2.0	50000	0.6214 ± 0.0022	0.6217	0.5629 ± 0.0014	0.5632
3.0	100000	0.5434 ± 0.0010	0.5433	0.5215 ± 0.0006	0.5214
5.0	20000	0.5059 ± 0.0005	0.5058	0.5028 ± 0.0003	0.5029

At the critical point $c = 1$, S converges slowly in p: 0.65574 ± 0.00126 ($p = 2 \cdot 10^4$, 6 seeds), 0.65784 ± 0.00043 ($2 \cdot 10^5$, 6 seeds) and 0.65796 (10^6, 1 seed) against 0.65803, like L (0.74857, 0.74972, 0.74989 against 0.75).

L at $c = 1$ converges as p grows: 0.7466 ($p = 2000$), 0.7486 ($2{\cdot}10^4$), 0.74976 ($2{\cdot}10^5$), 0.74992 (10^6); at $c = 2$, 0.62162 ± 0.00023 at $p = 10^6$ against 0.62170. The giant-SCC fraction matches x^2 (0.6353 against 0.6349 at $c = 2$). The second fixed point matters: with $d = 1$ the formula would give $S = 0.6312, 0.5766, 0.5246$ at $c = 1.5, 2, 3$, outside the error bars, whereas the fixed points $d = 0.485, 0.221, 0.061$ give 0.6163, 0.5632, 0.5214. Further exact rows: $c = 1.5$, $p = 10^5$, 32 graphs, $L = 0.69819 \pm 0.00024$ (formula 0.69810); $c = 2$, $p = 3.2 \cdot 10^5$, $L = 0.62170 \pm 0.00012$ (0.62170). The Barak–Erdős phase-transition paper cited by Reisach et al. (2026) concerns monotone-path reachability at the dense scale $\log n/n$ and is not used here.

O.1 Directed Erdős–Rényi summary graphs: full proof of Proposition 7

Status. Everything in this subsection is proved on paper. The only inputs used without proof are textbook facts (F1)–(F3) below, checked against the author's online version of van der Hofstad (2016) (theorem numbers are those of that version and may differ in print). Lemma 47 is the standard local-coupling argument for sparse random graphs and is written concisely. The deterministic witness lemmas are machine-checked; no probabilistic step is. No human probabilist has yet checked this subsection (see the AI-use statement).

Limits and their order. For a fixed summary graph, Theorem 2 gives $\lim_{T\to\infty} s_\rho(G_T) = L(G_S) = \frac{1}{2} + |E_\times|/(2|E_S|)$, and Theorem 5 gives $s_\rho^W = \frac{1}{2} + |E_\uparrow|/(2|E_S|)$ for every T. For $G_S \sim D(p, c/p)$ we take the limit $T \to \infty$ at fixed p first (this defines $L(G_S)$) and then let $p \to \infty$; for the stationary value no T-limit is needed. The regime of fixed T with $p \to \infty$ is not covered (the bound of Theorem 2 is vacuous for $T \le p$).

Model and notation. $D(p, c/p)$: each ordered pair (i, j), $i \neq j$, is a non-loop edge independently with probability $q = c/p$, and every node has a self-loop (loops change neither reachability nor Src). m is the number of non-loop edges, $|E_S| = p + m$, N the number of non-loop edges inside an SCC, so $|E_\times| = m - N$. out(v), in(v) are the sets of nodes reachable from v, resp. reaching v (v included), T_v^{out}, T_v^{in} their sizes; $v \rightsquigarrow w$ means $w \in \text{out}(v)$. $x = x(c)$ is the largest solution of $x = 1 - e^{-cx}$ (the survival probability of a Poisson(c) Galton–Watson tree), $\theta_K = \theta_K(c)$ the probability that its total progeny is at least K ($\theta_K \downarrow x$), and d the smallest fixed point in $[0, 1]$ of $f(d) = 1 + e^{-c} - e^{-c(1-x)d}$. A node is *in-isolated* if it has no non-loop parent. $\mathcal{G}$ denotes the *edge-rooted Poisson tree*: an edge (i, j); i has Poisson(c) parents and Poisson(c) further children, j Poisson(c) further parents and Poisson(c) children; every other node has Poisson(c) parents and Poisson(c) children besides the edge through which it was reached; all independent.

Textbook facts. (F1) Poisson coupling (van der Hofstad, 2016, Thm. 2.10): $d_{\text{TV}}(\text{Bin}(n, q), \text{Poi}(c)) \le nq^2 + |nq - c|$. (F2) Chernoff: for a sum X of independent Bernoulli variables with mean μ and $\delta \in (0, 1)$, $P(X \le (1-\delta)\mu) \le e^{-\delta^2\mu/2}$. (From $Ee^{-tX} \le \exp(\mu(e^{-t} - 1))$ with $t = -\log(1-\delta)$, Markov's inequality, and $(1-\delta)\log(1-\delta) \ge -\delta + \delta^2/2$; the upper-tail version $P(X \ge \mu + s) \le e^{-s}$ for $s \ge e^2\mu$ follows in the same way.) (F3) Galton–Watson facts (van der Hofstad, 2016, Thms. 3.1,

3.16): the extinction probability is the smallest root of $\eta = e^{c(\eta-1)}$, so the survival probability is $x(c)$, positive iff $c > 1$.

Deterministic lemmas (machine-checked, ErLemmas.lean; no loop hypothesis). A node z is a *source node* if every ancestor of z is reachable from z. By Theorem 5, $(i,j) \in E_\uparrow$ iff some source node reaches j but not i.

(W1) If $(i,j) \in E_\uparrow$, there are $k \notin \{i,j\}$ with $k \to j$ and a source node z with $z \rightsquigarrow k$, $z \not\rightsquigarrow i$ (upEdge_witness).

(W2) If $i \to j$, $k \to j$, and a source node reaches k but not i, then $(i,j) \in E_\uparrow$ (mem _upEdges_of_witness).

(W3) If $i \to j$, $k \to j$ and $\mathrm{in}(k) \cap \mathrm{in}(i) = \emptyset$, then $(i,j) \in E_\uparrow$ (mem_upEdges_of_disjoint _anc).

(W5) A non-loop edge (i,j) lies inside an SCC iff $j \rightsquigarrow i$. The event $\{j \rightsquigarrow i\}$ and the set $\mathrm{out}(j)$ do not change when the edge (i,j) is added or removed (a simple path from j to i does not use it). (Trivial; not formalised.)

Explorations and forced edges. Forward exploration of $\mathrm{out}(v)$ keeps explored, active and unseen sets X_t, A_t, U_t, $a_t = |A_t|$, $\mathrm{disc}_t = t + a_t$, $X_0 = \emptyset$, $A_0 = \{v\}$. Step $t+1$ takes the oldest active node w, reveals the pairs (w,u), $u \in U_t$, and makes the ξ_{t+1} heads of present pairs active; it stops at $T = T_v^{\mathrm{out}}$. Backward exploration of $\mathrm{in}(v)$ reveals the pairs (u,w) instead. Let F be a set of at most three ordered pairs and G^F the graph in which the pairs of F are present and all other pairs are independent $\mathrm{Be}(q)$. *Bookkeeping.* In a forward exploration a pair is revealed only when its tail is explored, and each node is explored once, so each pair is revealed at most once; the pairs revealed at step $t+1$ are disjoint from those revealed before and from all pairs whose tail is not in X_{t+1}. Hence, conditionally on the history $\mathcal{H}_t$, the non-forced pairs revealed at step $t+1$ are independent $\mathrm{Be}(q)$, and $\xi_{t+1} = \#\{u \in U_t : (w,u) \in F\} + \mathrm{Bin}(|U_t| - f_w, q)$ with $f_w \le 3$ the number of forced pairs from w into U_t. Thus, given $\mathcal{H}_t$, ξ_{t+1} stochastically dominates $\mathrm{Bin}(|U_t| - 3, q)$ and is dominated by $3 + \mathrm{Bin}(p,q)$. The same holds for backward explorations (with heads in place of tails), and, by the same argument, for a backward exploration run after a forward one, conditionally on the forward history, as long as it never discovers a node that the forward exploration discovered (only pairs with tail outside X_f and head explored backward are revealed, and none of them was revealed forward).

Fix $c > 1$ and $\varepsilon \in (0, 1/5)$ with $\beta = c(1-4\varepsilon) > 1$; put $n' = \lceil (1-4\varepsilon)p \rceil$, $\kappa = (\beta-1)^2/(8\beta)$, $\delta = (\beta-1)\varepsilon/(\beta+1)$, $t_0 = C_3 \log p$ with $C_3\kappa \ge 8$. Call an exploration *admissible* if, conditionally on its history, every offspring count ξ_{t+1} taken while $\mathrm{disc}_t < \varepsilon p$ dominates $\mathrm{Bin}(n', q)$ and every offspring count is dominated by $3 + \mathrm{Bin}(p,q)$. By the bookkeeping, forward and backward explorations in G^F are admissible for large p (since $|U_t| - 3 > (1-\varepsilon)p - 3 \ge n'$), and so is the backward exploration of Lemma 45 below.

Lemma 44 (dichotomy) *There is C (depending on c, ε) such that for large p every admissible exploration with final size T satisfies (a) $P(K \le T < \varepsilon p) \le Ce^{-\kappa K}$ for all $K \ge 1$; (b) $P\big(T \ge t_0$ and $[\mathrm{disc}_t < \varepsilon p$ for all t, or $a_\tau < \delta p$ at $\tau = \min\{t : \mathrm{disc}_t \ge \varepsilon p\}]\big) \le Cp^{-8}$.*

Consequently, with probability at least $1 - Cp^{-7}$*, every node* v *of* G *has, in both directions,* $T_v < t_0$ *or* $(T_v \geq \varepsilon p$ *and* $a_\tau \geq \delta p)$*.*

Proof Let ξ'_t be independent $\mathrm{Bin}(n', q)$ variables coupled so that $\xi_t \geq \xi'_t$ for $t \leq \tau$ (admissibility; after τ or after the exploration stops, ξ'_t uses external randomness), and $W_t = 1 + \sum_{s \leq t}(\xi'_s - 1)$. By induction $a_t \geq W_t$ for $t \leq \min(T, \tau)$. Since $tn'q \geq t\beta$ and $(\beta+1)/2 = (1 - \frac{\beta-1}{2\beta})\beta$, (F2) gives $P(W_t \leq (\beta-1)t/2) \leq P(\mathrm{Bin}(tn', q) \leq t(\beta+1)/2) \leq e^{-\kappa t}$, so $\mathrm{Bad}(t_1) = \{\exists t \geq t_1 : W_t \leq (\beta-1)t/2\}$ has probability at most $e^{-\kappa t_1}/(1 - e^{-\kappa})$. (a) If $T < \varepsilon p$ then $\tau = \infty$, so $0 = a_T \geq W_T$, which is $\mathrm{Bad}(K)$ when $T \geq K$. (b) On $\mathrm{Bad}(t_0)^c$ with $T \geq t_0$: for $t_0 \leq t \leq \min(T, \tau)$, $a_t \geq W_t > 0$, so the exploration cannot stop before τ, hence $\tau < \infty$. At τ, $a_\tau + \tau \geq \varepsilon p$, and if $\tau \geq t_0$ also $a_\tau > (\beta - 1)\tau/2$; minimising $\max((\beta-1)\tau/2, \varepsilon p - \tau)$ over τ gives $a_\tau \geq \delta p$; if $\tau < t_0$, $a_\tau \geq \varepsilon p - t_0 \geq \delta p$. $e^{-\kappa t_0} \leq p^{-8}$. The consequence is a union bound over p nodes and two directions. □

Lemma 45 (large out-sets reach large in-sets) *With probability at least* $1 - Cp^{-5}$*: for all* v, w*,* $T_v^{\mathrm{out}} \geq t_0$ *and* $T_w^{\mathrm{in}} \geq t_0$ *imply* $v \rightsquigarrow w$*.*

Proof Fix (v, w). Explore $\mathrm{out}(v)$ forward up to τ_f (history $\mathcal{H}_f$, discovered set $R_f = X_f \cup A_f$). Since every ξ is dominated by $3 + \mathrm{Bin}(p, q)$, $P(\max_t \xi_t \geq \varepsilon p) \leq e^{-\varepsilon p}$ for large p (F2, upper tail), so $|R_f| \leq 2\varepsilon p$ except on that event. Then explore $\mathrm{in}(w)$ backward, stopping at τ_b or when a node of $R_f \subseteq \mathrm{out}(v)$ is discovered; in the latter case $v \rightsquigarrow w$. Otherwise the backward exploration is the true exploration of $\mathrm{in}(w)$, its unseen set has at least $p - 2\varepsilon p - \varepsilon p - 3 \geq n'$ usable nodes while disc $< \varepsilon p$, and by the bookkeeping it is admissible conditionally on $\mathcal{H}_f$. By Lemma 44(b) for both explorations, outside probability $2Cp^{-8}$: if $T_v^{\mathrm{out}}, T_w^{\mathrm{in}} \geq t_0$ and no connection occurred, then $|A_f|, |A_b| \geq \delta p$. A_f and A_b are disjoint, and every pair $(a, a') \in A_f \times A_b$ is unrevealed (its tail is not in X_f, its head not in X_b), so given both histories these pairs are independent $\mathrm{Be}(q)$ (or forced present): $P(\text{no edge } A_f \to A_b) \leq (1-q)^{\delta^2 p^2 - 3} \leq e^{-c\delta^2 p/2}$. An edge $a \to a'$ gives $v \rightsquigarrow a \to a' \rightsquigarrow w$. Union bound over p^2 pairs. □

Let Ω be the intersection of the events of Lemma 44 (consequence) and Lemma 45, $P(\Omega^c) \leq Cp^{-5}$. Put $\mathcal{S} = \{v : T_v^{\mathrm{out}} \geq t_0\}$, $\mathcal{T} = \{v : T_v^{\mathrm{in}} \geq t_0\}$.

Corollary 46 (structure) *On* Ω*:* $\mathcal{S} \times \mathcal{T}$ *is contained in the reachability relation; every SCC other than the one containing* $\mathcal{S} \cap \mathcal{T}$ *has fewer than* t_0 *nodes; if* $|\mathcal{S} \cap \mathcal{T}| \geq t_0$*, this SCC equals* $\mathcal{S} \cap \mathcal{T}$*.*

Proof The first claim is Lemma 45. A node $u \notin \mathcal{S} \cap \mathcal{T}$ has $T_u^{\mathrm{out}} < t_0$ or $T_u^{\mathrm{in}} < t_0$, and its SCC lies in $\mathrm{out}(u) \cap \mathrm{in}(u)$. If u were in the SCC of $\mathcal{S} \cap \mathcal{T}$, that SCC (of size $\geq t_0$) would lie in $\mathrm{out}(u) \cap \mathrm{in}(u)$, a contradiction. □

(This re-proves the part of the Karp–Łuczak structure theorem we need; the statement is recorded in the introduction of Dovgal et al., 2020.)

Lemma 47 (local concentration; standard coupling) *Fix R and a function φ of the isomorphism class of the radius-R neighbourhood $B_R(e)$ of a non-loop edge e (all edges incident to nodes at undirected distance $< R$ from e, rooted at the oriented edge e), and let $\Phi_p = p^{-1}\sum_e \varphi(B_R(e))$, the sum over non-loop edges. (a) If φ is bounded, $E\Phi_p \to cE_{\mathcal{G}}\varphi$ and $\operatorname{Var}\Phi_p \to 0$; the same holds for node-rooted functionals with the node-rooted Poisson tree. (b) If $\varphi \geq 0$ and $\varphi(B_R(e)) \leq \operatorname{indeg}(j)$ for $e = (i,j)$, then $\limsup_p E\Phi_p \leq cE_{\mathcal{G}}[\min(\varphi, M)] + C/M$ for every M, with C not depending on p or M.*

Proof (a) First moment: $E\sum_e \varphi(B_R(e)) = \sum_{(i,j)} q\, E[\varphi(B_R((i,j); G \cup \{(i,j)\}))]$. Explore the undirected neighbourhood of (i,j) breadth first, revealing for each explored node w all pairs (w,u) and (u,w). On $\{|B_R| \leq M\}$, each explored node has $\operatorname{Bin}(|U|,q)$ new children and $\operatorname{Bin}(|U|,q)$ new parents among the unseen nodes, $|U| \geq p - M$, which (F1) couples with $\operatorname{Poisson}(c)$ up to an error $\leq Mq^2p + cM/p$ per count, and the pairs between already seen nodes are present with total probability $\leq M^2q$. So the truncated ball is within $C(M,R)/p$ in total variation of the truncated $\mathcal{G}$. Tightness: each exploration step is dominated by $\operatorname{Bin}(2p,q)$ (resp. $\operatorname{Poisson}(2c)$), so $E|B_R| \leq (1 + 2c + o(1))^R$ and $P(|B_R| > M) \leq C^R/M$ uniformly in p; let $p \to \infty$, then $M \to \infty$. Second moment: $E\Phi_p^2 = p^{-2}\big[\sum_e + \sum_{e,f \text{ sharing a node}} + \sum_{e,f \text{ node-disjoint}}\big]$. The first two sums are $O(p)$ since $E\sum_v d(v)^2 = O(p)$. For node-disjoint e, f, explore both balls jointly in $G \cup \{e,f\}$: on {both sizes $\leq M$} they are disjoint and non-adjacent except with probability $\leq C(M,R)/p$, and then the two explorations are coupled with independent copies of $\mathcal{G}$. Hence $p^{-2}\sum_{\text{disjoint}} \to (cE_{\mathcal{G}}\varphi)^2$ and $\operatorname{Var}\Phi_p \to 0$. (b) Apply (a) to $\min(\varphi, M)$ and bound the rest by $E\sum_{(i,j)} \operatorname{indeg}(j)\mathbf{1}[\operatorname{indeg}(j) > M] = E\sum_j \operatorname{indeg}(j)^2\mathbf{1}[\operatorname{indeg}(j) > M] \leq p\,E[D^3]/M$, $D \sim \operatorname{Bin}(p-1,q)$, and $E[D^3] \leq c^3 + 3c^2 + c$ for all p (uniform integrability). □

Local functionals. All of the following are functions of $B_{2K+1}(e)$ (a breadth-first search of the first K nodes of a set stays within distance $K-1$), are invariant under isomorphism, and count all qualifying nodes, so no labelling enters. $A_e^K = \mathbf{1}[T_j^{\text{out}} \geq K,\ T_i^{\text{in}} \geq K]$, with $E_{\mathcal{G}}A^K = \theta_K^2$ (the descendant tree of j and the ancestor tree of i are independent Poisson Galton–Watson trees). $\Psi_K(e) = \text{cl}_1 \vee \text{cl}_2$, with $\text{cl}_1 = [T_i^{\text{in}} < K$ and j has a parent $k \notin \{i,j\}]$ and $\text{cl}_2 = [$some in-isolated s has $T_s^{\text{out}} < K$, $j \in \text{out}(s)$, $i \notin \text{out}(s)]$. $\varphi_K^\beta(e) = \#\{k \notin \{i,j\} : k \to j,\ T_k^{\text{in}} < K,\ T_i^{\text{in}} < K,\ \text{in}(k) \cap \text{in}(i) \neq \emptyset\} \leq \operatorname{indeg}(j)$, with $E_{\mathcal{G}}\min(\varphi_K^\beta, M) = 0$ (in $\mathcal{G}$ the ancestor trees of i and of another parent of j are disjoint).

Proposition 48 $(L, c > 1)$ *For $c > 1$, $N/p \to cx^2$ in L^1 and in probability.*

Proof By (W5), $N = \sum_e Z_e$ with $Z_e = \mathbf{1}[j \rightsquigarrow i]$; compare with $\tilde{N}_K = \sum_e A_e^K$. (i) $Z = 1, A = 0$: $j \rightsquigarrow i$ and ($T_j^{\text{out}} < K$ or $T_i^{\text{in}} < K$). By (W5) these events do not depend on the presence of e, so the expected number of such edges is $p(p-1)q\,P(i \in \text{out}(j), T_j^{\text{out}} < K)$ plus the same with $\text{in}(i)$, and by exchangeability of the $p-1$ nodes other than j, $P(i \in \text{out}(j), T_j^{\text{out}} < K) = E[(T_j^{\text{out}} - 1)\mathbf{1}(T_j^{\text{out}} < K)]/(p-1) \leq K/(p-1)$. Total $\leq 2cK$. (ii) $Z = 0, A = 1$: on Ω, $T_j^{\text{out}} \geq t_0$ and $T_i^{\text{in}} \geq t_0$ would give $j \rightsquigarrow i$, so T_j^{out} or T_i^{in} lies in $[K, t_0) \subseteq [K, \varepsilon p)$. The expected number of such edges is at most $\sum_{(i,j)} q\,P_{G\cup\{e\}}(\cdot) \leq cp \cdot 2Ce^{-\kappa K}$ by Lemma 44(a)

in $G^{\{e\}}$; on Ω^c the count is at most m and $E[m\mathbf{1}_{\Omega^c}] \le (Em^2)^{1/2}P(\Omega^c)^{1/2} = O(p\cdot p^{-5/2})$. So $E|N-\tilde{N}_K| \le 2cK + 2cCpe^{-\kappa K} + o(1)$. By Lemma 47(a), $\tilde{N}_K/p \to c\theta_K^2$ in L^2; hence $\limsup_p E|N/p - cx^2| \le 2cCe^{-\kappa K} + c|\theta_K^2 - x^2| \to 0$ as $K\to\infty$. □

The same argument for the node functional $\mathbf{1}[T_v^{\text{out}} \ge K, T_v^{\text{in}} \ge K]$ gives $|\mathcal{S}\cap\mathcal{T}|/p \to x^2$, so by Corollary 46 the giant SCC has $(x^2+o_P(1))p$ nodes and all other SCCs fewer than $C_3\log p$, w.h.p.

Proposition 49 (the tree computation) *In $\mathcal{G}$, with Ψ_∞ defined by $T_i^{\text{in}} < \infty$ and $T_s^{\text{out}} < \infty$, $E_{\mathcal{G}}\Psi_K \uparrow P_{\mathcal{G}}(\Psi_\infty) = F_\uparrow$ as $K\to\infty$.*

Proof Both clauses increase in K to their $K=\infty$ versions (monotone convergence). Let $A(i)$ be the event that the ancestor tree of i is infinite and $D(j)$ that the descendant tree of j is infinite (probability x each, independent). The other parents of j are Poisson(c) many. (1) Not $A(i)$: Ψ_∞ holds iff j has another parent. Clause 1 needs exactly that; clause 2 needs it too, since a path from s to j ends with an edge $k\to j$ and $k=i$ would put i into out(s). Probability $(1-x)(1-e^{-c})$. (2) $A(i)$: clause 1 fails. Clause 2 needs (a) not $D(j)$, since out(s) contains the descendant tree of j (probability $1-x$), and (b) another parent k of j and a chain $k=u_0\leftarrow u_1\leftarrow\cdots\leftarrow u_r=s$ ending at an in-isolated s, in which every u_t has only finite descendant trees below its children other than u_{t-1} ($u_{-1}=j$); such an s does not reach i, because paths into i stay in the ancestor tree of i. The probability that all other children of a node have finite descendant trees is $E[(1-x)^{\text{Poi}(c)}] = e^{-cx} = 1-x$. Let $Q(v)$ be the event that v is in-isolated or some parent u of v has all other children finite and $Q(u)$ holds, and $d^* = P(Q)$. Parents are Poisson(c) many and each is *good* (other children finite and Q) independently with probability $(1-x)d^*$, so by Poisson thinning $d^* = e^{-c}+1-e^{-c(1-x)d^*} = f(d^*)$. Q is the increasing union of its depth-n truncations Q_n, with $P(Q_0)=e^{-c}=f(0)$ and $P(Q_{n+1})=f(P(Q_n))$; as f is increasing and continuous, $P(Q_n)=f^{n+1}(0)$ increases to the smallest fixed point d of f (if d' is a fixed point, $f^n(0)\le f^n(d')=d'$). So $d^*=d$. The number of good other parents of j is Poisson($c(1-x)d$), so (b) has probability $1-e^{-c(1-x)d}$, independently of $A(i)$ and $D(j)$. Adding (1) and (2) gives $F_\uparrow$. □

(An independent Monte Carlo of the event itself, not of the closed form, gives 0.38756 ± 0.00077 against 0.38773 at $c=1.5$ and 0.19049 ± 0.00062 against 0.18962 at $c=2$.)

Proposition 50 (reduction of $E_\uparrow$ to Ψ_K, $c>1$) $E\sum_e|\mathbf{1}[e\in E_\uparrow]-\Psi_K(e)| \le (\delta_1(K)+o(1))p$ *with $\delta_1(K) = C(c^2+K^2)e^{-\kappa K}+o_K(1)$, where $o_K(1)\to 0$ as $K\to\infty$ and C does not depend on K or p.*

Proof Let $e=(i,j)$ be a non-loop edge. *Clause 2 never errs*: an in-isolated s is a source node, so $j\in$ out(s), $i\notin$ out(s) give $e\in E_\uparrow$. *Case I: $T_i^{\text{in}} < K$.* (I-a) If $e\in E_\uparrow$, (W1) gives another parent of j, so clause 1 holds: no error. (I-b) Clause 1 holds and $e\notin E_\uparrow$. Then $\mathbf{1}[\text{I-b}] \le \sum_{k\notin\{i,j\}}\mathbf{1}[e,(k,j)\text{ present},\ T_i^{\text{in}}<K,\ e\notin E_\uparrow]$ (a union over *all* other parents; no parent is singled out). Fix the triple (i,j,k) and force $F=\{(i,j),(k,j)\}$. By (W3), $e\notin E_\uparrow$ forces $\text{in}(k)\cap\text{in}(i)\neq\emptyset$, and by (W2) every source node reaching k reaches i. Split on T_k^{in}:

(α_1) $K \le T_k^{\text{in}} < \varepsilon p$: probability $\le Ce^{-\kappa K}$ (Lemma 44(a), backward, in G^F). (α_2) $T_k^{\text{in}} \ge \varepsilon p$: the backward exploration of $\text{in}(k)$ explores at least $\varepsilon p/2$ nodes. When a node $y \ne j$ is explored, the pairs (u, y) with u unseen are revealed (all absent with conditional probability $\ge (1-q)^p$), and the pairs (u, y) with u already seen are never revealed by this exploration and are independent of it and of each other (distinct heads); so, given the history, y is in-isolated with probability $\ge (1-q)^p \ge e^{-c}/2$, and the number of in-isolated nodes among the first $\varepsilon p/2$ explored nodes dominates $\text{Bin}(\varepsilon p/2 - 1, e^{-c}/2)$. By (F2) at least K of them are in-isolated except with probability $e^{-\Omega(p)}$. Each in-isolated node of $\text{in}(k)$ is a source node reaching k, hence reaches i, so lies in $\text{in}(i)$; but $|\text{in}(i)| < K$. So (α_2) has probability $e^{-\Omega(p)}$. (β) $T_k^{\text{in}} < K$ and $\text{in}(k) \cap \text{in}(i) \ne \emptyset$. Summed over k this is $\varphi_K^\beta(e)$. The number of triples is p^3 and each has both pairs present with probability q^2, so cases (α_1), (α_2) contribute at most $c^2 p(Ce^{-\kappa K} + e^{-\Omega(p)})$; case ($\beta$) contributes $E\sum_e \varphi_K^\beta(e) \le (C/M + o(1))p$ for every M by Lemma 47(b), i.e. $o(p)$. *Case II:* $T_i^{\text{in}} \ge K$. Clause 1 fails. (II-0) $K \le T_i^{\text{in}} < \varepsilon p$: at most $cpCe^{-\kappa K}$ such edges in expectation (Lemma 44(a) in $G^{\{e\}}$). (II-1) $T_i^{\text{in}} \ge \varepsilon p$, so $i \in \mathcal{T}$. Work on $\Omega' = \Omega \cap \{\max_v \text{indeg}(v) \le C \log p\}$; $P(\Omega'^c) \le Cp^{-2}$, and the edges counted on Ω'^c contribute at most $E[m\mathbf{1}_{\Omega'^c}] \le (Em^2)^{1/2} P(\Omega'^c)^{1/2} = O(1)$. An error here is an edge $e \in E_\uparrow$ with $\Psi_K(e) = 0$. Take a witness source node z of (W1), in a source SCC σ. On Ω, $z \in \mathcal{S}$ would give $z \rightsquigarrow i$ (Lemma 45), so $T_z^{\text{out}} < t_0$. (A) $\sigma = \{z\}$ (i.e. z in-isolated) and $T_z^{\text{out}} < K$: clause 2 holds with $s = z$; no error. (B1) z in-isolated and $K \le T_z^{\text{out}} < t_0$. The number of such edges is at most $\sum_z \mathbf{1}[K \le T_z^{\text{out}} < t_0] \sum_{j \in \text{out}(z)} \text{indeg}(j)$. Given $\text{out}(z) = O$ with $|O| = r$ (a completed forward exploration), the pairs with tail outside O and head in O are unrevealed, so the expected number of edges into O is at most $r^2 + rpq = r^2 + cr$. With $P(T_z^{\text{out}} = r) \le Ce^{-\kappa r}$ (Lemma 44(a)), the expectation is at most $p\sum_{r \ge K} Ce^{-\kappa r}(r^2 + cr) \le C' pK^2 e^{-\kappa K}$. (B2) $|\sigma| \ge 2$. Then $|\sigma| < t_0$ (as $\sigma \subseteq \text{out}(z)$), every node of σ has out-set $\text{out}(z)$ of size $< t_0$, and the number of edges with head in $\text{out}(z)$ is at most $t_0 \cdot C \log p$ on Ω'. Let N_σ be the number of source SCCs with $2 \le |\sigma| < t_0$. For a set R of size r, "R strongly connected" and "no edge enters R" depend on disjoint sets of pairs; the first requires every node of R to have a parent in R (probability $\le (rq)^r$, distinct heads), the second has probability $(1-q)^{r(p-r)}$. Hence $EN_\sigma \le \sum_{r=2}^{t_0} \binom{p}{r} (rq)^r (1-q)^{r(p-r)} \le \sum_{r \ge 2} \left(ce^{1-c} e^{cr/p}\right)^r = O(1)$, using $\binom{p}{r} \le p^r/r!$, $r^r/r! \le e^r$, $r \le t_0 = O(\log p)$ and $ce^{1-c} < 1$ for $c \ne 1$. So the (B2) errors have expectation $O(1) \cdot t_0 C \log p = O(\log^2 p) = o(p)$: an expectation on the event Ω' plus the Cauchy–Schwarz bound for Ω'^c above. Adding all cases gives the claim. □

Theorem 51 (S, $c > 1$) *For $c > 1$, $|E_\uparrow|/p \to cF_\uparrow$, and therefore $S = \frac{1}{2} + |E_\uparrow|/(2(p+m)) \to \frac{1}{2} + cF_\uparrow/(2(1+c))$, in L^1 and in probability.*

Proof $E\big||E_\uparrow|/p - cF_\uparrow\big| \le E\sum_e |\mathbf{1}[e \in E_\uparrow] - \Psi_K(e)|/p + E|\Phi_p^{\Psi_K} - cE_{\mathcal{G}}\Psi_K| + c|E_{\mathcal{G}}\Psi_K - F_\uparrow|$; the terms are bounded by Prop. 50, Lemma 47(a) and Prop. 49; let $p \to \infty$, then $K \to \infty$. $m/p \to c$ in L^2 ($m \sim \text{Bin}(p(p-1), q)$), and S and L are bounded continuous functions of $(|E_\uparrow|/p, m/p)$ and $(N/p, m/p)$ on $[0, \infty)^2$. □

Proposition 52 ($c \le 1$) *(a) For $c < 1$, $EN \le c^2/(1-c)$. (b) For $c \le 1$, $N/p \to 0$ in L^1, so $L \to \frac{1}{2} + \frac{c}{2(1+c)}$ ($L(1) = 3/4$). (c) For $c \le 1$, $|E_\uparrow|/p \to c(1-e^{-c})$ in L^1, so $S \to \frac{1}{2} + \frac{c(1-e^{-c})}{2(1+c)}$ ($S(1) = 0.65803$); this is $F_\uparrow$ at $x = 0$.*

Proof (a) A non-loop edge inside an SCC lies on a simple directed cycle, and the expected number of pairs (simple cycle of length k, edge on it) is $k \cdot \frac{(p)_k}{k} q^k \le c^k$; sum over $k \ge 2$. (b) $c < 1$ from (a). $c = 1$: $D(p, 1/p)$ is a subgraph of $D(p, c'/p)$ for every $c' > 1$ under the monotone coupling, and an edge inside an SCC of the subgraph is inside an SCC of the larger graph, so $\limsup_p EN/p \le c' x(c')^2 \to 0$ as $c' \downarrow 1$ (Prop. 48). (c) Let $\Psi^1_K = \mathrm{cl}_1$. Case I ($T^{\mathrm{in}}_i < K$): (I-a) as before, no error; (I-b) with the same union over other parents k and forced $F = \{(i,j),(k,j)\}$, (W3) forces $\mathrm{in}(k) \cap \mathrm{in}(i) \ne \emptyset$, so either $T^{\mathrm{in}}_k \ge K$ or the event of φ^{β}_K holds. The expected number of triples with $T^{\mathrm{in}}_k \ge K$ is $E \sum_e \#\{k \notin \{i,j\} : k \to j,\ T^{\mathrm{in}}_k \ge K\}$, a local functional dominated by $\mathrm{indeg}(j)$, so by Lemma 47(b) it is at most $(c \cdot c\theta_K + C/M + o(1))p$; the φ^{β}_K part is $o(p)$ as before. Case II ($T^{\mathrm{in}}_i \ge K$): at most $E \sum_e \mathbf{1}[T^{\mathrm{in}}_i \ge K] = (c\theta_K + o(1))p$ (Lemma 47(a)). Hence $\limsup_p E \sum_e |\mathbf{1}[e \in E_\uparrow] - \Psi^1_K(e)|/p \le c^2\theta_K + c\theta_K + C/M$, which tends to 0 as $K, M \to \infty$ because $\theta_K \to x(c) = 0$ for $c \le 1$ (F3). Finally $E_{\mathcal{G}} \Psi^1_K = (1 - \theta_K)(1 - e^{-c}) \to 1 - e^{-c}$. □

What is not proved. Rates of convergence; the critical window $c = 1 + \lambda p^{-1/3}$ (Łuczak and Seierstad, recorded in Dovgal et al., 2020; not used); almost sure convergence; the regime of fixed T with $p \to \infty$. The proof does not use the structure theorems of Karp and of Łuczak (not opened by us); it re-proves the part it needs.

Numerical checks of the intermediate claims (fixed seeds, exact computation; supplement `x1proof/verify`). (W4) $E_\uparrow \subseteq$ edges into confluences: 0 violations in 48 graphs. Dichotomy (Lemma 44): at $c = 2$, $p = 10^5$, the mass of $K \le T < 0.02p$ is 0.0097, 0.0008, 0 for $K = 5, 10, 20$; at $c = 1$ it is 0.37, 0.25, 0.17 (critical, no dichotomy, as expected). Structure (Corollary 46): for $c \ge 1.5$ the second-largest SCC never had more than 2 nodes; giant fraction 0.63439 ($p = 10^6$, $c = 2$) against $x^2 = 0.63491$. Prop. 50 edge by edge: at $c = 2$, $p = 2 \cdot 10^5$, $|E_\uparrow| = 76{,}220$ and $\sum_e \Psi_{40}(e) = 76{,}220$ with no edge in either difference; clause 2 never erred in any run. Concentration: the standard deviation of N/p times $\sqrt{p}$ is 2.88–3.07 for $p = 5 \cdot 10^3$ to $3.2 \cdot 10^5$ ($c = 2$), as Lemma 47 requires.

Appendix P Accuracy Interpretation of s_ρ

An edge $\{x, y\}$ of a DAG is shown in a uniformly random order together with $(\rho_{\mathrm{first}}, \rho_{\mathrm{second}})$. A *pairwise count rule* is $f(a,b) \in [0,1]$, the probability of declaring "first is the tail"; the *sort rule* is $f_{\mathrm{sort}} = 1, \frac{1}{2}, 0$ for $a <, =, > b$. For the true edge $x \to y$ the accuracy is $\frac{1}{2} f(\rho_x, \rho_y) + \frac{1}{2}(1 - f(\rho_y, \rho_x))$.

Proposition 53 (elementary) *On every DAG, (i) the accuracy of f_{sort} is exactly s_ρ; (ii) every pairwise count rule has accuracy at most s_ρ; (iii) f_{sort} never errs on an edge with $\rho_x \ne \rho_y$.*

Lean: accRule_sortRule, accRule_sortRule_rho, accRule_le_sOf, accRule_le_sRho, edgeAcc_sortRule_strict, rho_not_gt_of_edge.

Proof By Reisach et al.'s Theorem 1, $\rho_x \le \rho_y$ on an edge $x \to y$. If $\rho_x < \rho_y$ the accuracy is at most 1, attained by the sort rule; if $\rho_x = \rho_y = a$ it is $\frac{1}{2} f(a,a) + \frac{1}{2}(1 - f(a,a)) = \frac{1}{2}$ for

every f, because the random order forces the tie decision to $1/2$. These are the edge scores. (i) restates the definition of s_ρ; (ii) is a three-line consequence of their Theorem 1. □

Combined with Theorem 5, the best accuracy of any pairwise count rule on the stationary unrolled DAG is $\frac{1}{2}+|E_\uparrow|/(2|E_S|)$, at most 0.665 for ER summary graphs and 0.75 for river-like in-forests ($\mu \leq 1$). The restriction to counts is necessary (exhaustive enumeration over labelled DAGs, not proved):

nodes p	2	3	4	5
Bayes rule on the two counts (= edge-pooled s_ρ)	0.5	0.5625	0.592262	0.605547
Bayes rule on counts and the two node degrees	0.5	0.625	0.660714	0.669238

For the skeleton $a - c - b$ the collider has counts $(2, 2, 3)$ (all edges strict) while the chain and the fork have all counts equal to 3; among the tied graphs the centre c is the tail with probability $2/3$ under a uniform prior, so a degree-aware rule beats the coin. So s_ρ is the Bayes accuracy of pairwise count rules only. This section is interpretive, not a new theorem.

Appendix Q Benchmark Experiment B1: Design, Results and Deviations

Process. The analysis plan (`b1/PLAN.md`) was committed before any simulation that produced an algorithm score; only graph-only exact values were computed before it. Two deviations were recorded in the plan after the validation run (seeds 9000–9004, used only for debugging and timing) and before the test run: D1, the burn-in was raised from 50 to 100 steps because the validation mixing check failed (error $2.8 \cdot 10^{-3}$, threshold 10^{-3}; $1.4 \cdot 10^{-10}$ after the change); D2, SID is undefined when PC returns a cyclic directed part, so PC is dropped from the SID comparison in those cells (930 instead of 1,190 tests). D3: a container restart killed the first test run after about half of one shard; no results file had been written and no partial score was inspected. The test run was redone from scratch with the same deterministic seeds. No threshold or hypothesis was changed. As with the other studies, the plan is internally time-stamped and agent-executed (Appendix J).

Design. Summary graphs with a self-loop at every node, $p \in \{5, 10\}$, 30 test graphs (seeds 1000–1029) per family: symmetrised ER (expected degree 2) and the directed ring (strongly connected), the chain, random recursive out- and in-trees, and directed ER $G(p, c/p)$ with $c \in \{1, 2\}$. Linear Gaussian VAR(1) with every weight $\sim$ Unif$(0.5, 1)$, rescaled to spectral radius 0.8 if larger, so relatives in the population equal graph relatives; noise s.d. $\sim$ Unif$(0.5, 1)$ per node (R-het, primary), equal noise (R-eq), and R-het standardised (R-std). G1: x_1 is noise, then $T - 1$ steps, truth G_T. G2: 100 latent burn-in steps, then the window of T slices, truth the window copy of G_T; the burn-in makes slice-B nodes latent common parents of the first window slice, so we also report SHD excluding pairs inside the first slice. G2s: exact stationary start via the discrete Lyapunov equation (replication of G2). Methods with

columns shuffled: rel-SortnRegress (Reisach et al. Algorithms 1–2, $\alpha = 0.05$), R^2-SortnRegress, Var-SortnRegress, a random order followed by the same regression step, FLOP (`flopsearch` 0.3.0, BIC penalty 2, 10 restarts), PC (causal-learn, Fisher z, $\alpha = 0.01$; not run at $p = 10$, $T = 10$ for time), and a labels-visible reference that sorts by time. DAGMA, NOTEARS and GOLEM were run in the separately planned extension of Appendix S. SID by `gadjid`.

Pre-specified hypotheses and verdicts. H1 (start helps rel-SortnRegress on non-strongly-connected families; at least 4/5 Holm-significant and pooled mean $\geq$ 0.02): supported, 5/5, pooled +0.125 $[0.116, 0.134]$; tracking (Spearman of cell means against the exact gap ≥ 0.5): 0.80; the plan's $p = 1.2 \cdot 10^{-10}$ treated the 42 cells as independent, and a graph-cluster permutation gives $p \leq 10^{-4}$ (post hoc, Appendix S); OLS $d = 0.53\,\text{gap} - 0.027$. H2a (gain below 0.03 in every strongly connected cell with $T \geq p$, the review's expectation): fails, as the exact gaps predicted before the run (e.g. ring $p = 5$, $T = 5$: +0.217; only symER $p = 5$, $T = 10$ is small, +0.019). H2b (non-negligible gain at $T < p$): holds, +0.13 to +0.31 in all six cells. H2c (symER gain decreases with T): holds, +0.195, $p = 1.2 \cdot 10^{-11}$. H3 (some Holm-significant SHD rank swap): supported, 11 swaps: rel- vs R^2-SortnRegress on chains ($p = 10$, $T = 3, 5$), Var-SortnRegress vs FLOP (3 cells) and vs PC (5 cells) on in-trees, and R^2-SortnRegress vs the random order on one ring cell. SID: 9 swaps; orientation score: 9; SHD without first-slice pairs: 9. Swap counting is a crude summary: the post hoc interaction analysis of Appendix S finds Holm-significant G1×G2 interactions in 294 of 595 pair×cell SHD tests. Median Kendall τ between the G1 and G2 rankings: 0.73. FLOP is best on average under both generators (normalised SHD 0.27 vs 0.46).

Two artefacts. Pooled G1−G2 order accuracy over all seven families: rel-SortnRegress +0.133 (the H1 value +0.125 is over the five non-strongly-connected families; unchanged by standardisation, since it is scale invariant); Var-SortnRegress +0.202 raw, +0.244 with equal noise, +0.002 standardised. Variance accumulates from the noise-only first slice, so the fixed start is var-sortable; standardising removes this but not the relatives artefact. Only the rel- vs R^2-SortnRegress swaps follow the mechanism of Theorems 2 and 5.

Cost of the burn-in. The labels-visible reference has normalised SHD 0.044 (G1) and 0.273 (G2), but 0.041 vs 0.048 without first-slice pairs (G2s identical): stationary-start benchmarks should score the window without first-slice pairs or as an ADMG.

Exact curves and the Δ bound. Figure Q1 shows exact $s_\rho(G_T)$ for $T = 2, \ldots, 20$. At $T = 3$ it lies between 0.81 and 1.00; at $T = 20$ symER is still at 0.55–0.57 and the ring with $p = 10$ at 0.74. The Δ form of Theorem 2 was checked against brute-force relatives enumeration on 3,000 random summary graphs × 7 horizons: no violation of either bound, equality in 3,164 cases, strictly sharper than the $p-1$ bound in 10,186. For Theorem 5, $B = \Delta$ gave the stationary value in 726 of 726 graphs and $B = \Delta - 1$ did not in 445, so Δ cannot be lowered in general.

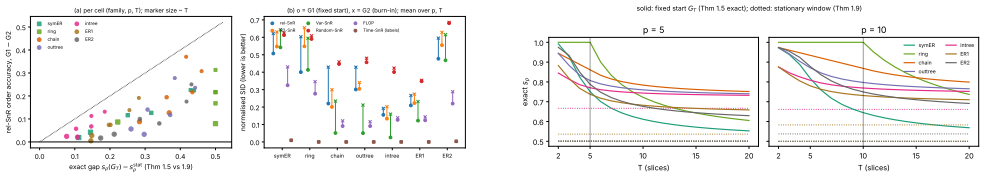


Fig. Q1 Left: per-cell G1−G2 gain in rel-SortnRegress order accuracy against the exact population gap (all 42 cells below the diagonal and above 0), and normalised SID per method under G1 and G2. Right: exact $s_\rho(G_T)$ against T, mean over the 30 test graphs per family and p.

Appendix R B1-NL: Replication under Nonlinear Dynamics, Signed Weights and Uniform Noise

Process. A separate plan (`b1nl/PLAN.md`) was committed before any algorithm score; two notes (a faster but output-identical diagnostic, and the limited power of sanity rule S1) were added after the validation run and before the test run. No threshold, hypothesis or seed changed. The test run (5,040 data sets) was run once.

Design. As B1 (Appendix Q) with the B1 code imported unchanged: seven families, $p \in \{5, 10\}$, $T \in \{3, 5, 10\}$, $n = 1000$, G1 against G2 (burn-in 100), R-het and R-std. Three parametrisations. NL: additive VAR with a linear self-loop and cross terms $w\,g(x)$, g drawn from tanh, sin and $(x + x^2/2)/(1 + x^2/4)$, positive w, stability by scaling the Lipschitz-bound matrix to spectral radius at most 0.8. SGN: linear, all weights $\pm\text{Unif}(0.5, 1)$. SGNU: SGN with uniform noise. New seeds (25 graphs per family and p for NL and SGN, 10 for SGNU); PC and R-eq dropped and G2s replaced by a 500-step burn-in, to fit one core (stated in the plan).

Results. H1 (same rule as B1) supported in all three: 5/5 families Holm-significant ($p \le 1.4 \cdot 10^{-9}$); pooled gain +0.108 [0.100, 0.116] (NL), +0.137 [0.128, 0.146] (SGN), +0.137 [0.123, 0.152] (SGNU); positive in all 126 cells. Tracking: Spearman 0.68, 0.82, 0.78 (slopes 0.34, 0.56, 0.56). Under NL the correlation-count estimator recovers about 65% of the graph relatives (recall 0.67 under G1, 0.62 under G2), which explains the smaller slope, as stated before the run. Under signed weights sample relatives can deviate from graph relatives (out-trees under G2: estimated rel-sortability 0.455), yet the gain is as large as in B1. H3: one Holm-significant SHD swap under NL (R^2-SortnRegress against the random order, chains, $p = 10$, $T = 10$), none under SGN, not testable for SGNU (10 seeds, stated in the plan). By the plan's rule H3 is therefore supported under NL, but on a substantively minor swap, and we do not count it as a replication of B1's swaps; B1's rel- against R^2-SortnRegress chain swap was not detected on SHD at this replication's power, although the interaction replicates (Appendix S). Without PC, 5 of B1's 11 swap types cannot occur. FLOP is the best label-free method run there under both generators in every configuration. The var/rel split holds: Var-SortnRegress gains +0.19, +0.24, +0.24 raw and about 0 standardised; rel-SortnRegress +0.11, +0.15, +0.14. Sanity rules pass (S1 at most 0.015; mixing error at most $5.6 \cdot 10^{-5}$).

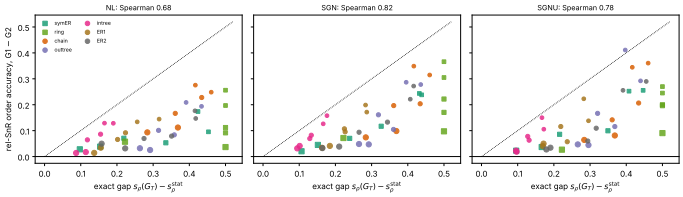


Fig. R2 B1-NL: per-cell G1−G2 gain in rel-SortnRegress order accuracy against the exact population gap, for NL, SGN and SGNU.

Appendix S Competitive Learners, Re-Analysis, Detectability Oracle and Generator Survey

Material in `b1x/` and `lit3/` of the supplement. The learner extension is a planned experiment; the re-analysis and the oracle are post hoc and use no new data.

Learner extension: process and design. The plan (`b1x/PLAN.md`), code and a validation timing were committed and pushed before any score (Appendix J); the test run took 4.21 h on one core and was analysed once, with no deviation (the size rule kept 12 seeds after a projected 4.35 h). B1's generators and SEM, new seeds: families ring, chain, in-tree and ER(c=2), $p \in \{5, 10\}$, $T \in \{3, 5\}$, 12 graphs per (family, p), i.e. 96 summary graphs and 384 data sets (G1 and G2), each analysed raw and standardised with time labels hidden. DAGMA linear (`dagma` 1.1.1, $\lambda = 0.03$), NOTEARS linear (the authors' reference code, $\lambda = 0.1$), GOLEM-EV/NV (a numpy port of the authors' code with the same score, optimiser, defaults and post-processing, gradient-checked; $p = 5$ only, as planned), all thresholded at 0.3; FLOP, the SortnRegress baselines and the empty graph as comparators; PCMCI+ (tigramite, ParCorr, $\tau_{\max} = 1$) on the time-indexed data as a labels-using reference. Metrics: SHD/$|E|$ (empty graph = 1), normalised SID, ancestral order accuracy; unit of inference the summary graph. T= 10 and the families symER, out-tree and ER(c=1) were excluded for run time.

Verdicts. HX1 (some raw continuous method Holm-significantly favoured by the fixed start): supported; all four raw variants improve on all three metrics (DAGMA SHD/$|E|$ -0.548 $[-0.663, -0.438]$, SID -0.147, order accuracy $+0.197$; NOTEARS -0.419, -0.127, $+0.156$; GOLEM-EV -0.491, -0.182, $+0.225$; GOLEM-NV -0.409, -0.193, $+0.232$; Holm $\leq 4 \cdot 10^{-8}$). HX2 (opposite-sign, both Holm-significant rank change against FLOP on SHD/$|E|$, 64 tests): supported, two changes, both on in-trees: DAGMA−FLOP -0.218 under G1 (Holm 0.012) and $+0.306$ under G2 ($1.4 \cdot 10^{-5}$); NOTEARS−FLOP -0.176 (0.0011) and $+0.193$ (0.020); the same two appear on SID and order accuracy, and DAGMA's persists without first-slice pairs (0.08 / 0.73 against FLOP's 0.30 / 0.37). HX3 (standardisation removes the gain): DAGMA's SHD gain removed (-0.005 $[-0.027, 0.016]$), all other pairs reduced but not removed by the plan's rule; on order accuracy the standardised effect is significantly negative for DAGMA (-0.064), GOLEM-EV (-0.093) and GOLEM-NV (-0.097), and NOTEARS keeps $+0.036$. rel-SortnRegress replicates ($+0.187$ $[0.169, 0.205]$), and B1's chain swap of rel- against R^2-SortnRegress recurs on the new seeds. No fit failed; NOTEARS on

standardised data returned a cyclic graph in 15.6% of data sets (SID missing for 60 of 384). Best label-free method by mean SHD/$|E|$ per (family, p): FLOP in 8/8 cells under G2; under G1 DAGMA on chains with $p = 5$ (0.09 vs 0.10) and on in-trees with $p = 10$ (0.06 vs 0.33), NOTEARS on in-trees with $p = 5$ (0.09 vs 0.27). Raw continuous methods beat the empty graph on average under both generators but are worse than it on chains under G2 (1.14–1.25). Standardised DAGMA and GOLEM orient most tree edges backwards under G1 (order accuracy 0.26–0.27). The gains and the reversals are largest on chains and in-trees, where G1 makes the window almost perfectly var-sortable (Var-SortnRegress 0.95–0.96), and absent or reversed on rings and ER(c=2). This is consistent with a var-sortability effect, but the experiment does not establish it.

Predictions of the plan and what the attribution rests on. The plan stated three predictions. For HX1 it predicted favourable effects on raw data for all continuous methods; this held. For HX2 it predicted no rank change against FLOP on SHD/$|E|$; this failed (two changes). For HX3 it predicted "reduced, not removed" for DAGMA and NOTEARS on order accuracy, because some scale-free gain should survive standardisation. By the plan's rule both were labelled "reduced, not removed", but for DAGMA the standardised effect is negative (-0.064), so no gain survives; for NOTEARS a gain survives ($+0.036$ order accuracy, -0.077 SHD/$|E|$, -0.073 SID, all Holm-significant). The standardised runs are a weak test of the attribution to var-sortability. Standardised DAGMA and GOLEM have order accuracy below 1/2 under both generators (DAGMA 0.35 / 0.41, GOLEM 0.31–0.33 / 0.41–0.42), so they cannot show a scale-free effect of either sign. NOTEARS's remaining gain is a scale-free start effect on a competitive method. The fixed start also raises R^2-sortability, which is scale free and outside our theory, and we did not separate it. In B1 the gain in Var-SortnRegress order accuracy is the same on rings ($+0.224$) and chains ($+0.231$), while the learners gain on chains and not on rings; the explanation by levels (0.73 against 0.93 under G1) was made after the results. A control that rescales each data set to the other generator's marginal variances would test the attribution; it has not been run. GOLEM did not overtake FLOP on in-trees (G1 differences -0.011 and -0.042, Holm 1 and 0.46). Table 3 omits GOLEM-EV (SHD/$|E|$ 0.44 / 0.93 raw) and standardised GOLEM (0.97 / 1.30 EV, 0.95 / 0.99 NV).

Re-analysis of B1 and B1-NL (post hoc). Graph-cluster bootstrap CIs (10,000 resamples, the three horizons of a graph kept together) are narrower than data-set bootstrap CIs, because T is a fixed design factor; they are used in Table 2. Tracking: cell-level Spearman 0.80 $[0.76, 0.82]$, unit-level 0.62 $[0.59, 0.64]$, slope 0.53 $[0.50, 0.56]$ in B1; 0.68/0.44/0.34 (NL), 0.82/0.64/0.56 (SGN), 0.78/0.62/0.56 (SGNU); graph-cluster permutation $p \leq 10^{-4}$ in all four (the floor). Interactions $I = [m_1 - m_2](\text{G1}) - [m_1 - m_2](\text{G2})$ per graph: Holm-significant in 294/595 (B1), 113/420 (NL), 232/420 (SGN) pair×cell SHD tests and 0 for SGNU (10 graphs cannot pass Holm). In-tree Var-SortnRegress−FLOP on SHD: -68.4 $[-77.0, -60.4]$ (B1), -49.9 (NL), -75.5 (SGN), -75.5 (SGNU); SID -0.12 in all four. Chain rel-$-R^2$-SortnRegress: SID and orientation interactions replicate at 60–80% of B1's size; the family-pooled SHD interaction is about 0 in B1-NL, but in B1's swap cell ($p = 10$, $T = 3$) it keeps its sign (-52, -16, -32). Empty-graph reference (B1, all families, G1 / G2, SHD/$|E|$):

rel- 1.05/1.70, R^2- 1.33/1.96, Var- 0.61/1.39, random 1.72/2.04, FLOP 0.27/0.46, PC 0.40/0.64; the empty graph has normalised SID 0.50. Order accuracy without edges leaving the first window slice (non-SC families): rel-SortnRegress +0.089 $[0.079, 0.100]$ (B1), +0.067 (NL), +0.087 (SGN), +0.095 (SGNU). Multiplicity: each confirmatory claim of B1, B1-NL, B3 and the extension gets one claim-level p (the k-th smallest Holm p for "k of m families" rules, the best swap for swap rules, the cluster permutation for tracking, the plan's Holm p for B3); Holm over the 21 claims at family-wise 0.05 keeps all but B1-NL H3-SGN ($p = 0.108$); the closest survivors are B1 H3 (Holm 0.0079), B1-NL H3-NL and HX2 (both 0.033). Several claim p are at resolution floors (10^{-4} for permutations, 10^{-5} for B3's sign flips).

Detectability calculation ("oracle"; post hoc). For every B1 test graph, parameter set and horizon we propagated the exact window covariance (G1: $\Sigma_{t+1} = A^\top \Sigma_t A + D$ from $\Sigma_1 = D$; G2: after the same 100 burn-in steps). Population relatives ($|\rho| > 10^{-20}$) reproduce the exact s_ρ of Theorems 2 and 5 in all 2,520 cases. The oracle replaces each sample correlation in rel-SortnRegress's count by the probability that the Fisher-z statistic at $n = 1000$ exceeds the algorithm's threshold ($\varepsilon = 0.062$) and predicts each edge's order accuracy as $P(c_i < c_j) + \frac{1}{2}P(c_i = c_j)$, treating the indicators as independent. It uses no simulated data. Against the observed G1−G2 gain: cell level Spearman 0.99, $R^2 = 0.97$, slope 1.03, intercept −0.010; unit level $R^2 = 0.61$ against 0.34 for the exact gap (difference +0.27 $[0.23, 0.30]$); in a joint regression the gap's coefficient is −0.05. Predicted G1 / G2 levels: chain 0.756/0.528 (observed 0.749/0.532), ring 0.702/0.503 (0.687/0.507), in-tree 0.746/0.677 (0.743/0.673). Without slice-1 edges the non-SC gain is predicted to fall from +0.130 to +0.096 (observed +0.125 to +0.089). What this shows and does not show: the calculation is the expectation of rel-SortnRegress under the exact covariances of each generator, so the agreement says that the simulated gain has the size those covariances imply and that the independence approximation is adequate. It uses the full start-dependent covariances, so it includes every way in which the start changes pairwise correlations: pairs that are relatives under the fixed start only (the mechanism of Section 3) and pairs that are relatives under both starts but differ in correlation strength (a transient that needs no graph argument). We did not decompose the predicted gain into these two parts, so the calculation does not show that the gain is due to the first. The exact gap therefore predicts sign and ranking only.

Generator survey. We read the source code of tigramite 5.2.10.1 (`structural_causal_process`: zeros plus noise, burn-in $\lfloor 0.2T \rfloor$; `var_process`: i.i.d. $N(0, 1)$ start, no burn-in), the CDML generator (Lawrence et al., 2021) (zeros, $2\tau_{\max}$ steps dropped), CausalTime (Cheng et al., 2024) (rollouts of 20 steps from a real-data window, no burn-in, in the demo), UUMCdata (Herman et al., 2025) (exact stationary start), causalnex 0.12.1 (zeros, burn-in 100 or more; data with lag labels), the NeuralGC VAR and Lorenz-96 simulators (Tank et al., 2021) (burn-in 100 and 1000) and gCastle 1.0.4 (no continuous time-series simulator; its Hawkes simulator starts with an empty history). cdt, causaldag, causal-learn and lingam ship no time-series benchmark generator. Lohse and Wahl (2025) state a burn-in without its length. The time-unrolled construction of Reisach et al. (2026) starts at $t = 1$ and gives every slice to an i.i.d. method as a separate variable; its code is not public, so only the graph-level start is

verified. CauseMe's documentation could not be opened with TLS verification, so we make no claim about it. File and line references are in `lit3/GENERATOR_SURVEY.md`. Reading of exposure: the burn-in lengths above can be compared with $\Delta\tau_{\max}$, which concerns the graph artefact only. The variance artefact needs the start to be far from the stationary distribution relative to the mixing time of the process, which depends on the spectral radius of the generated coefficients; we computed no mixing time for any of these generators, so we do not grade them on it. All of them except the time-unrolled construction produce a single record with time order for time-series methods, so neither artefact applies to them as shipped; they would be exposed only if windows from their first steps were given to a method that ignores time order.

Appendix T B3: Discriminating Test of the Source-SCC Rule on Held-Out Data

Process. The plan (`b3/PLAN.md`, with an OSF-formatted copy) was pushed before any byte of the held-out data was read; zip CRC-32 values were taken from the remote archive directory before the plan and SHA-256 hashes at download. The plan is internally time-stamped only: it was not posted to an external registry before data access, and results were committed 3–4 minutes after the plan. A disclosed development check on study-3 data from the same device had shown every colour edge to be detectable, so the primary outcome was expected.

Data. Causal Chambers experiments not previously downloaded by the project. H1 (primary): `lt_camera_walks_v1`, experiments `ar_1_uniform_{ref, pol_1_90, pol_12_30}`: five independent stationary AR(1) actuators drive the light sensors, so the colour block is one weak component with three source SCCs; $3 \times 30{,}000$ steps, 30 units of 3,000. H2 (co-primary): `lt_camera_v1/scm_5_red` and `scm_5_blue`, i.i.d. data from the collider R→G←B, 20 units. Secondary: `scm_5_green`, `scm_4_green`, `scm_4_blue`, and `scm_4_red` as a single-source control.

Test. E1's estimator unchanged. Competitors: constant 1/2, the weak-component-size rule, the theorem on the design graph G_D, and E1's eight naive rules. Primary set D^*: instances where the theorem and the weak-component rule disagree (56% of instances for H1, 47% for H2, from graph-only computation before the plan); on D^* the prediction is the same on G_{lin} and G_D. Failure of the estimator pushes observed values towards the tie rules, so a pass on D^* cannot come from ties. Sign-flip tests over units, Holm, family $\alpha = 0.025$; pass requires $\Delta \leq -0.10$ against every competitor that disagrees on D^*.

Results. H1: on 2,361 instances, counts sorted as predicted in 2,356; $\Delta = -0.498$ against both tie rules, 30/30 units, $p \approx 10^{-5}$; non-relatives declared dependent 0.054, relatives detected 0.990. H2: $\Delta = -0.50$ against the tie rules and the sensor indicator, 20/20 units; R→G sorted 264/264 when B is observed and tied 123/123 when not. S1 (predicted tie, detected edge; 1,251 instances): theorem beats the eight naive rules ($\Delta = -0.43$, 30/30 units) and the theorem on G_D narrowly ($\Delta = -0.21$, 21/30 units, Holm $p = 0.021$). No kill rule triggered.

Failures. `scm_4_blue`: the tie rule is significantly better ($\Delta = +0.115$, 0/10 units, reverse $p = 0.001$); post hoc, the intervened blue channel ranges over 1–13

of 255 levels, so none of its edges is detected, and the pooled detection prerequisite (0.837) did not catch this. The pooled secondary analysis still passes, in 20/30 units. In `ar_1_uniform_pol_1_90` the polariser linear-zero refinement of Appendix K fails (polarisers correlate with ir_3, vis_3 at $|r| \approx 0.2$–0.28; G_D matches 0.959 against 0.041 for G_{lin}); H1 does not depend on it. Deviations: the inputs of two experiments do not reproduce from the published generator seeds (ranges match; not gating).

Appendix U Proofs for Sections 4.2 and 4.3

Proposition 8 is proved in the main text. The proofs below are paper proofs; none is machine-checked. Notation as in Section 4.3.

Proof of Proposition 9 (a) Every edge of G_T goes from a slice t to slice $t+1$, so a directed path ending in $v(t)$ has at most $t-1$ edges, and the self-chain $v(1) \to \cdots \to v(t)$ has exactly $t-1$. Graph isomorphisms preserve height, so two labellings of the same column DAG as unrollings with self-loops put the same columns in slice t, and T is the largest height and p the number of columns of height one. (b) The column DAG of the ring $i \to i+1$ with self-loops has the edges $(i,t) \to (i,t{+}1)$ and $(i,t) \to (i{+}1,t{+}1)$. With the new unit label $i' = i - t \pmod p$ the first family reads $i' \to i'-1$ and the second $i' \to i'$, which is the unrolling of the reversed ring with self-loops. (c) Without loops the unrolling of $0 \to 1$ has the edges $0(t) \to 1(t{+}1)$, $t < T$, which are pairwise disjoint; every one of them has a tail of height one and a head of height two. □

Proof of Proposition 10 Two parents $i(t)$ and $k(t)$ of a node $j(t{+}1)$ lie in the same slice and are not adjacent, so every node with at least two parents is an unshielded collider, and all its incoming edges have the same direction in every member of the class. The remaining edges are those into nodes with exactly one parent; since every node $j(t{+}1)$ has the parent $j(t)$, these are the self-chain edges of loop-only units. Fix such a unit j and a member of the class. On the chain $j(1), \ldots, j(T)$ the member has no collider (a collider at a chain node would be unshielded, as chain neighbours at distance two are not adjacent, and G_T has none there), so the chain is oriented away from one of its nodes $j(s)$. Conversely, re-rooting the chain at any $j(s)$ keeps the skeleton, creates no collider (every chain node still has at most one parent, because a loop-only unit has no parent outside its chain), removes none, and creates no cycle (a directed cycle would have to enter the chain from outside, and no edge does). The chains of different loop-only units are disjoint and can be re-rooted independently, which gives T^{m_0} members. □

Proof of Proposition 11 (a) Let G' be the column DAG of a sparsest representation. Σ is Markov to G', because the representation is a linear structural equation model on G' with independent noise terms (the start covariance is diagonal under the fixed-start law). G' has $(T-1)|\operatorname{supp} A'|$ edges, at most as many as G_T, because the true parameter is itself a representation. As the distribution is faithful to G_T, a DAG to which it is Markov has at least as many edges as G_T, with equality only if it is Markov equivalent to G_T (Chickering, 2002, Theorem 4). By Proposition 10, G' is G_T with the chains of some loop-only units re-rooted. G' is also an unrolling with self-loops, so by Proposition 9(a) every edge of G' joins consecutive heights, and every node of height at least two has a self-parent, with different nodes having different self-parents. Re-root the chain of a loop-only unit j at $j(s)$. If $1 < s < T$, then $j(s)$ has no parent in G', and $j(s{-}1)$ and $j(s{+}1)$ both have $j(s)$ as their only parent, so

both would have the self-parent $j(s)$: impossible. If $s = T$, then $j(1)$ has height T in G'; if j has a child $k \neq j$ in G_S, the edge $j(1) \to k(2)$ is an edge of G' (it enters a node with two parents or more, namely $j(1)$ and $k(1)$), which gives $k(2)$ the height $T + 1$: impossible. So only isolated units can be reversed, and only as a whole. (b) Let v_t be the variance of $x_j(t)$, $v_{t+1} = a^2 v_t + d$. The reversed chain is a representation iff the regression of $x_j(t)$ on $x_j(t+1)$ has the same coefficient $a v_t / v_{t+1}$ and the same residual variance $d\, v_t / v_{t+1}$ for all $t \in \{1, \ldots, T-1\}$, i.e. iff $v_t / v_{t+1} = r$ does not depend on t. For $T = 2$ there is one transition and nothing to compare. For $T \geq 3$, $v_3 - v_2 = a^2(v_2 - v_1)$ gives $\frac{1}{r}(\frac{1}{r} - 1) = a^2(\frac{1}{r} - 1)$, so $r = 1$, which is stationarity, or $1/r = a^2$, which forces $d = 0$ and is excluded. If $r = 1$ for the first two transitions then v_t is constant for all t. (c) follows from (a) and (b). □

Proof of Proposition 12 (a) In the truth, $a_2 = \alpha a_1 + \varepsilon_a$ and $b_2 = \beta a_1 + \gamma b_1 + \varepsilon_b$, with an arbitrary covariance of (a_1, b_1). Take the units $a' = (a_1, a_2)$ and $b' = (b_2, b_1)$, i.e. the first slice $\{a_1, b_2\}$ and the second slice $\{a_2, b_1\}$. Then $a_2 = \alpha a_1 + 0 \cdot b_2 + \varepsilon_a$ and $b_1 = \beta' a_1 + \gamma' b_2 + \eta$, where η is the residual of the linear regression of b_1 on (a_1, b_2). By construction η is uncorrelated with the new first slice; it is a linear function of $(a_1, b_1, \varepsilon_b)$ and therefore uncorrelated with ε_a, and ε_a is uncorrelated with (a_1, b_2). So the new innovation covariance is diagonal and uncorrelated with the new first slice, whose covariance is free. The new self-loop coefficients are $\alpha \neq 0$ and γ', which is non-zero whenever $\gamma \neq 0$ (the partial correlation of b_1 and b_2 given a_1 is non-zero); the support of the new transition matrix has three entries when $\beta' \neq 0$ and two otherwise, so the new representation has no more edges than the truth. (b) A stationary Gaussian AR(1) process has a symmetric autocovariance, so the reversed window has the same covariance, and an isolated unit is uncorrelated with all other columns. (c) $\Gamma = \sigma^2 (I - A^2)^{-1}$ solves $\Gamma = A \Gamma A^\top + \sigma^2 I$ and commutes with A. The time-reversed stationary process has transition matrix $\Gamma A^\top \Gamma^{-1} = A$ and innovation covariance $\Gamma - A \Gamma A^\top = \sigma^2 I$, so the reversed window is a representation with the same support. □

On Conjecture 13. Let H be G_T plus a complete DAG on the first slice. Under the stated faithfulness, the argument of Proposition 11(a) shows that the DAG of a sparsest representation (its unrolling plus a complete DAG on its first slice) is Markov equivalent to H, hence has the same skeleton U. In H the edges between slices 1 and 2 are not fixed by colliders, because the parents of a slice-2 node lie in the first-slice clique. If the only partition of the nodes of U into T layers of size p, with edges inside layer 1 or between consecutive layers and a perfect matching between consecutive layers, is the true one, the conjecture follows; the reversed layering is excluded when slice T is an independent set of U and slice 1 is not. We checked uniqueness by hand for the summary edge $a \to b$ with self-loops and $T = 3$; Proposition 12(a) shows that it fails for $T = 2$. We have no proof for general graphs, and the case in which the first slice is not a clique in U is not covered.

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